

General Linear Models using SPSS

@ Kolling Institute and
Northern Clinical School

Presented by

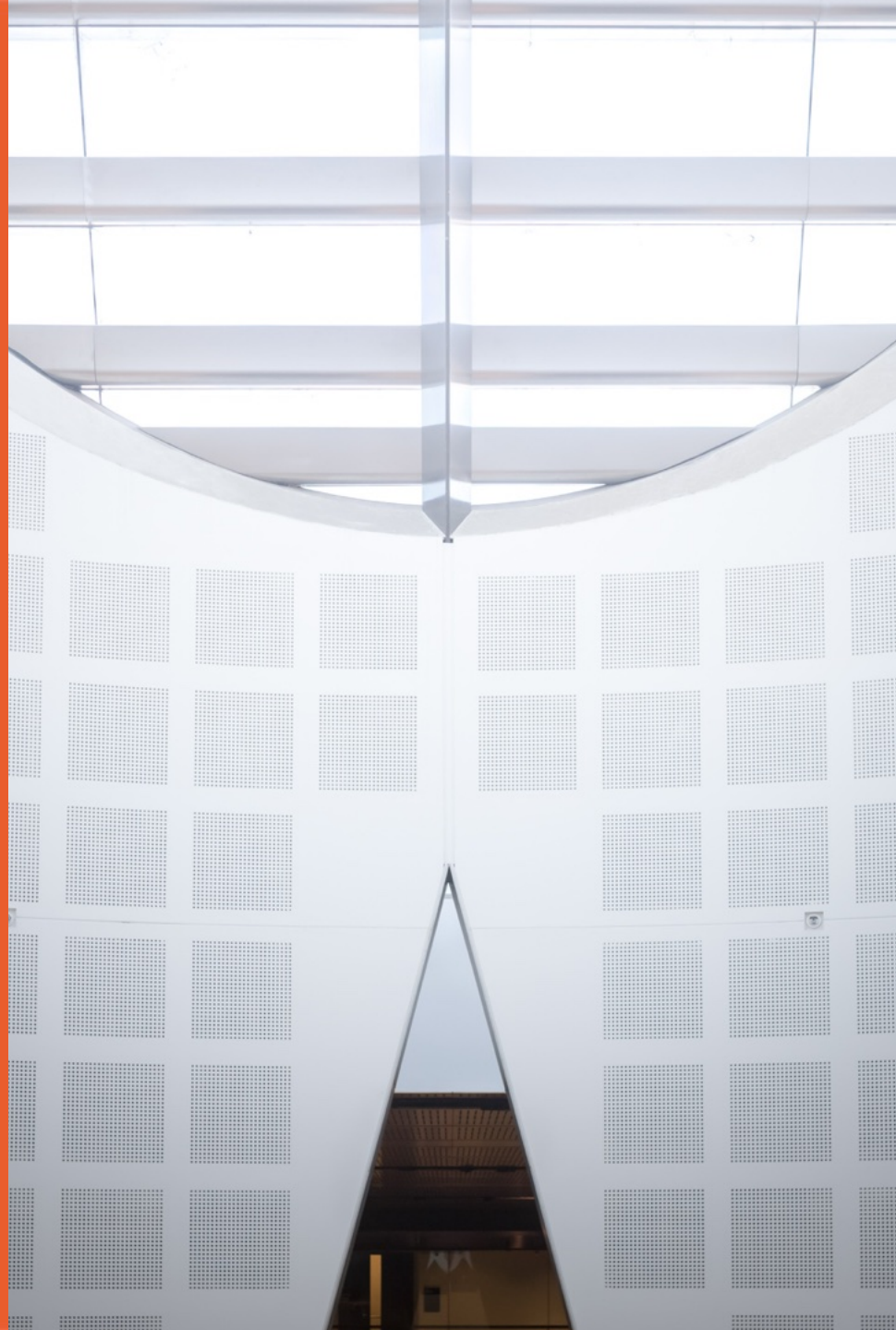
Jim Matthews

Senior Consultant: Statistics
Sydney Informatics Hub
Core Research Facilities
The University of Sydney

December 2019



THE UNIVERSITY OF
SYDNEY



Outline

- General Linear Model – introduction
- Introduction to SPSS
- Workshop examples
 - Two-way ANOVA
 - Repeated measures using GLM Repeated Measures
 - Linear regression
 - Binary Logistic Regression
- SPSS Syntax
- Overview of Generalized Linear Models in SPSS

What is the General Linear Model

The General Linear Model has many forms, varieties and names. Some you might have heard of are:

- ANOVA
- ANCOVA
- MANOVA
- Mixed Model
- Repeated Measures ANOVA
- Factorial Model
- Random Effects Model
- Generalised Linear Model (encompasses GLM)

Introduction to SPSS

Open SPSS

- Open the first example file: “grocery_1 month.sav”
- you will find the sample files on your PC with your installed copy of SPSS, eg. Mac location is

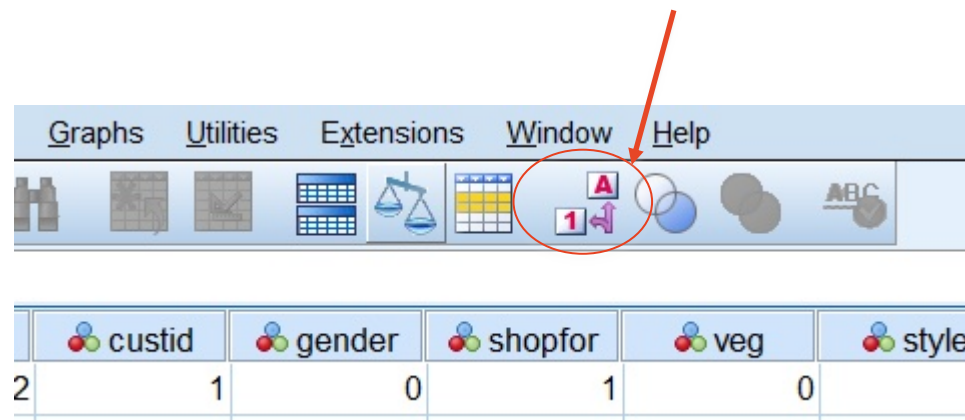
Apple SSD > Applications > IBM > SPSS > Statistics > 25 > Samples >

- Have a look at the “Data View”. Note the variable types in the column headings
- Eg Scale  amtspent Ordinal  size Nominal  storeid

Introduction to SPSS

Open SPSS

- Switch to “Variable View”. Note the variable properties including “Name”, “Label” and “Measure”
- Switch back to Data View. Select “Value Labels” to change the view to display value labels.



Introduction to SPSS

Variables

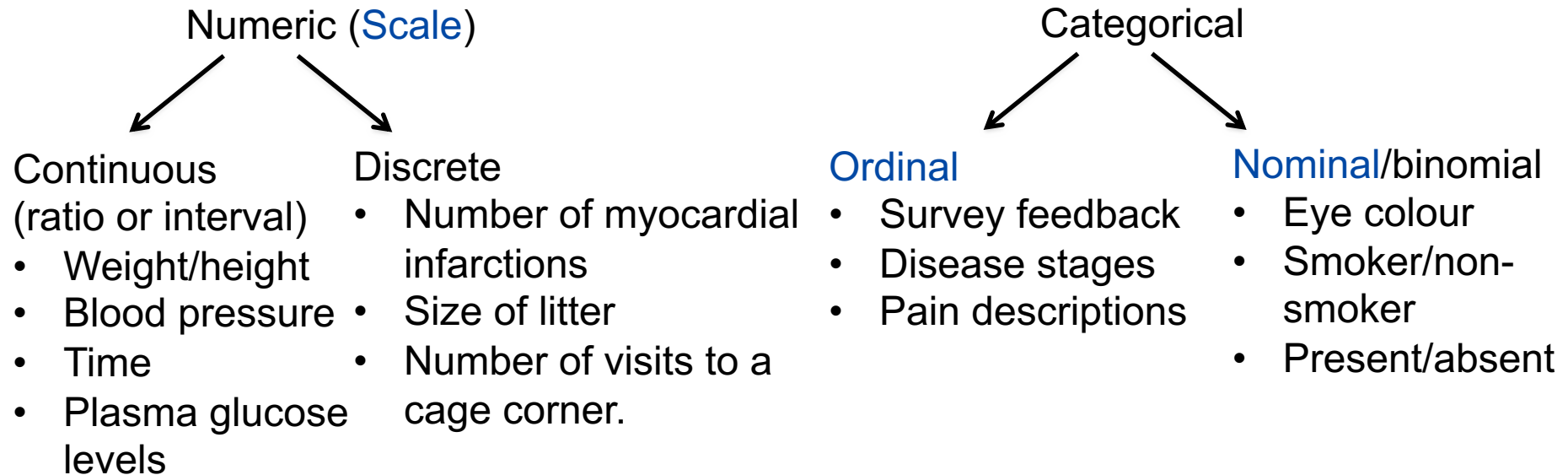
Dependent Variable – these are measured or observed to determine whether they vary in response to the independent variables.

In SPSS – could also be referred to as: Response variable

Independent Variable – a variable that may be used to explain variation in the dependent variable

In SPSS could also be referred to as: Factor, Fixed Factor, Random Factor, Covariate, Predictors, Categorical Covariate, Subjects

SPSS Types of variables



In SPSS the choices are: **Scale**, **Ordinal** and **Nominal**

- Discrete is a Scale variable with 0 decimal places

SPSS User interface

SPSS has three main windows:

- Data Editor (self explanatory)
 - Data view
 - Variable view
- Viewer Window (where the analysis output is shown)
- Syntax Editor (command line for entering syntax)
- This workshop was based on SPSS v24, but has since been updated for v25.

SPSS Data entry and characteristics

Entry

- Data can be entered into the Data Editor directly
- Data can be imported from Excel or other format (best to use the .csv format)

Format

- Wide Format – each row represents one entity's data
- Long Format – repeated measures for a single entity appear on multiple rows
- The format can be switched from wide to long if necessary

Measures

- Scale, Ordinal, Nominal

Coding Variables

- For nominal variables we can assign numeric values to value labels

File format

- *.sav

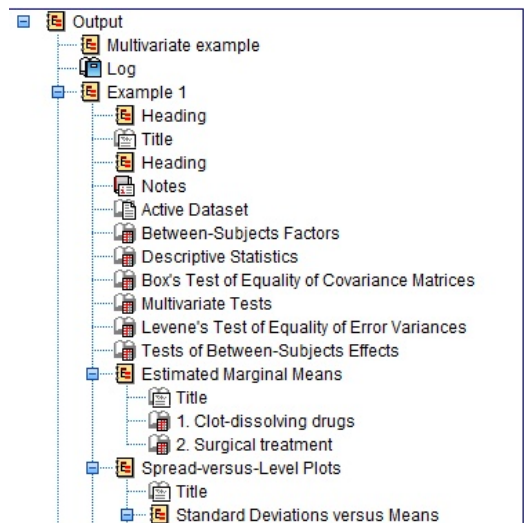
SPSS Viewer Window

Tree Diagram

- Left hand side of viewer
- Shows the heading of each procedure, analysis, graph, etc
- Item descriptions can be edited.

Output

- Output is displayed on RHS
- Double-click each item to edit it
- The corresponding syntax is also shown in the “log” items
- Viewer file saved as *.spv format

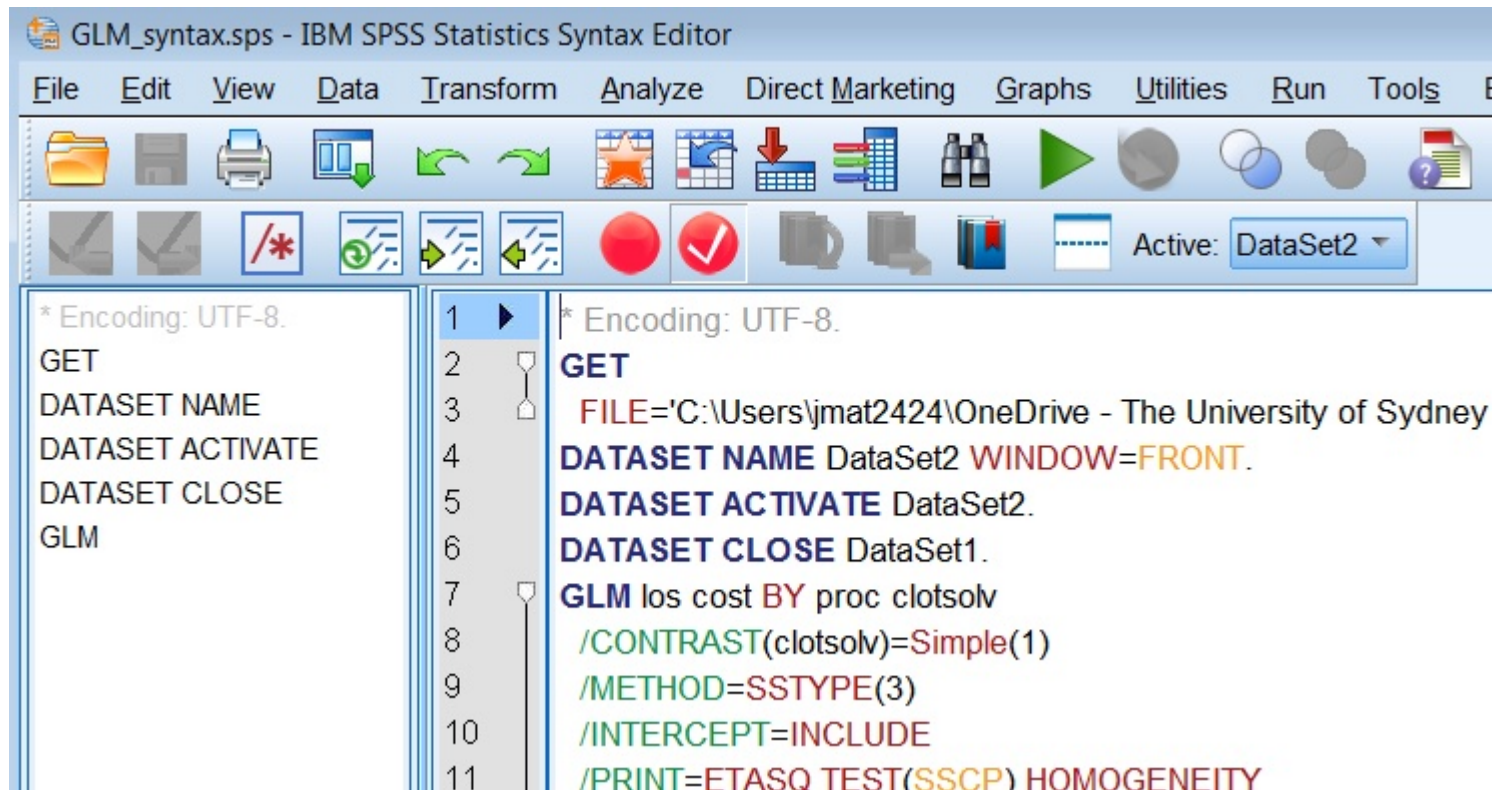


Tests of Between-Subjects Effects					
Dependent Variable: Treatment costs					
Source		Type III Sum of Squares	df	Mean Square	F
Intercept	Hypothesis	1643850.165	1	1643850.165	6.021
	Error	548051.533	2.007	273013.267 ^a	
proc	Hypothesis	574495.237	2	287247.618	2092.758
	Error	895.761	6.526	137.258 ^b	
clotsolv	Hypothesis	3773.589	3	1257.863	5.036
	Error	1504.069	6.021	249.796 ^c	
proc * clotsolv	Hypothesis	1555.807	6	259.301	22.604
	Error	91609.633	7986	11.471 ^d	

a. .949 MS(proc) + .501 MS(clotsolv) - .482 MS(proc * clotsolv) + .032 MS(Error)
b. .508 MS(proc * clotsolv) + .492 MS(Error)
c. .962 MS(proc * clotsolv) + .038 MS(Error)
d. MS(Error)

SPSS Syntax Editor Window

Not necessary for many situations
(but useful to keep as a record of your analysis!)



Workshop examples

1. Two-way ANOVA
2. GLM Repeated Measures
3. Linear regression
4. Binary logistic regression

We will use the SPSS Case Studies in the help files as the basis for demonstration.

1. Two way ANOVA

We will follow the tutorial example looking at purchasing habits in a grocery store chain. Data file is: “grocery_1 month”

The dependent variable is “amount spent” (numeric/scale)

The independent variables are “gender” (2 levels) and “shopping style” (3 levels)

Therefore we have a 2x3 Factorial model, or 2-way ANOVA.

What question are we asking?

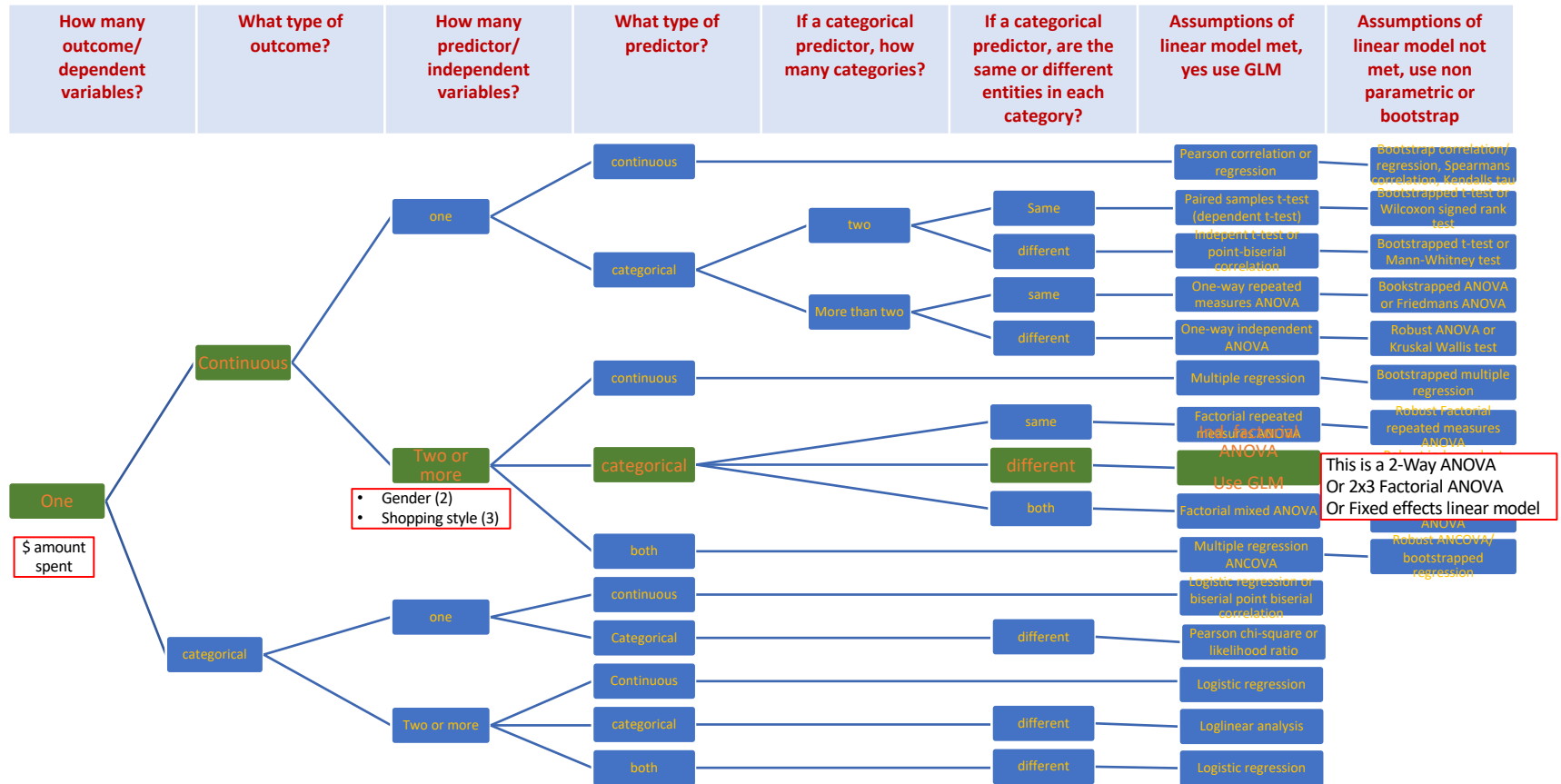
We want to see if shopping style is related to the total monthly spend, and whether there are sex (gender) differences.

1. Two way ANOVA

In SPSS this ANOVA procedure is found under:
Analyze > General Linear Model > Univariate

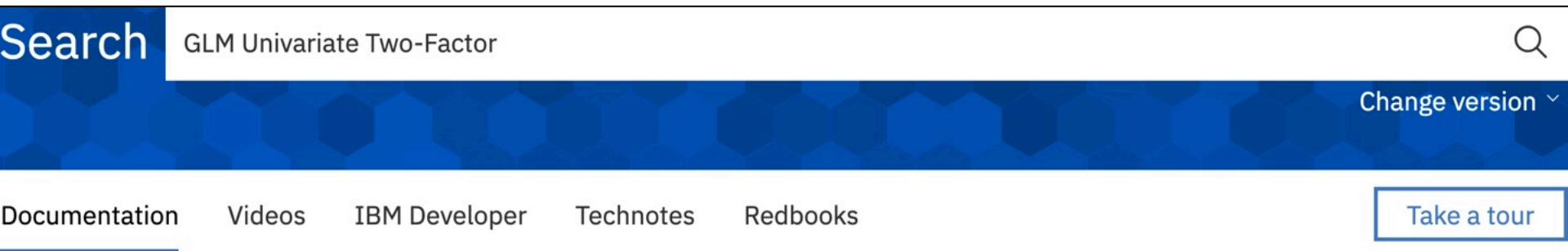
The roadmap shows how to get there.

1. Two way ANOVA



1. Two way ANOVA

1. Open SPSS
2. Open another file > choose “grocery_1 month”
Go to help/topics, then search “GLM univariate Two-Factor”
3. Choose “Using GLM Univariate to Perform a Two-Factor Analysis of Variance”



Search GLM Univariate Two-Factor

Change version

Documentation Videos IBM Developer Technotes Redbooks

Take a tour

226 results

+ Using **GLM Univariate** to Perform a **Two-Factor** Analysis of Variance

Using **GLM Univariate** to Perform a **Two-Factor** Analysis of Variance A grocery store chain surveyed a set of customers concerning their purchasing habits. Given the survey results and how much each customer spent in the previous month, the store wants to see if the frequency with which customers shop is

1. Two way ANOVA

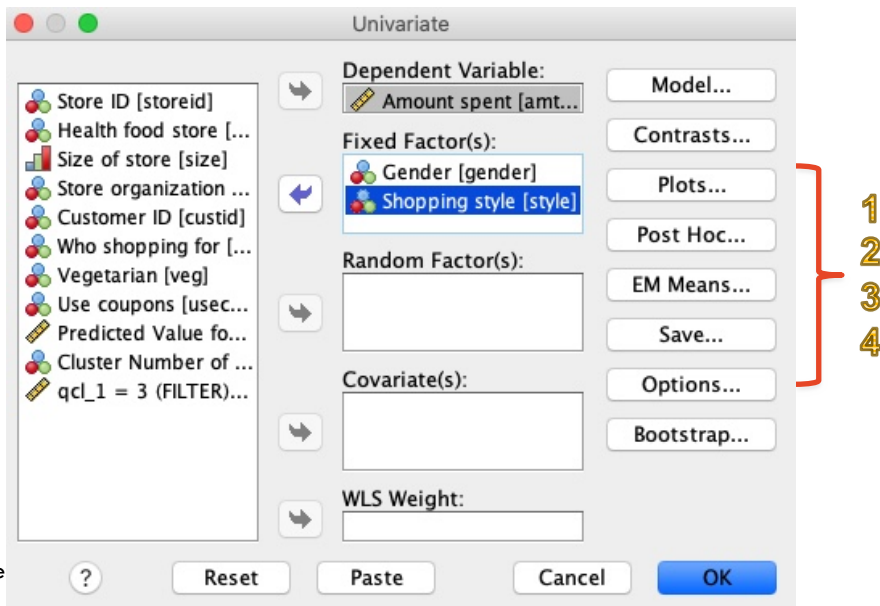
1. Keep tutorial window open alongside your SPSS dataset window for reference.
2. Now follow the steps in the tutorial.

Step 1: Running the Analysis

1. To run a GLM Univariate analysis, from the menus choose:

Analyze > General Linear Model > Univariate...

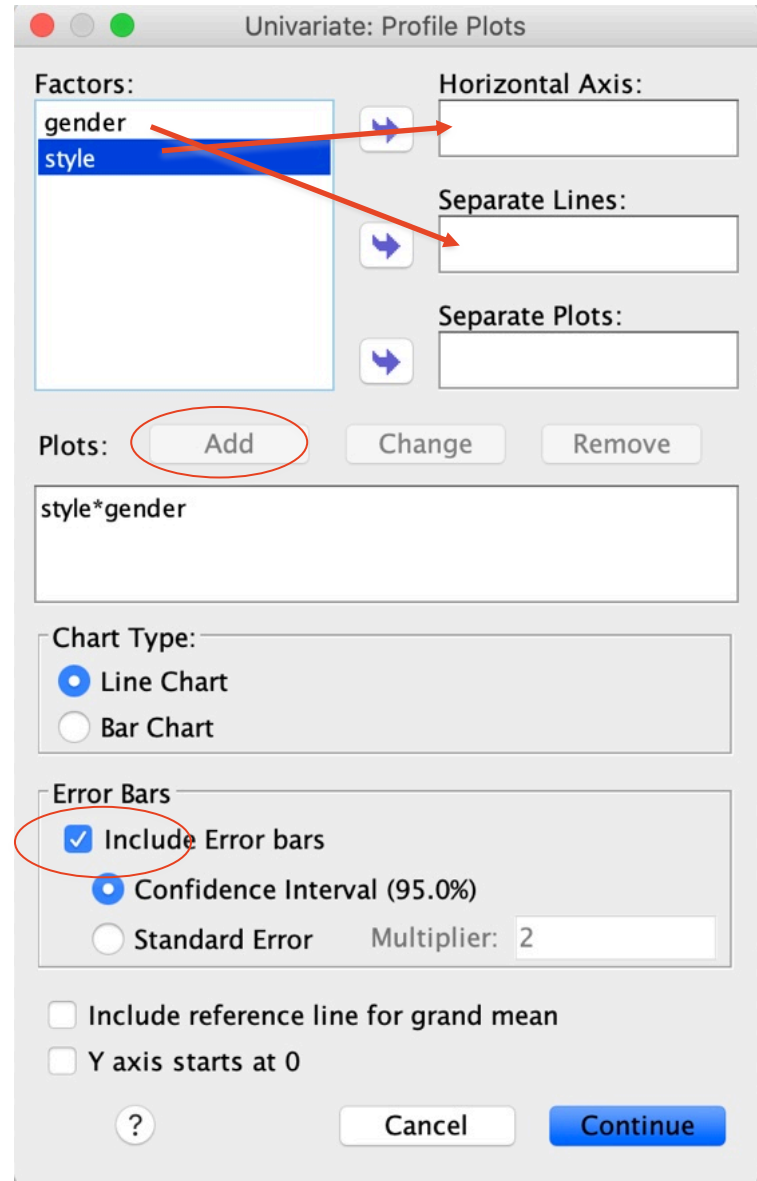
Figure 1. GLM Univariate dialog box



1. Two way ANOVA

Step 1: Running the analysis

- Define required Plots



The image shows the 'Univariate: Profile Plots' dialog box in SPSS. The 'Factors' list on the left contains 'gender' and 'style', with 'style' selected. Red arrows point from 'style' to the 'Horizontal Axis' field and from 'gender' to the 'Separate Lines' field. The 'Plots' section has an 'Add' button circled in red, with 'style*gender' entered in the text box below it. The 'Chart Type' section has 'Line Chart' selected. The 'Error Bars' section has 'Include Error bars' checked and circled in red, with 'Confidence Interval (95.0%)' selected and a 'Multiplier' of 2. Other options like 'Include reference line for grand mean' and 'Y axis starts at 0' are unchecked. The 'Continue' button is highlighted in blue.

Univariate: Profile Plots

Factors:

gender
style

Horizontal Axis:

Separate Lines:

Separate Plots:

Plots: Add Change Remove

style*gender

Chart Type:

☒ Line Chart
☐ Bar Chart

Error Bars

☒ Include Error bars
☒ Confidence Interval (95.0%)
☐ Standard Error Multiplier: 2

☐ Include reference line for grand mean
☐ Y axis starts at 0

? Cancel Continue

1. Two way ANOVA

Step 1: Running the analysis

- Define required Post Hoc tests

Univariate: Post Hoc Multiple Comparisons for Observed Means

Factor(s):
gender
style

Post Hoc Tests for:
style

Equal Variances Assumed

☐ LSD	☐ S-N-K	☐ Waller-Duncan
☐ Bonferroni	☒ Tukey	Type I/Type II Error Ratio: 100
☐ Sidak	☐ Tukey's-b	☐ Dunnett
☐ Scheffe	☐ Duncan	Control Category: Last
☐ R-E-G-W-F	☐ Hochberg's GT2	Test
☐ R-E-G-W-Q	☐ Gabriel	☒ 2-sided ☐ < Control ☐ > Control

Equal Variances Not Assumed

☐ Tamhane's T2	☐ Dunnett's T3	☐ Games-Howell	☐ Dunnett's C
---------------------------------------	---------------------------------------	---------------------------------------	--------------------------------------

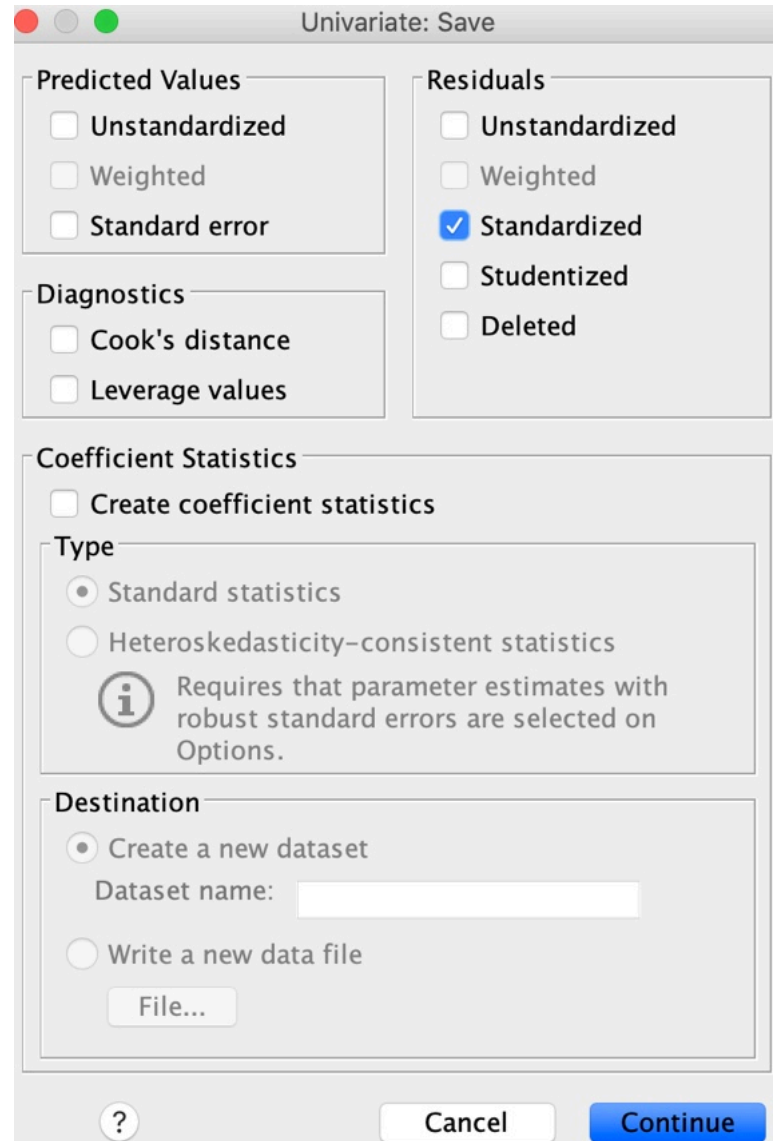
?

Cancel Continue

1. Two way ANOVA

Step 1: Running the analysis

- Save residuals (for normality test – not in the SPSS tutorial)



The image shows the 'Univariate: Save' dialog box in SPSS. The dialog is titled 'Univariate: Save' and has a standard macOS window title bar with red, yellow, and green buttons. It is divided into several sections with expandable/collapsible headers. The 'Predicted Values' section is expanded, showing three options: 'Unstandardized' (unchecked), 'Weighted' (unchecked), and 'Standard error' (unchecked). The 'Residuals' section is also expanded, showing four options: 'Unstandardized' (unchecked), 'Weighted' (unchecked), 'Standardized' (checked with a blue checkmark), and 'Studentized' (unchecked). Below 'Standardized' is a 'Deleted' option (unchecked). The 'Diagnostics' section is expanded, showing two options: 'Cook's distance' (unchecked) and 'Leverage values' (unchecked). The 'Coefficient Statistics' section is expanded, showing 'Create coefficient statistics' (unchecked). Below this is a 'Type' section with two radio buttons: 'Standard statistics' (selected) and 'Heteroskedasticity-consistent statistics' (unselected). Below the radio buttons is an information icon (i) with a text box that reads: 'Requires that parameter estimates with robust standard errors are selected on Options.' The 'Destination' section is expanded, showing two radio buttons: 'Create a new dataset' (selected) and 'Write a new data file' (unselected). Below the radio buttons is a text field labeled 'Dataset name:' followed by a white input box. Below the input box is a 'File...' button. At the bottom of the dialog are three buttons: a question mark icon, a 'Cancel' button, and a 'Continue' button.

Univariate: Save

Predicted Values

- ☐ Unstandardized
- ☐ Weighted
- ☐ Standard error

Residuals

- ☐ Unstandardized
- ☐ Weighted
- ☒ Standardized
- ☐ Studentized
- ☐ Deleted

Diagnostics

- ☐ Cook's distance
- ☐ Leverage values

Coefficient Statistics

- ☐ Create coefficient statistics

Type

- ☒ Standard statistics
- ☐ Heteroskedasticity-consistent statistics

 Requires that parameter estimates with robust standard errors are selected on Options.

Destination

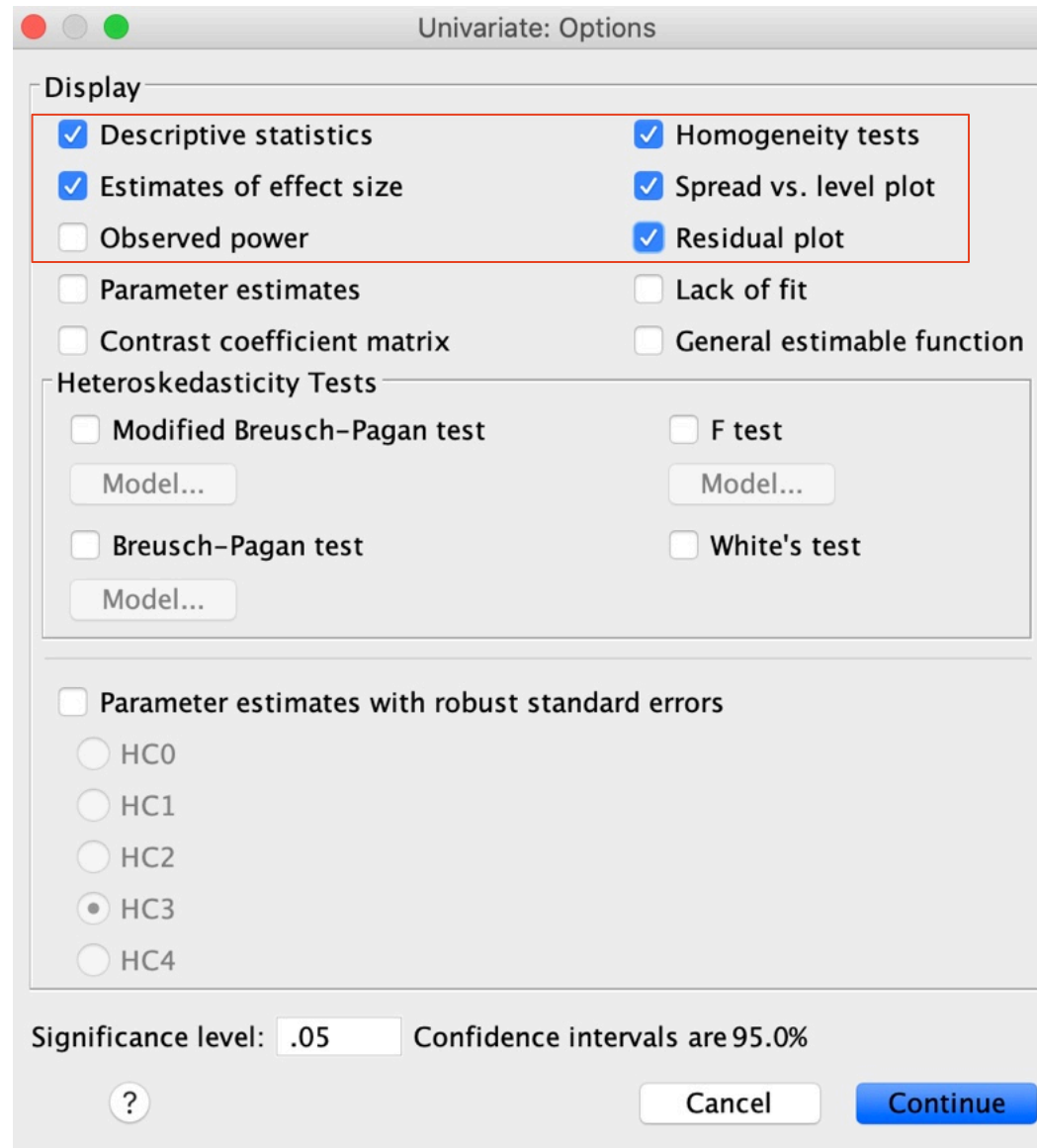
- ☒ Create a new dataset
- ☐ Write a new data file

Dataset name:

1. Two way ANOVA

Step 1: Running the analysis

- Define other options including tests, plots and parameter estimates



The image shows a screenshot of the 'Univariate: Options' dialog box in a statistical software package. The dialog is titled 'Univariate: Options' and has a standard macOS-style title bar with red, yellow, and green window control buttons. The 'Display' section is expanded, showing a list of options. A red rectangular box highlights the first six options: 'Descriptive statistics', 'Estimates of effect size', 'Observed power', 'Homogeneity tests', 'Spread vs. level plot', and 'Residual plot'. The first three options have checkboxes, while the last three have checkmarks. Below the highlighted section, there are more options: 'Parameter estimates', 'Contrast coefficient matrix', 'Lack of fit', and 'General estimable function'. The 'Heteroskedasticity Tests' section is also expanded, showing 'Modified Breusch-Pagan test', 'Breusch-Pagan test', 'F test', and 'White's test'. Each of these tests has a checkbox and a 'Model...' button. At the bottom, there is a section for 'Parameter estimates with robust standard errors' with radio buttons for HC0, HC1, HC2, HC3 (selected), and HC4. The 'Significance level' is set to .05, and 'Confidence intervals are 95.0%'. At the very bottom, there is a question mark icon, a 'Cancel' button, and a 'Continue' button.

Univariate: Options

Display

- ☒ Descriptive statistics
- ☒ Estimates of effect size
- ☐ Observed power
- ☒ Homogeneity tests
- ☒ Spread vs. level plot
- ☒ Residual plot
- ☐ Parameter estimates
- ☐ Contrast coefficient matrix
- ☐ Lack of fit
- ☐ General estimable function

Heteroskedasticity Tests

- ☐ Modified Breusch-Pagan test
Model...
- ☐ Breusch-Pagan test
Model...
- ☐ F test
Model...
- ☐ White's test

☐ Parameter estimates with robust standard errors

- ☐ HC0
- ☐ HC1
- ☐ HC2
- ☒ HC3
- ☐ HC4

Significance level: .05 Confidence intervals are 95.0%

? Cancel Continue

1. Two way ANOVA

Step 2: Descriptive Statistics

Figure 1. Descriptive statistics

Dependent Variable: Amount spent

Gender	Shopping style	Mean	Std. Deviation	N
Male	Biweekly; in bulk	413.0657	90.86574	35
	Weekly; similar items	440.9647	98.23860	120
	Often; what's on sale	407.7747	69.33334	30
	Total	430.3043	93.47877	185
Female	Biweekly; in bulk	343.9763	100.47207	35
	Weekly; similar items	361.7205	90.46076	102
	Often; what's on sale	405.7269	80.57058	29
	Total	365.6671	92.64058	166
Total	Biweekly; in bulk	378.5210	101.25839	70
	Weekly; similar items	404.5552	102.48440	222
	Often; what's on sale	406.7681	74.42114	59
	Total	399.7352	98.40821	351

This table displays descriptive statistics for each combination of factors

1. Two way ANOVA

Step 2: Descriptive Statistics

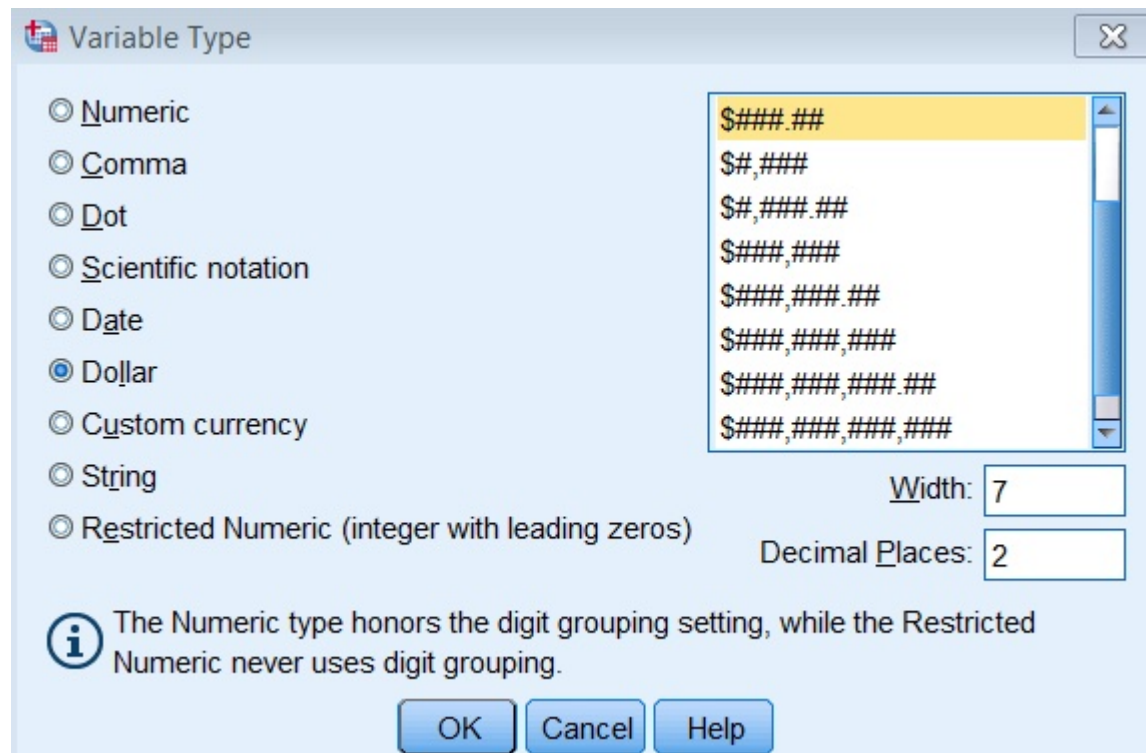
Double click to edit, highlight cells for mean and SD, right-click Cell Properties, Format value as currency with 2 decimals.

Descriptive Statistics				
Dependent Variable: Amount spent				
Shopping style	Gender	Mean	Std. Deviation	N
Biweekly; in bulk	Male	\$413.07	\$90.87	35
	Female	\$343.98	\$100.47	35
	Total	\$378.52	\$101.26	70
Weekly; similar items	Male	\$440.96	\$98.24	120
	Female	\$361.72	\$90.46	102
	Total	\$404.56	\$102.48	222
Often; what's on sale	Male	\$407.77	\$69.33	30
	Female	\$405.73	\$80.57	29
	Total	\$406.77	\$74.42	59
Total	Male	\$430.30	\$93.48	185
	Female	\$365.67	\$92.64	166
	Total	\$399.74	\$98.41	351

1. Two way ANOVA

Step 2: Descriptive Statistics

Note: you can also change the variable type to \$####.## but the output will still have too many decimal places.



1. Two way ANOVA

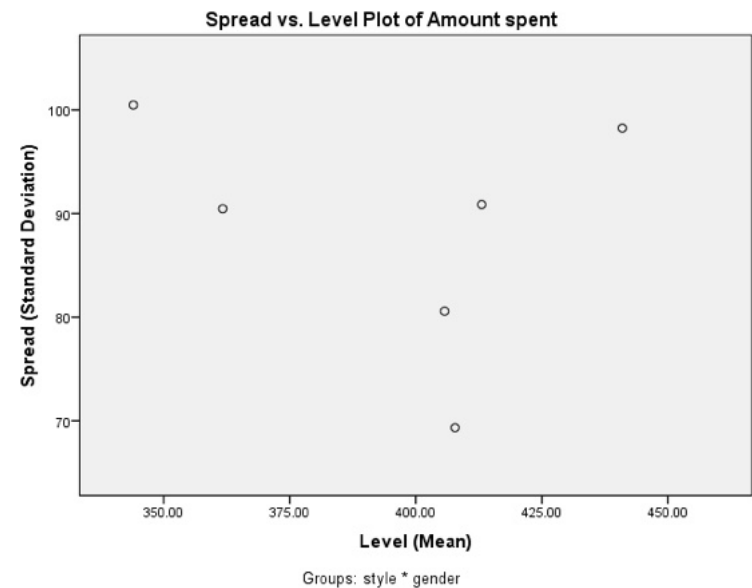
Step 3: Testing Homogeneity of the Variances

Figure 1. Levene's test of equality of the error variances

Dependent Variable: Amount spent

F	df1	df2	Sig.
1.157	5	345	.330

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

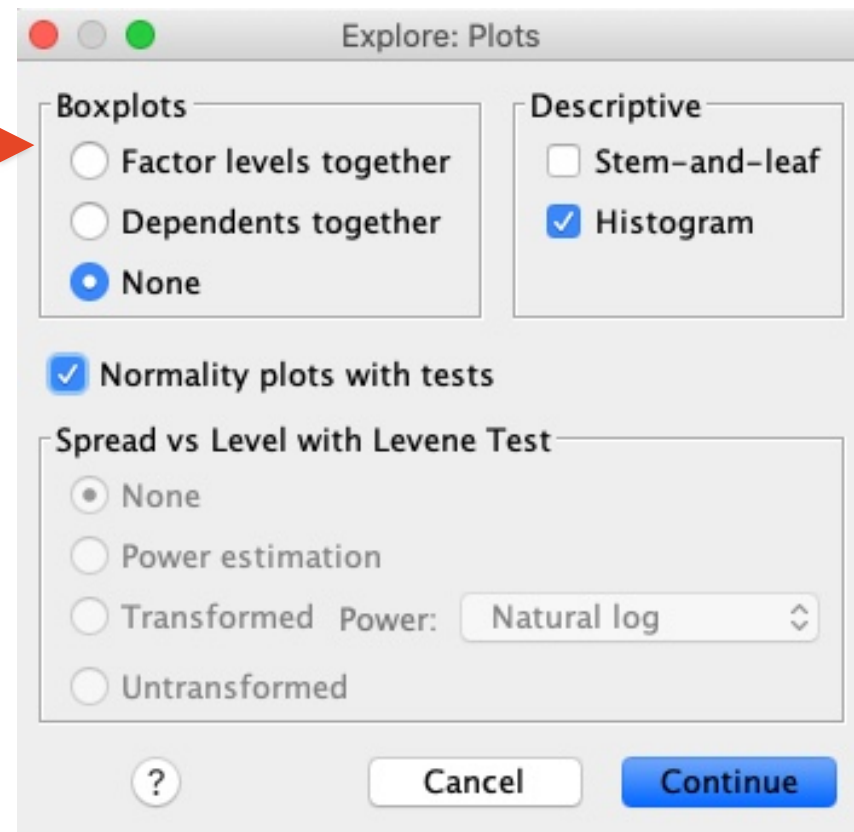
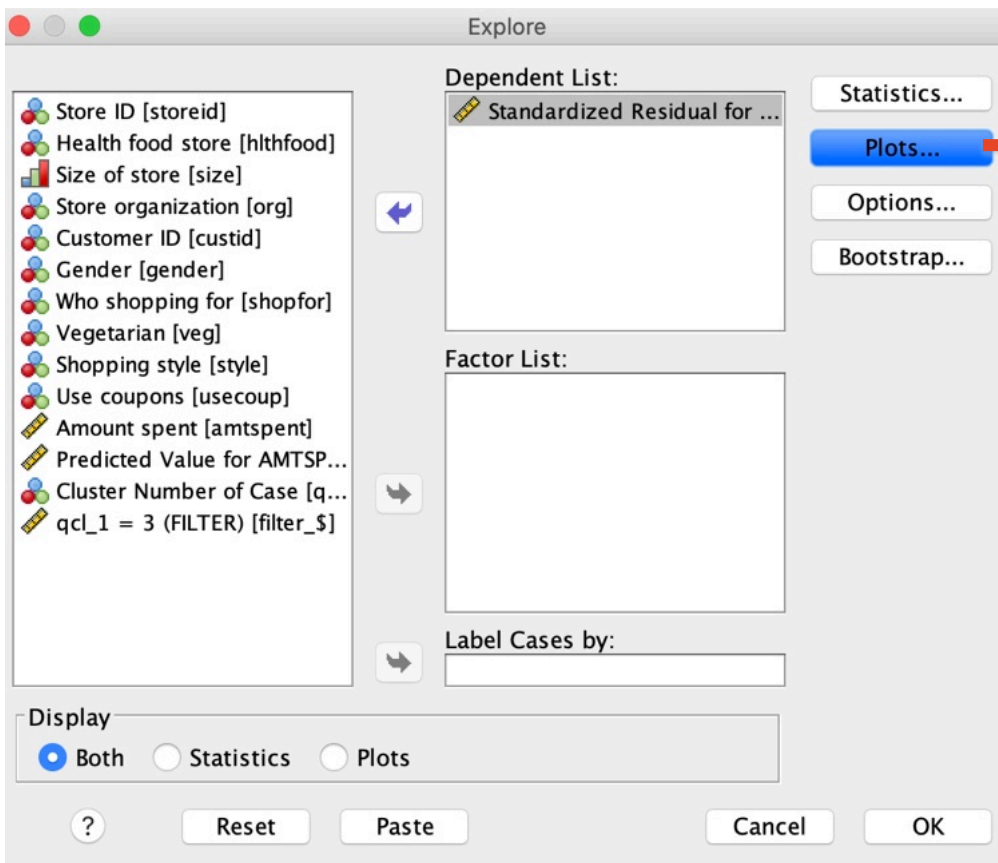


No problem with the assumption of equal variance across groups

1. Two way ANOVA

Step 3: Test for normality “of residuals” (not in tutorial)

Analyze>Descriptive Statistics>Explore



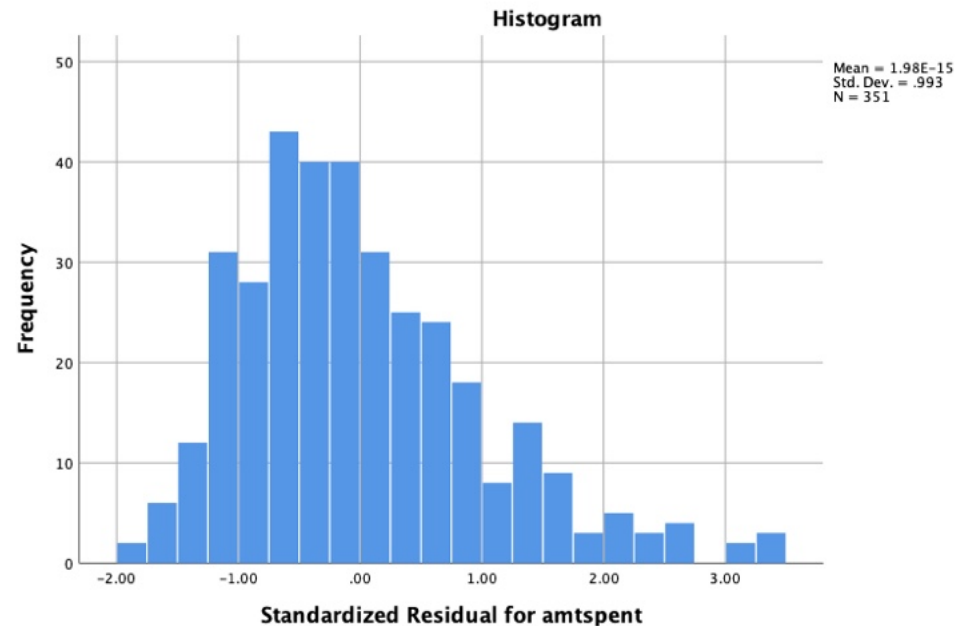
1. Two way ANOVA

Step 3: Test for normality
(of residuals; not in tutorial)

- Normality fails both tests
- Shapiro-Wilk and K-S tests very sensitive (note $df=351$)
- Probably OK to continue with 2-way ANOVA model, but note the skew

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Standardized Residual for amtspent	.079	351	.000	.949	351	.000

a. Lilliefors Significance Correction



1. Two way ANOVA

Step 4: Post Hoc tests

Multiple Comparisons

Dependent Variable: Amount spent

Tukey HSD

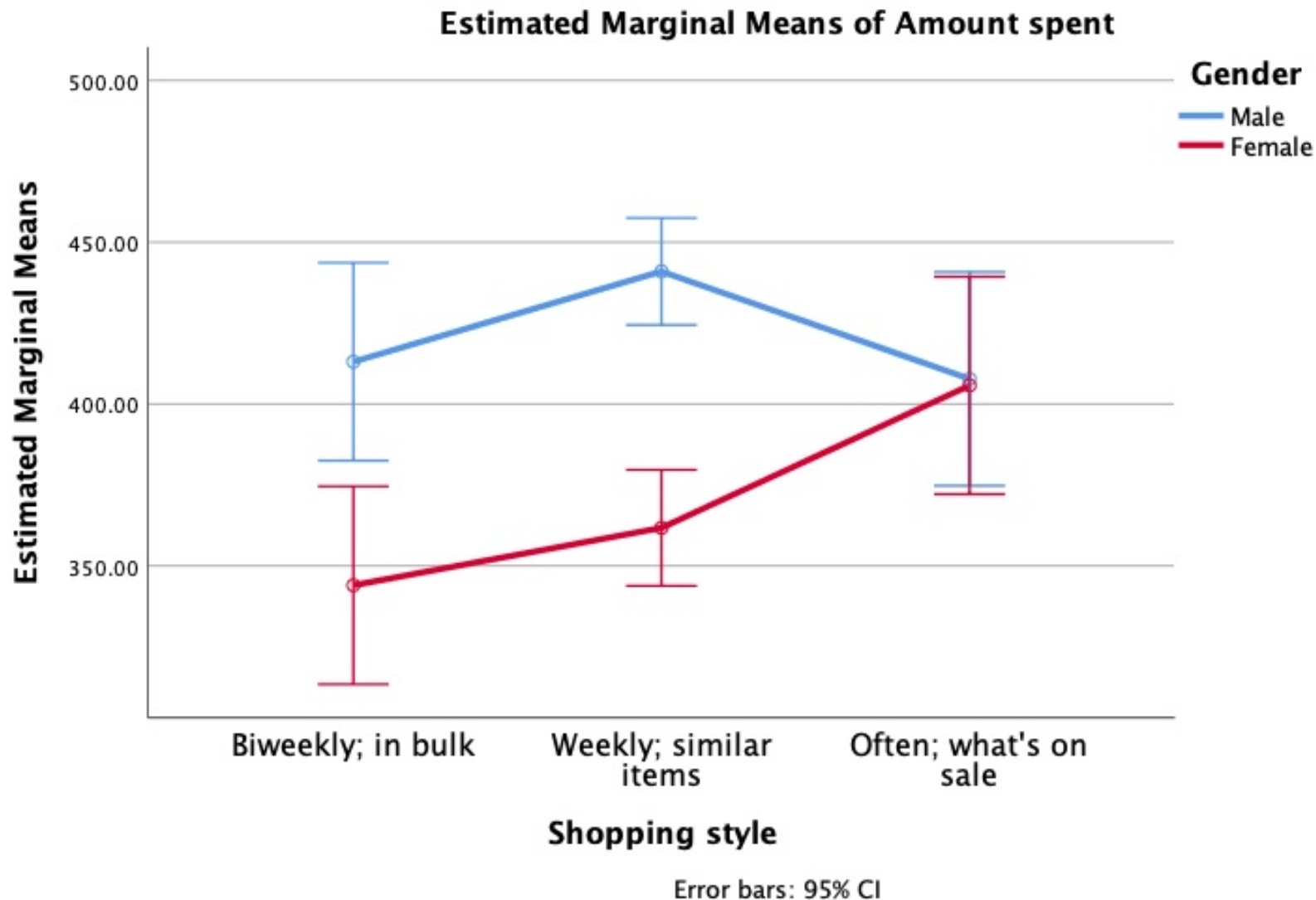
(I) Shopping style	(J) Shopping style	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Biweekly; in bulk	Weekly; similar items	-26.0342	12.61108	.099	-55.7191	3.6507
	Often; what's on sale	-28.2471	16.25946	.193	-66.5198	10.0256
Weekly; similar items	Biweekly; in bulk	26.0342	12.61108	.099	-3.6507	55.7191
	Often; what's on sale	-2.2130	13.47525	.985	-33.9320	29.5061
Often; what's on sale	Biweekly; in bulk	28.2471	16.25946	.193	-10.0256	66.5198
	Weekly; similar items	2.2130	13.47525	.985	-29.5061	33.9320

Based on observed means.

The error term is Mean Square(Error) = 8463.939.

Normally this step would come after the ANOVA (test of between subject effects)
In this case we did not choose pairwise comparisons for Gender

1. Two way ANOVA



1. Two way ANOVA

Step 6: Tests of Between-Subjects Effects

Figure 1. Tests of between-subjects effects

Dependent Variable: Amount spent

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	469402.996 ^a	5	93880.599	11.092	.000	.138
Intercept	39359636.4	1	39359636	4650.3	.000	.931
GENDER	158037.442	1	158037.44	18.672	.000	.051
STYLE	33506.210	2	16753.105	1.979	.140	.011
GENDER * STYLE	69858.325	2	34929.163	4.127	.017	.023
Error	2920058.824	345	8463.939			
Total	59475118.4	351				
Corrected Total	3389461.820	350				

a. R Squared = .138 (Adjusted R Squared = .126)

1. Two way ANOVA

Step 7: Summary

In this example, the post hoc tests did not reveal a difference in spending between customers who shopped biweekly and those who shopped more often. However, the estimated marginal means and profile plots revealed an interaction between the two factors, suggesting that male customers who shop once a week are more profitable than those who shop more often, while the pattern is reversed for female customers. The significance of this interaction effect was confirmed by the results of the ANOVA table.

1. Two way ANOVA

What to report?

- Type of analysis: 2-way ANOVA
- Descriptive Statistics – double-click then edit the table, choose appropriate formatting
- ANOVA assumptions checked

Descriptive Statistics

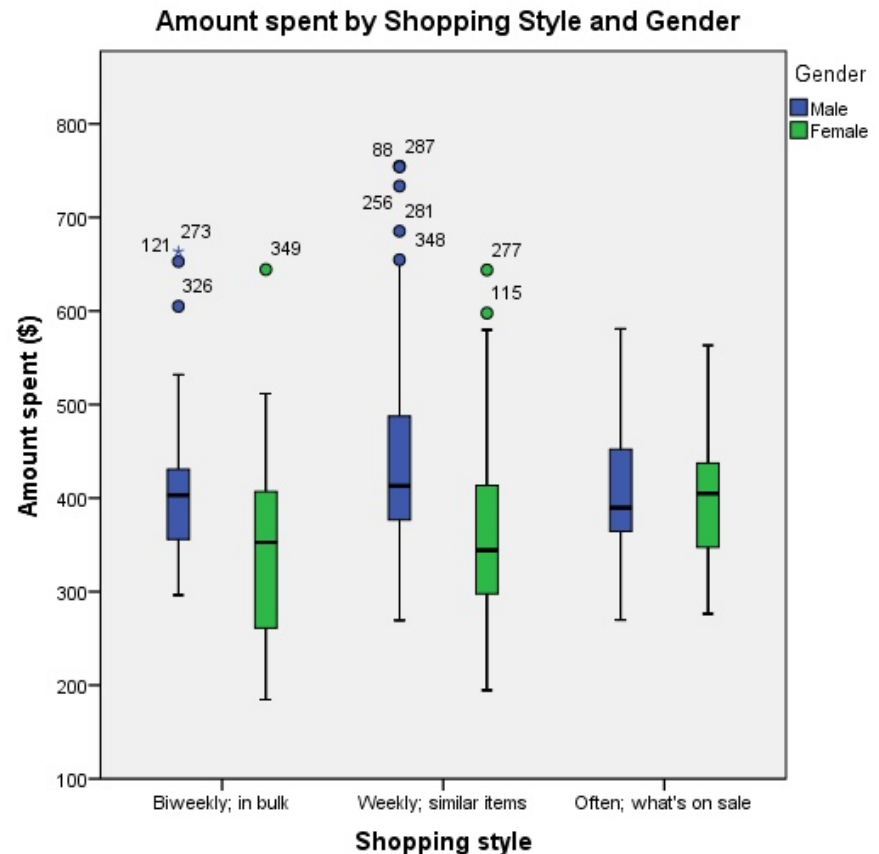
Dependent Variable: Amount spent

Gender	Shopping style	Mean	Std. Deviation	N
Male	Biweekly; in bulk	\$413	91	35
	Weekly; similar items	\$441	98	120
	Often; what's on sale	\$408	69	30
	Total	\$430	93	185
Female	Biweekly; in bulk	\$344	100	35
	Weekly; similar items	\$362	90	102
	Often; what's on sale	\$406	81	29
	Total	\$366	93	166
Total	Biweekly; in bulk	\$379	101	70
	Weekly; similar items	\$405	102	222
	Often; what's on sale	\$407	74	59
	Total	\$400	98	351

1. Two way ANOVA

What to report?

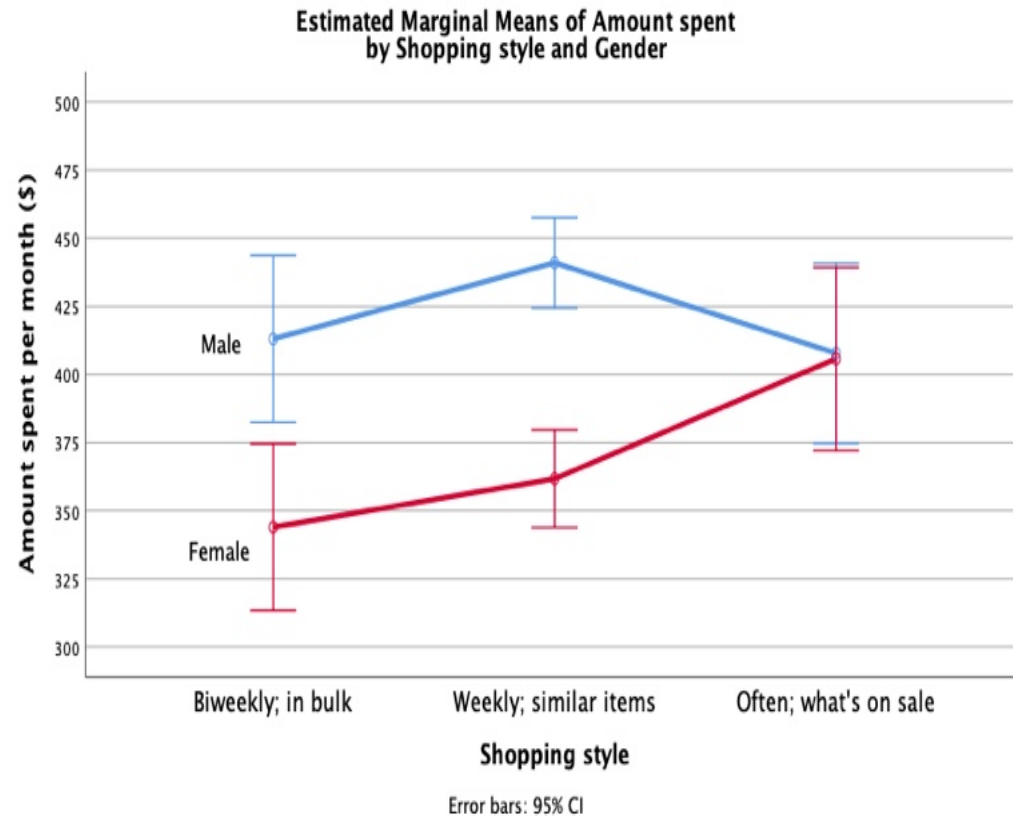
- Visual representation of data
- Use Chart Builder to create a Boxplot
- Note: this should have been done before the analysis!!



1. Two way ANOVA

What to report?

- Prepare Profile plot – double click then edit, remove legend, add labels, remove unnecessary decimal places, etc
- Add comments regarding the apparent interaction



1. Two way ANOVA

What to report?

- Table of Estimated marginal means with std errors and confidence intervals (optional)

Gender * Shopping style

Dependent Variable: Amount spent

Gender	Shopping style	Mean	Std. Error	95% Confidence Interval	
				Lower Bound	Upper Bound
Male	Biweekly; in bulk	\$413	\$15.55	382	444
	Weekly; similar items	\$441	\$8.40	424	457
	Often; what's on sale	\$408	\$16.80	375	441
Female	Biweekly; in bulk	\$344	\$15.55	313	375
	Weekly; similar items	\$362	\$9.11	344	380
	Often; what's on sale	\$406	\$17.08	372	439

1. Two way ANOVA

What to report?

- Tests of between subject effects (the ANOVA table)
- Comment on significant main effects and interactions
- Include a graph to illustrate the interactions

Tests of Between-Subjects Effects

Dependent Variable: Amount spent

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	469403 ^a	5	93881	11.092	.000	.138
Intercept	39359636	1	39359636	4650.274	.000	.931
gender	158037	1	158037	18.672	.000	.051
style	33506	2	16753	1.979	.140	.011
gender * style	69858	2	34929	4.127	.017	.023
Error	2920059	345	8464			
Total	59475118	351				
Corrected Total	3389462	350				

a. R Squared = .138 (Adjusted R Squared = .126)

1. Two way ANOVA

What to report?

- Statistical tests for specific comparisons of interest, eg Shopping Style
- Edit Post Hoc Tests table: hide the 3 duplicate rows, format the data, etc
- Add comments on the observed statistical (and scientific) significance for each desired comparison

Multiple Comparisons

Dependent Variable: Amount spent

Tukey HSD

(I) Shopping style	(J) Shopping style	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Biweekly; in bulk	Weekly; similar items	-\$26.03	12.61	.099	-55.72	3.65
	Often; what's on sale	-\$28.25	16.26	.193	-66.52	10.03
Weekly; similar items	Often; what's on sale	-\$2.21	13.48	.985	-33.93	29.51

Based on observed means.

The error term is Mean Square(Error) = 8463.939.

1. Two way ANOVA - complete

That wasn't so bad!



2. Repeated measures using GLM Repeated Measures

One continuous dependent variable

One or more categorical independent variables

One or more repeated measures (within subjects design)

In SPSS we use the procedure:

General Linear Model – Repeated Measures

We will follow the SPSS case study (**testmarket.sav**) assessing market sales.

The dependent variable is “sales”

The independent variables are “market ID” and “Promotion”

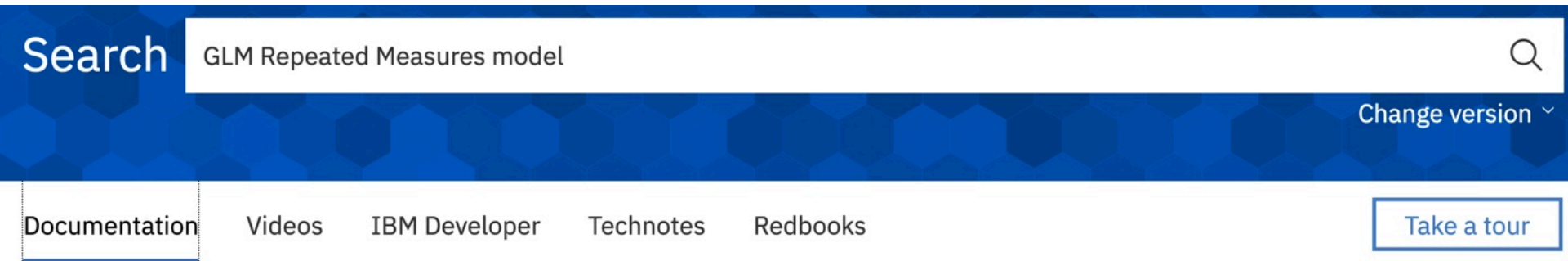
2. Repeated measures using GLM Repeated Measures



Adapted from "Discovering Statistics using IBM SPSS Statistics"
by Andy Field

2. Repeated measures using GLM Repeated Measures

1. Open SPSS
2. Open another file > choose “testmarket.sav”
3. Go to help/topics, then search “GLM Repeated Measures Model”
4. Choose “The GLM Repeated Measures Model”



169 results

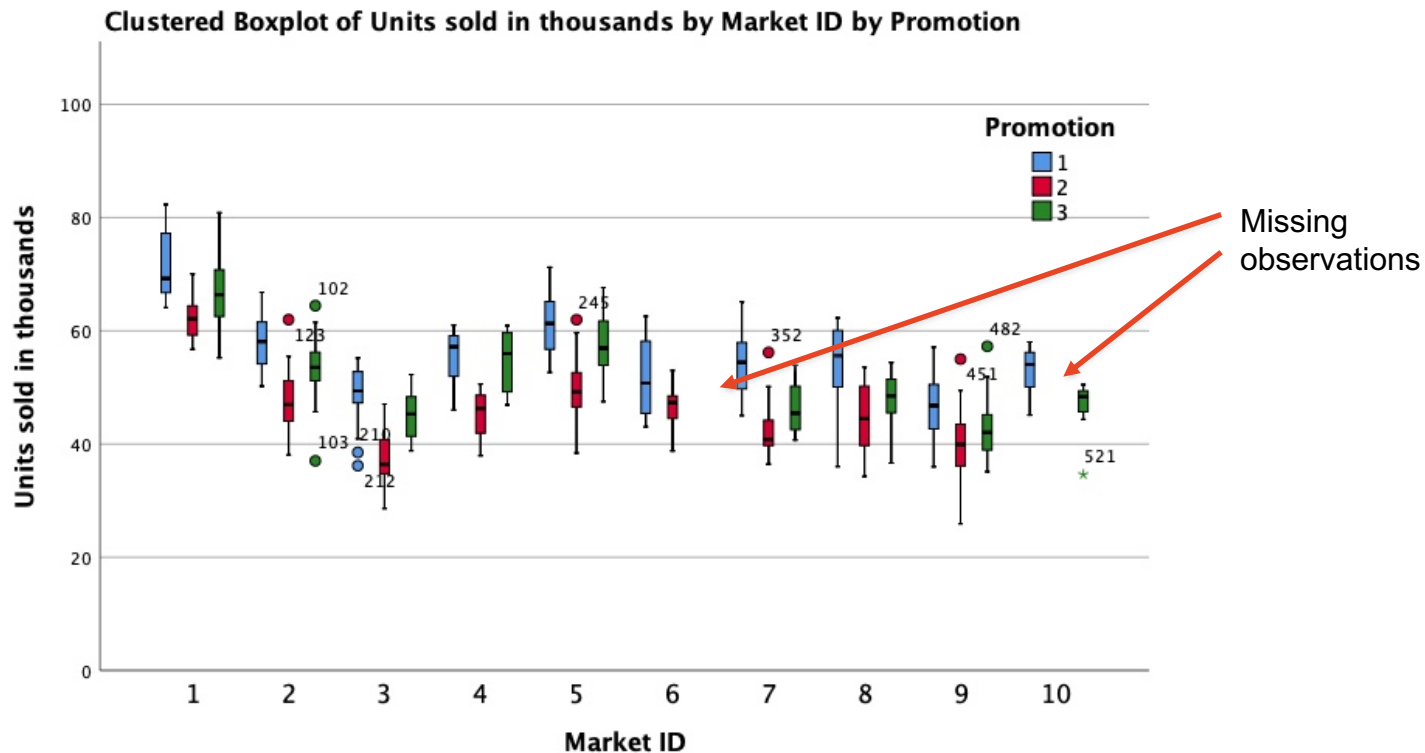
+ The GLM Repeated Measures Model

The **GLM Repeated Measures** procedure is based on the general linear **model**, in which factors and covariates are assumed to have linear relationships to the dependent variables. Factors. Categorical predictors should be selected as factors in the **model**. Levels of a between-subjects

2. Repeated measures using GLM Repeated Measures

Take a look at the data

Create boxplot of Market ID by promotion



2. Repeated measures using GLM Repeated Measures

1. Keep the tutorial window open for reference.

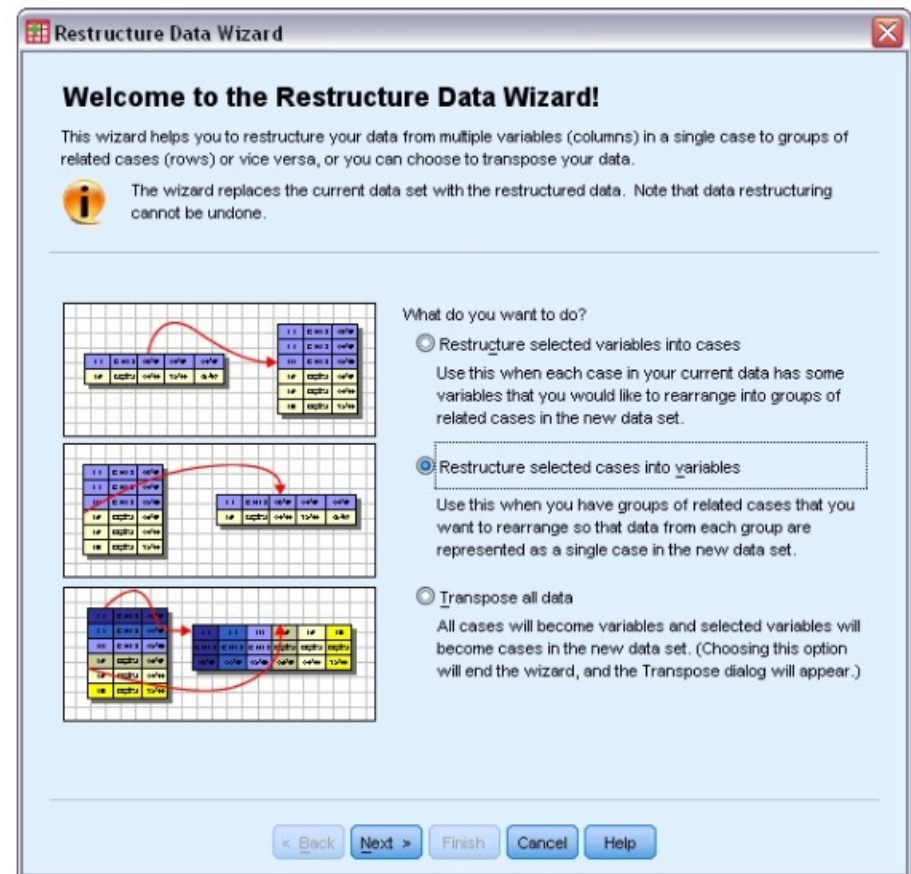
2. Follow the steps in the tutorial.

Step 1: Preparing the Data file

1. To restructure the data from cases to variables, from the menus choose:

Data > Restructure...

Figure 1. Restructure Data Wizard



2. Repeated measures using GLM Repeated Measures

Step 1: Preparing the Data file Select Variables

Restructure Data Wizard - Step 2 of 5

Cases to Variables: Select Variables

Data from case groups in the current file will be restructured into single cases in the new file. Choose variables that identify case groups by moving those variables into the Identifier Variable list. Optionally you can also choose Index Variables.

i The variables that remain in the list of Variables in the Current File either contain data that vary within a case group or data that do not vary. A variable with data that vary will become a group of new variables in the restructured file. A variable with data that do not vary will be copied into the new file.

Variables in the Current File:

- Market ID [marketid]
- Market size [mktsize]
- Age of store location [ageloc]
- Promotion [promo]
- Units sold in thousands [sales]

Identifier Variable(s):

- Location ID [locid]

Index Variable(s):

- Week [week]

< Back Next > Finish Cancel Help

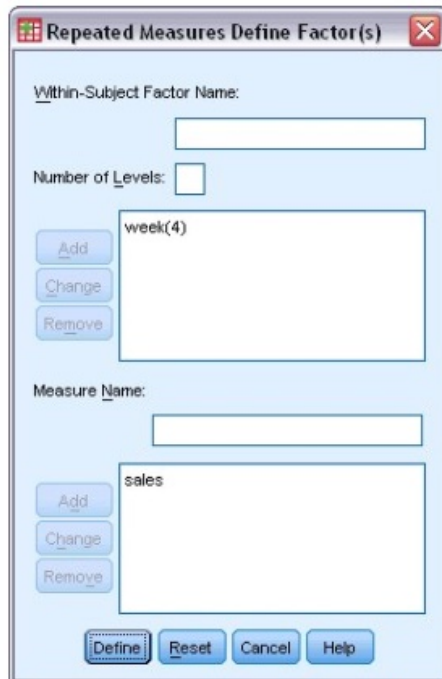
2. Repeated measures using GLM Repeated Measures

Step 2: Running the analysis

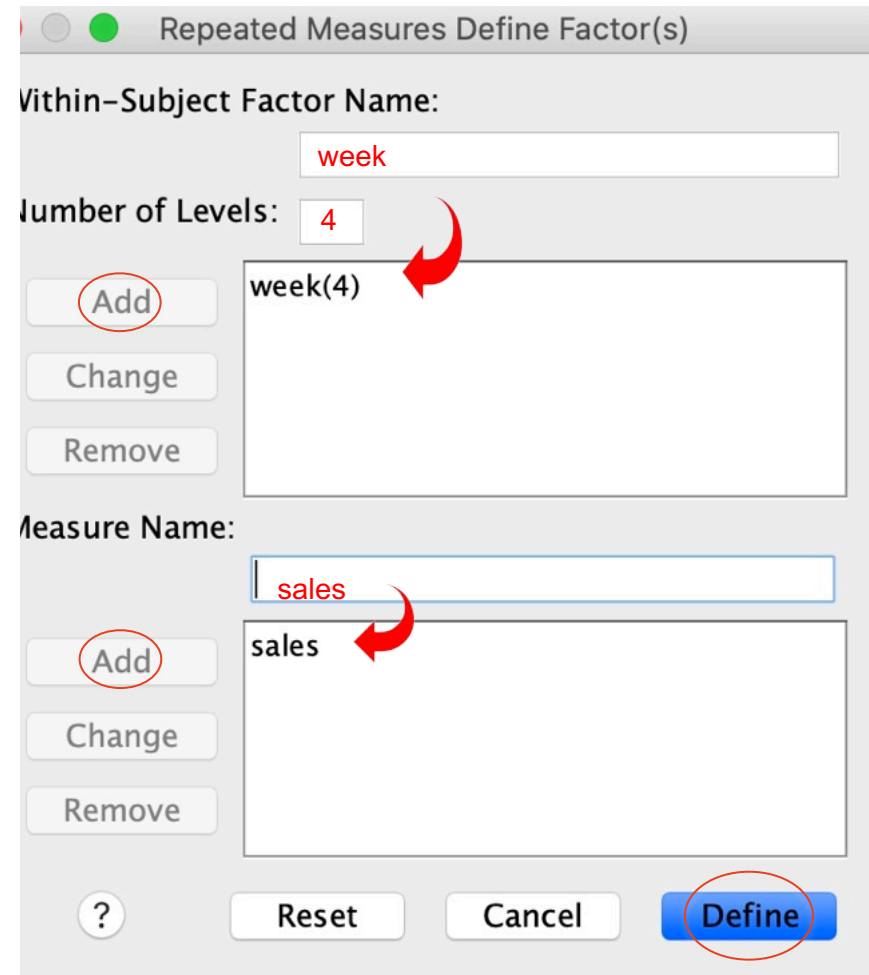
1. To run a GLM Repeated Measures analysis, from the menus choose:

Analyze > General Linear Model > Repeated Measures...

Figure 1. GLM Repeated Measures Define Factors dialog box



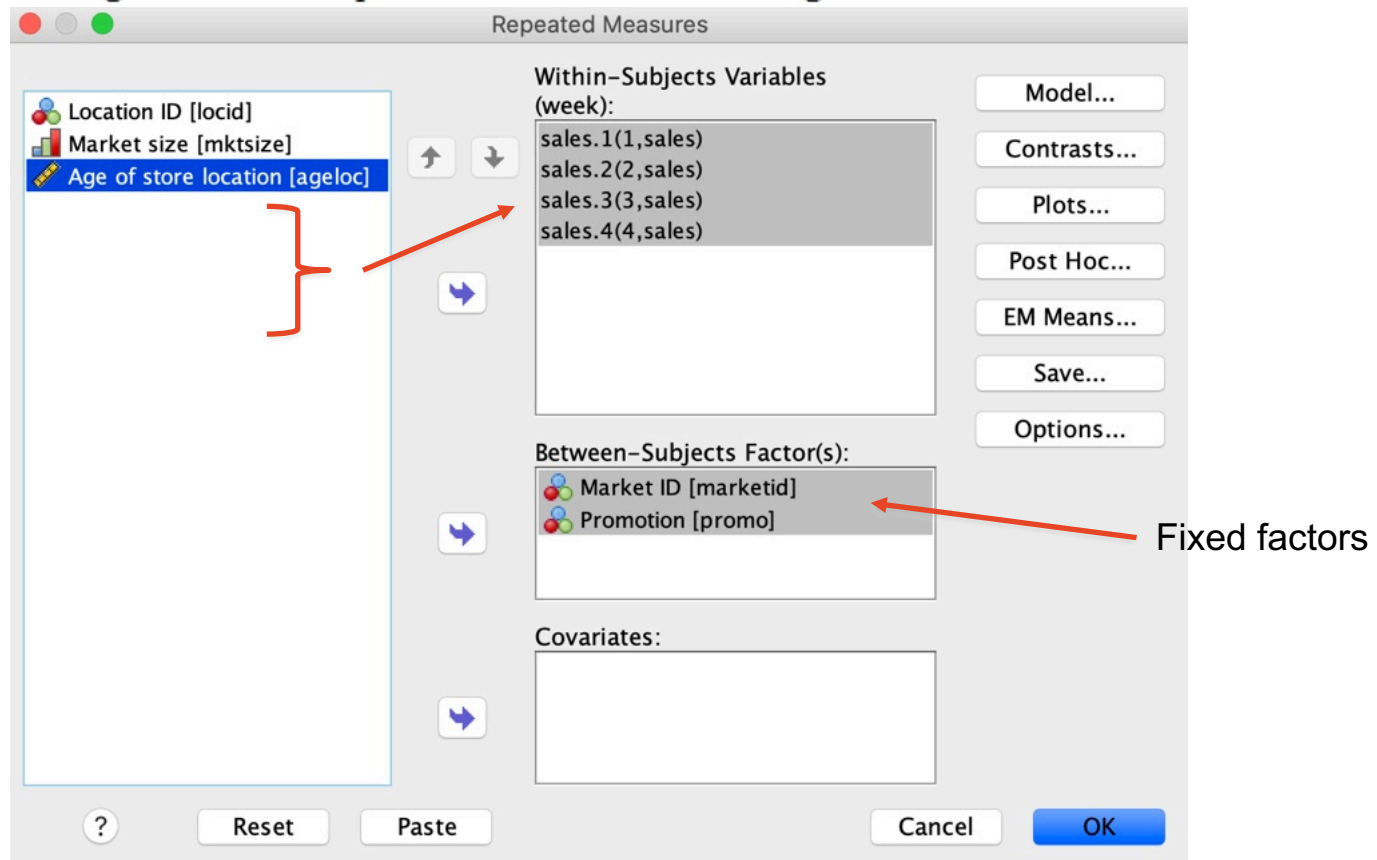
2. Type **week** as the within-subject factor name.
3. Type **4** as the number of levels.
4. Click **Add**.
5. Type **sales** as the measure name.
6. Click **Add**.
7. Click **Define**.



2. Repeated measures using GLM Repeated Measures

Step 2: Running the analysis

Figure 2. GLM Repeated Measures main dialog box



8. Select *SALES.1: Units sold in thousands* through *SALES.4: Units sold in thousands* (when variables are listed in file order) as within-subjects variables.

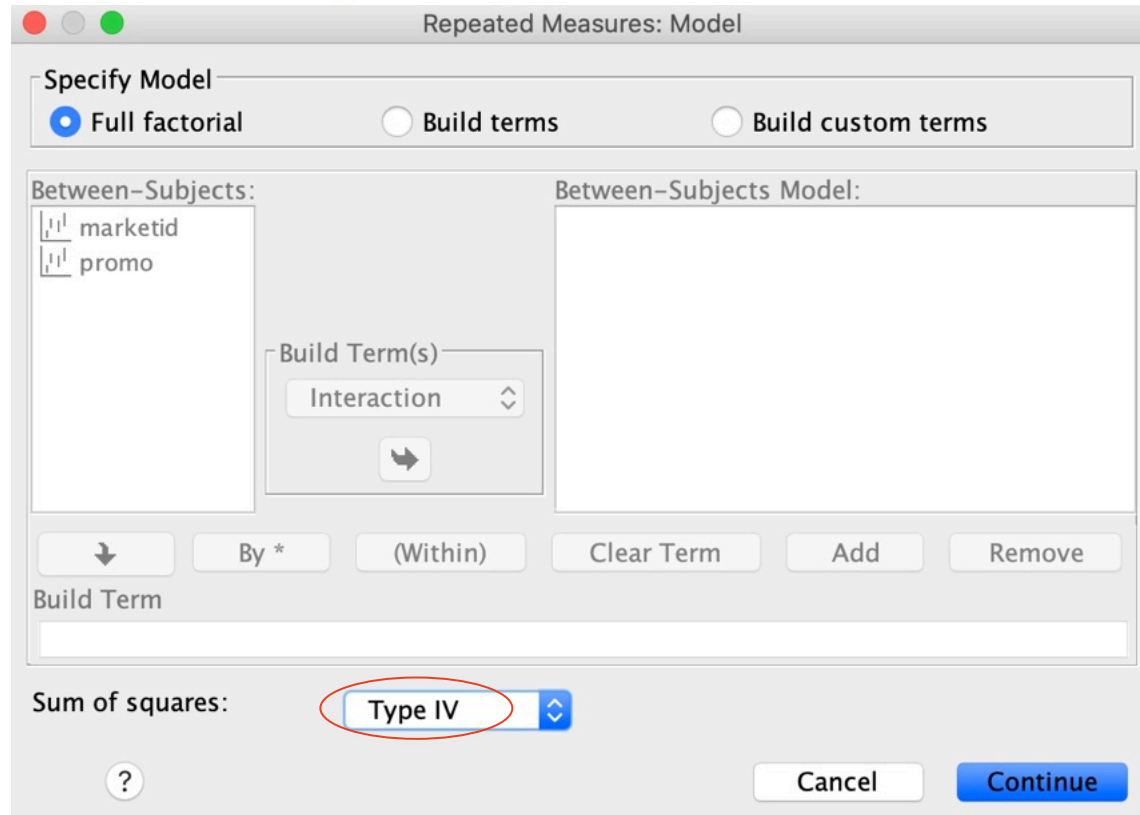
9. Select *Market ID* and *Promotion* as between-subjects factors.

2. Repeated measures using GLM Repeated Measures

Step 2: Running the analysis

10. Click **Model** in the GLM Repeated Measures dialog box.

Figure 3. Model dialog box



11. In the Model dialog box, select **Type IV** from the Sum of squares dropdown list.

The cell frequencies for the layout of *Market ID* and *Promotion * Market ID* are unbalanced, so using the Type IV sums of squares is recommended. Further, *Promotion * Market ID* has empty cells, which will result in reduced degrees of freedom for this effect.

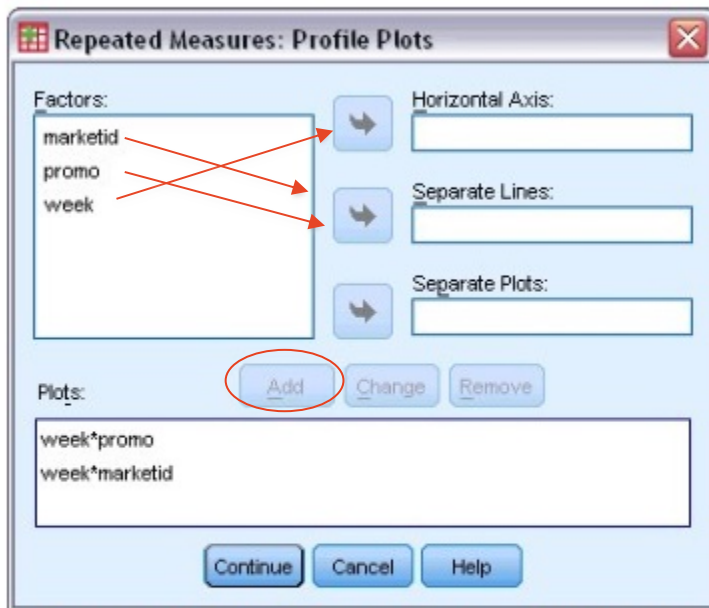
12. Click **Continue**.

2. Repeated measures using GLM Repeated Measures

Step 2: Running the analysis

13. Click **Plots** in the GLM Repeated Measures dialog box.

Figure 4. Profile Plots dialog box



14. Select *week* as the factor to place on the horizontal axis.

15. Select *promo* as the factor to define separate lines in the plot.

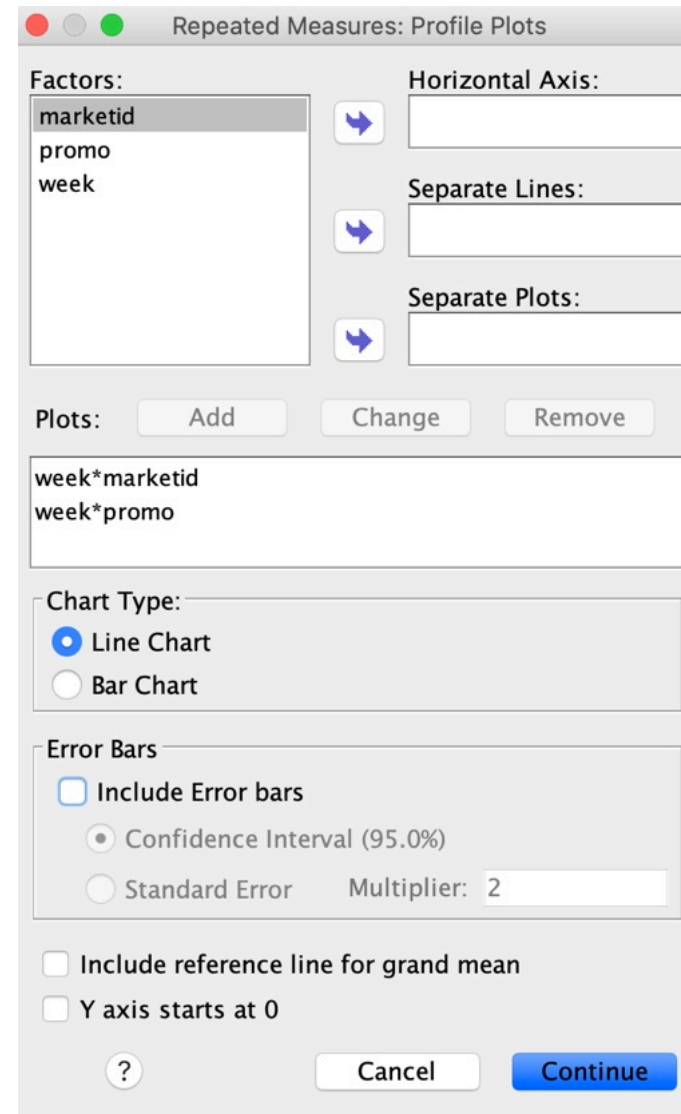
16. Click **Add**.

17. Select *week* as the factor to place on the horizontal axis.

18. Select *marketid* as the factor to define separate lines in the plot.

19. Click **Add**.

20. Click **Continue**.

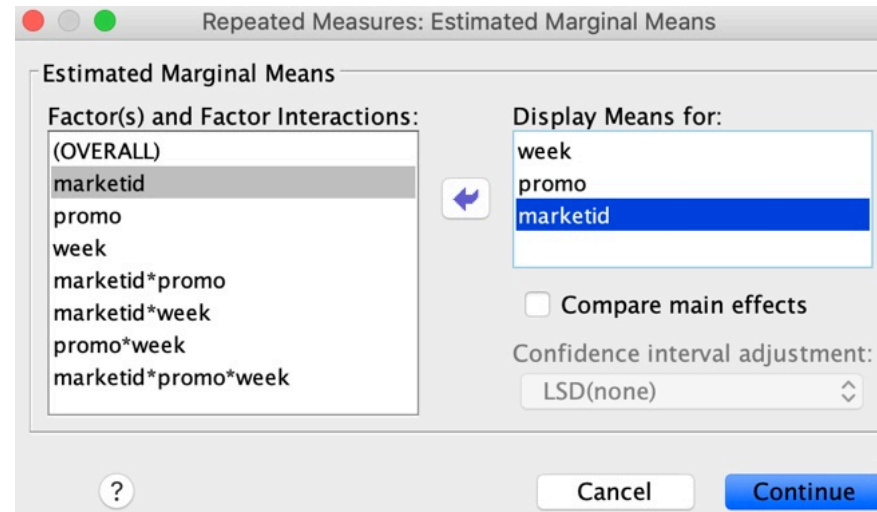
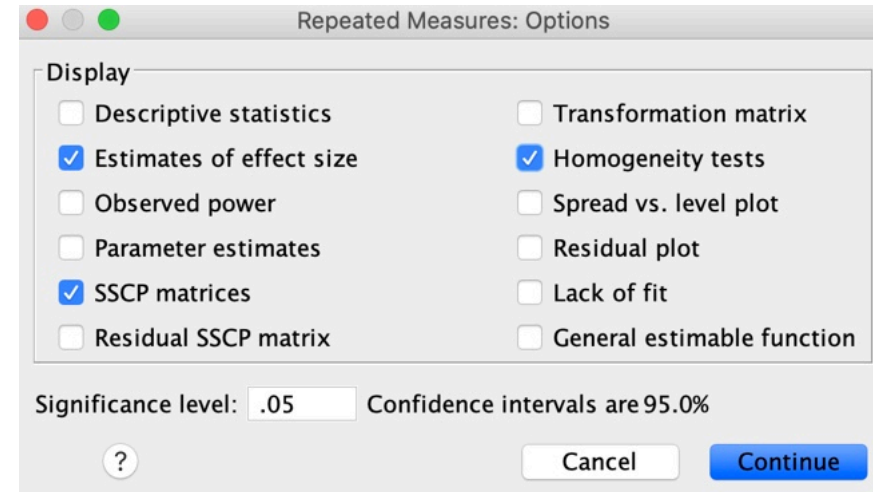
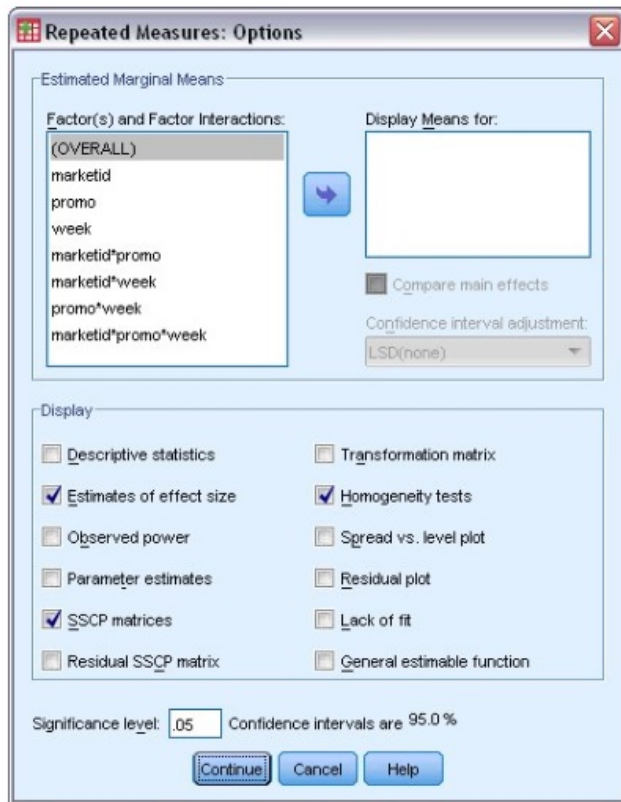


2. Repeated measures using GLM Repeated Measures

Step 2: Running the analysis

21. Click **Options** in the GLM Repeated Measures dialog box.

Figure 5. Options dialog box



22. In the Options dialog, select **Estimates of effect size**, **SSCP matrices**, and **Homogeneity tests** in the Display group.

23. Click **Continue**.

24. Click **OK** in the GLM Repeated Measures dialog box.

2. Repeated measures using GLM Repeated Measures

Step 3: Between-subjects effects

ANOVA table

Figure 1. Between-subjects SSCP matrix

			SALES
Hypothesis	Intercept	SALES	888161.1
	PROMO	SALES	5622.897
	MARKETID	SALES	19721.223
	PROMO * MARKETID	SALES	577.884
Error		SALES	2607.022

Based on Type IV Sum of Squares

Figure 2. Tests of between-subjects effects

Measure: SALES
Transformed Variable: Average

Source	Type IV Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Intercept	888161.056	1	888161	35771.4	.000	.997
PROMO	5622.897 ^a	2	2811.45	113.233	.000	.683
MARKETID	19721.223 ^a	9	2191.25	88.254	.000	.883
PROMO * MARKETID	577.884	16	36.118	1.455	.131	.181
Error	2607.022	105	24.829			

a. The Type IV testable hypothesis is not unique.

2. Repeated measures using GLM Repeated Measures

Step 4: Profile Plots

Figure 1. Profile plot for Promotion

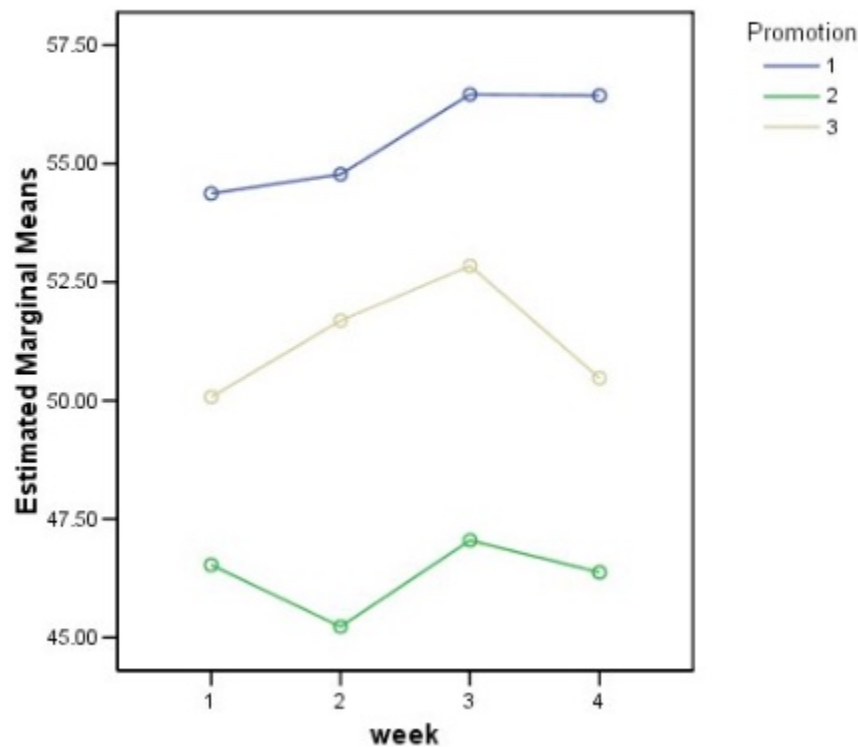
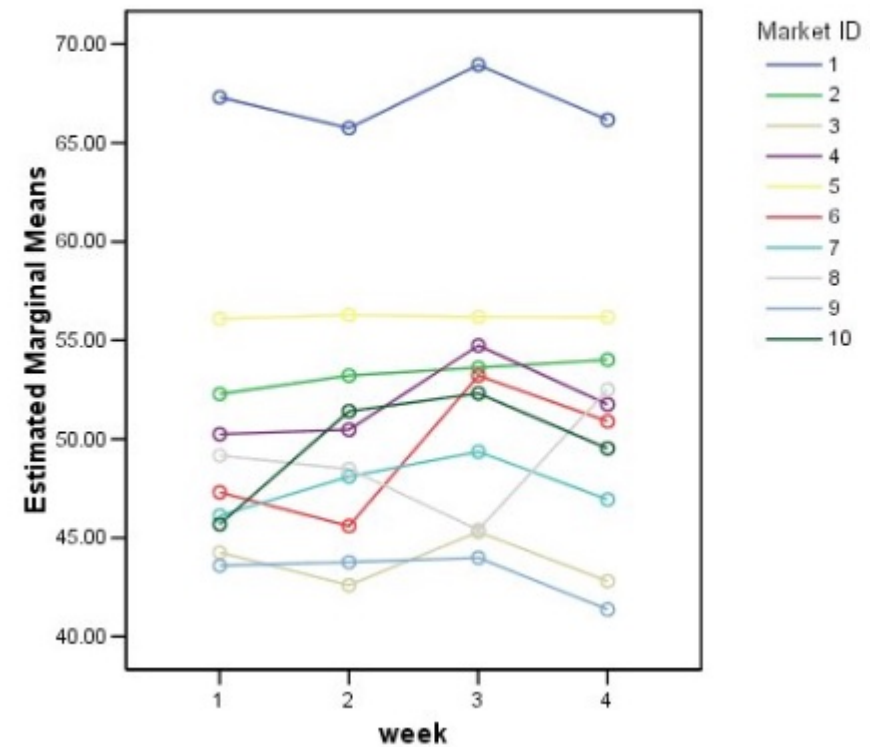


Figure 2. Profile plot for Market ID



2. Repeated measures using GLM Repeated Measures

Step 5: Within-subjects SSCP matrices

Figure 1. Within-subjects SSCP matrix

			WEEK : Column		
			Linear	Quadratic	Cubic
Hypothesis	Intercept	Linear	68.112	-46.798	-65.545
		Quadratic	-46.798	32.154	45.034
		Cubic	-65.545	45.034	63.075
	PROMO	Linear	3.572	13.789	9.615
		Quadratic	13.789	68.715	13.314
		Cubic	9.615	13.314	62.475
	MARKETID	Linear	123.224	.755	-77.768
		Quadratic	.755	320.267	65.502
		Cubic	-77.768	65.502	260.493
	PROMO * MARKETID	Linear	234.148	-.065	-142.181
		Quadratic	-.065	400.964	-103.383
		Cubic	-142.181	-103.383	989.186
Error	Linear	2232.956	-338.165	391.898	
	Quadratic	-338.165	3305.203	-100.125	
	Cubic	391.898	-100.125	3998.129	

Based on Type IV Sum of Squares

Used as input to the
multivariate and
univariate tests

2. Repeated measures using GLM Repeated Measures

Step 6: Multivariate tests of within-subjects effects

Multivariate Tests^a

Effect		Value	F	Hypothesis df	Error df	Sig.	Partial Eta Squared
week	Pillai's Trace	.056	2.033 ^b	3.000	103.000	.114	.056
	Wilks' Lambda	.944	2.033 ^b	3.000	103.000	.114	.056
	Hotelling's Trace	.059	2.033 ^b	3.000	103.000	.114	.056
	Roy's Largest Root	.059	2.033 ^b	3.000	103.000	.114	.056
week * marketid	Pillai's Trace	.210 ^c	.877	27.000	315.000	.646	.070
	Wilks' Lambda	.803 ^c	.870	27.000	301.455	.656	.070
	Hotelling's Trace	.229 ^c	.863	27.000	305.000	.666	.071
	Roy's Largest Root	.109 ^c	1.270 ^d	9.000	105.000	.262	.098
week * promo	Pillai's Trace	.038 ^c	.678	6.000	208.000	.668	.019
	Wilks' Lambda	.962 ^c	.672 ^b	6.000	206.000	.672	.019
	Hotelling's Trace	.039 ^c	.666	6.000	204.000	.677	.019
	Roy's Largest Root	.026 ^c	.885 ^d	3.000	104.000	.452	.025
week * marketid * promo	Pillai's Trace	.409	1.037	48.000	315.000	.413	.136
	Wilks' Lambda	.638	1.043	48.000	307.142	.403	.139
	Hotelling's Trace	.495	1.049	48.000	305.000	.393	.142
	Roy's Largest Root	.295	1.935 ^d	16.000	105.000	.025	.228

a. Design: Intercept + marketid + promo + marketid * promo
Within Subjects Design: week

b. Exact statistic

c. The Type IV testable hypothesis is not unique.

d. The statistic is an upper bound on F that yields a lower bound on the significance level.

None of the within-subjects effects are significant

2. Repeated measures using GLM Repeated Measures

Step 7: Testing homogeneity of covariance matrices

Figure 1. Box's test of equality of covariance matrices

Box's M	179.513
F	1.043
df1	120
df2	3687.015
Sig.	.357

← passes test for homogeneity
– required for multivariate
approach

2. Repeated measures using GLM Repeated Measures

Step 8: Mauchly's test of sphericity

(this is required for repeated measures – univariate approach)

Figure 1. Mauchly's test of sphericity

Measure: SALES

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Epsilon		
					Greenhouse-Geisser	Huynh-Feldt	Lower-bound
WEEK	.889	12.231	5	.032	.934	1.000	.333

↑
Typically use this correction value

2. Repeated measures using GLM Repeated Measures

Step 9: Univariate tests of within-subjects effects

Tests of Within-Subjects Effects

Measure: sales

Source		Type IV Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
week	Sphericity Assumed	163.341	3	54.447	1.798	.147	.017
	Greenhouse-Geisser	163.341	2.802	58.287	1.798	.151	.017
	Huynh-Feldt	163.341	3.000	54.447	1.798	.147	.017
	Lower-bound	163.341	1.000	163.341	1.798	.183	.017
week * marketid	Sphericity Assumed	703.984 ^a	27	26.073	.861	.668	.069
	Greenhouse-Geisser	703.984 ^a	25.221	27.912	.861	.661	.069
	Huynh-Feldt	703.984 ^a	27.000	26.073	.861	.668	.069
	Lower-bound	703.984 ^a	9.000	78.220	.861	.562	.069
week * promo	Sphericity Assumed	134.762 ^a	6	22.460	.742	.616	.014
	Greenhouse-Geisser	134.762 ^a	5.605	24.044	.742	.607	.014
	Huynh-Feldt	134.762 ^a	6.000	22.460	.742	.616	.014
	Lower-bound	134.762 ^a	2.000	67.381	.742	.479	.014
week * marketid * promo	Sphericity Assumed	1624.297	48	33.840	1.118	.285	.146
	Greenhouse-Geisser	1624.297	44.838	36.226	1.118	.290	.146
	Huynh-Feldt	1624.297	48.000	33.840	1.118	.285	.146
	Lower-bound	1624.297	16.000	101.519	1.118	.349	.146
Error(week)	Sphericity Assumed	9536.287	315	30.274			
	Greenhouse-Geisser	9536.287	294.248	32.409			
	Huynh-Feldt	9536.287	315.000	30.274			
	Lower-bound	9536.287	105.000	90.822			

Agrees with multivariate tests.
None of the within-subjects effects are significant

2. Repeated measures using GLM Repeated Measures

Step 10: Contrast Results

Tests of Within-Subjects Contrasts

Measure: sales

Source	week	Type IV Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	
week	Linear	68.112	1	68.112	3.203	.076	.030	Marginally significant
	Quadratic	32.154	1	32.154	1.021	.314	.010	
	Cubic	63.075	1	63.075	1.656	.201	.016	
week * marketid	Linear	123.224 ^a	9	13.692	.644	.757	.052	not significant
	Quadratic	320.267 ^a	9	35.585	1.130	.348	.088	
	Cubic	260.493 ^a	9	28.944	.760	.653	.061	
week * promo	Linear	3.572 ^a	2	1.786	.084	.920	.002	?
	Quadratic	68.715 ^a	2	34.358	1.091	.339	.020	
	Cubic	62.475 ^a	2	31.238	.820	.443	.015	
week * marketid * promo	Linear	234.148	16	14.634	.688	.800	.095	
	Quadratic	400.964	16	25.060	.796	.687	.108	
	Cubic	989.186	16	61.824	1.624	.075	.198	
Error(week)	Linear	2232.956	105	21.266				
	Quadratic	3305.203	105	31.478				
	Cubic	3998.129	105	38.077				

a. The Type IV testable hypothesis is not unique.

2. Repeated measures using GLM Repeated Measures

Note that you will have estimates of means for the repeated measure “week”.

Estimates				
Measure: Sales				
week	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
1	50.471 ^a	.593	49.295	51.648
2	50.717 ^a	.594	49.539	51.894
3	52.275 ^a	.594	51.098	53.452
4	51.288 ^a	.555	50.189	52.388

a. Based on modified population marginal mean.

2. Repeated measures using GLM Repeated Measures

Step 11: Summary

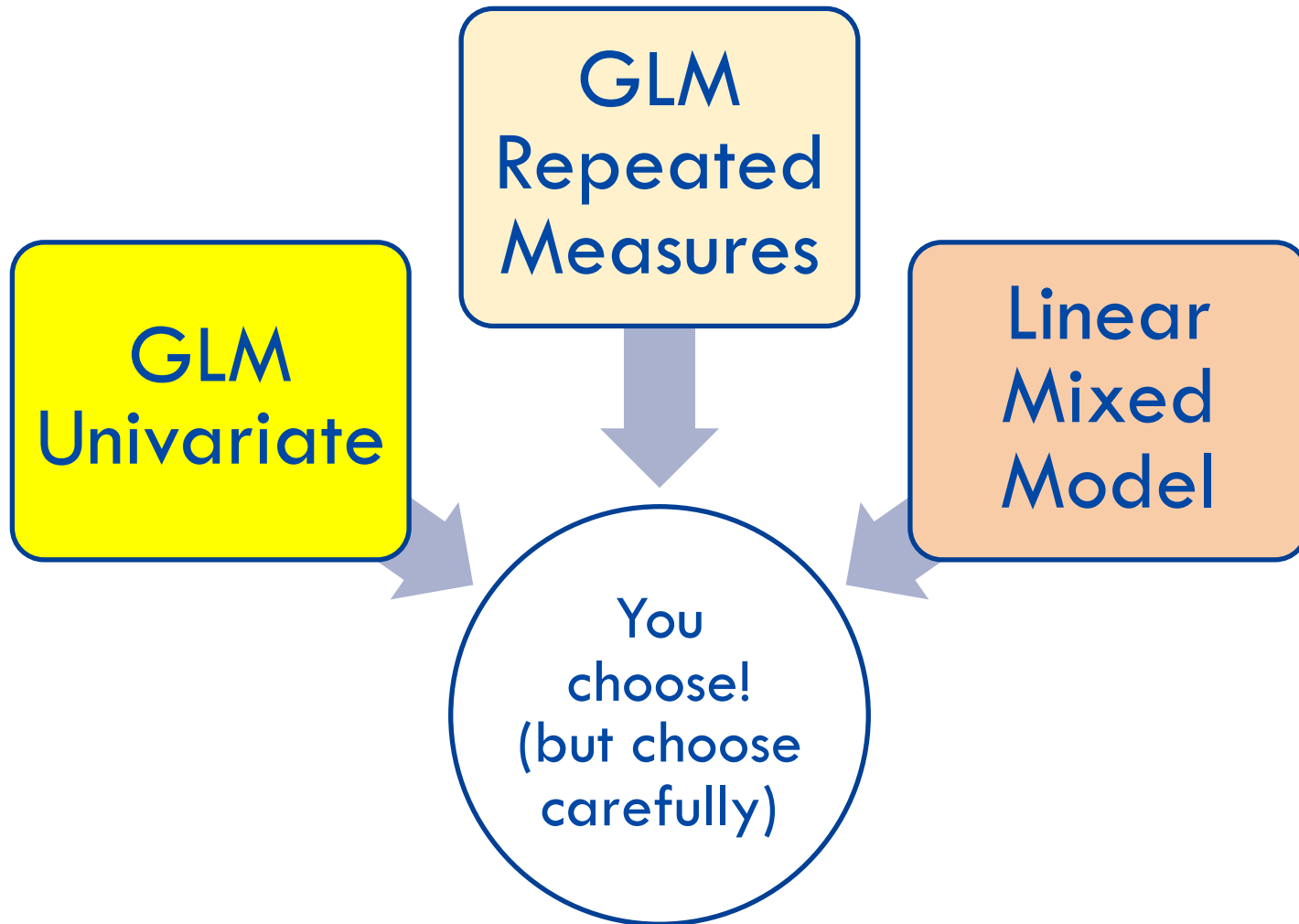
Using the GLM Repeated Measures procedure, you have found that there is considerable market-to-market variation in sales. Also, the profile plots suggest that the first promotion is more effective than the others. In order to generalize these results to other markets, you would like to be able to specify *Market ID* as a random effect. While the GLM Repeated Measures procedure does not allow for random effects, the GLM Univariate procedure does not allow for within-subjects effects.

However, another of your findings was that the within-subjects factor *Week* might not be contributing to the model. If you choose to ignore the possibility of week-to-week variation, you could fit *Market ID* as a random effect.

Another solution is to use the Linear Mixed Models procedure. That procedure will allow you to specify *Market ID* as a random effect and give you greater control over the specification of the covariance matrix for the within-subjects factor.

In any case, the *Promotion * Market ID* interaction should be removed, because it does not contribute to the model.

Choice of GLM procedure for repeated measures



3. Linear regression

Linear regression is useful where you have several (or many) independent variables that may be useful in modelling a dependent variable. The process of model building will show which independent variables are significant as well as their strength of influence on the dependent outcome.

- The independent variables must be numeric (ordinal or scale in SPSS)

3. Linear regression

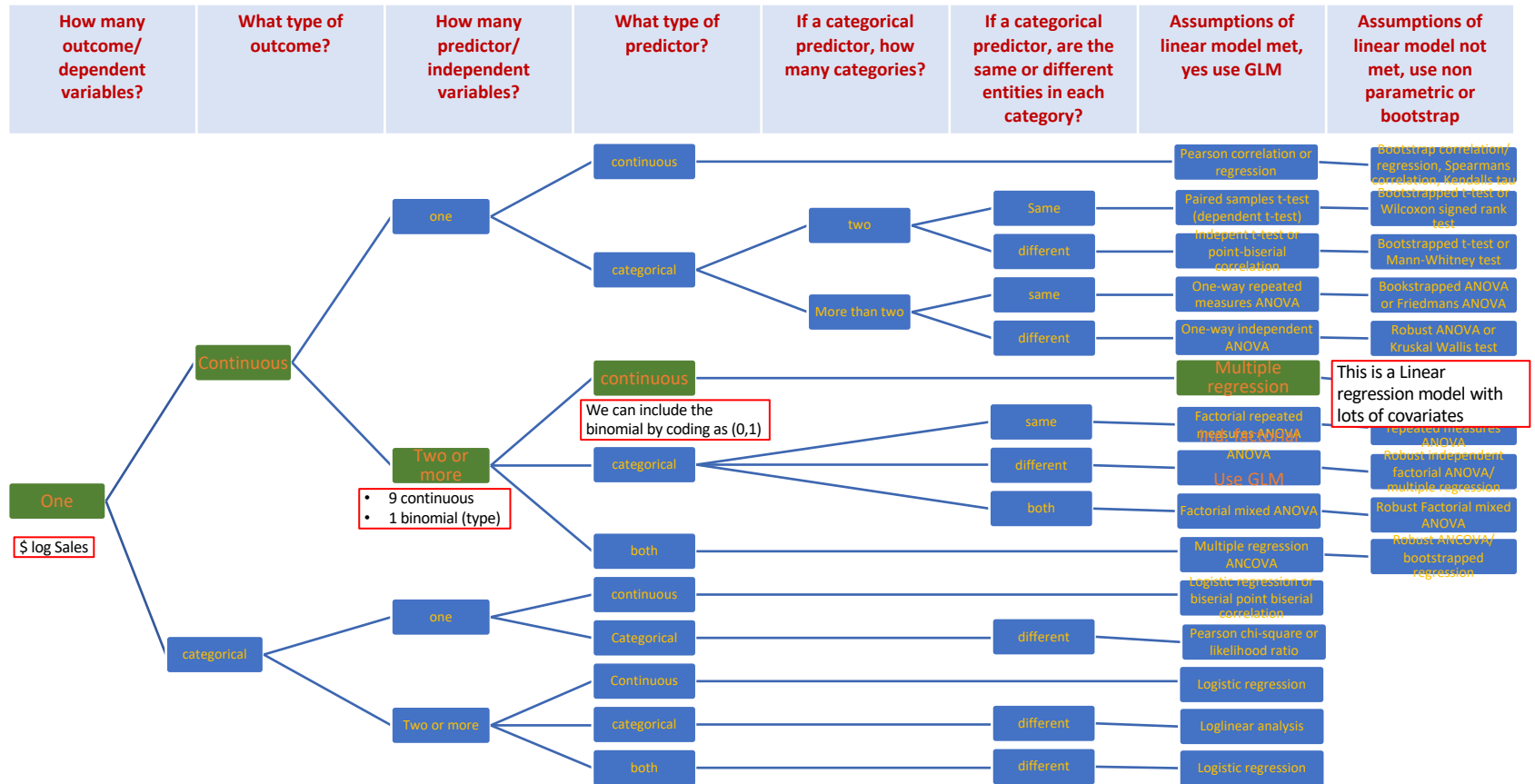
SPSS tutorial example using `car_sales.sav` file

We are interested to find out the relationship between vehicle sales and vehicle characteristics

Two approaches

1. Follow tutorial example including factor analysis
2. Simple model building approach

3. Linear regression



3. Linear regression

1. Open SPSS
2. Open another file > choose “**car_sales.sav**” (saved on USB)
3. Go to help/topics, then search “model vehicle sales”
4. Choose “**Using linear regression to model vehicle sales**”

Search

model vehicle sales



Change version ▾

Documentation

Videos

IBM Developer

Technotes

Redbooks

Take a tour

154 results

+ Using linear regression to model vehicle sales

An automotive industry group keeps track of the **sales** for a variety of personal motor **vehicles**. In an effort to be able to identify over- and underperforming **models**, you want to establish a relationship between **vehicle sales** and **vehicle** characteristics. Information

3. Linear regression

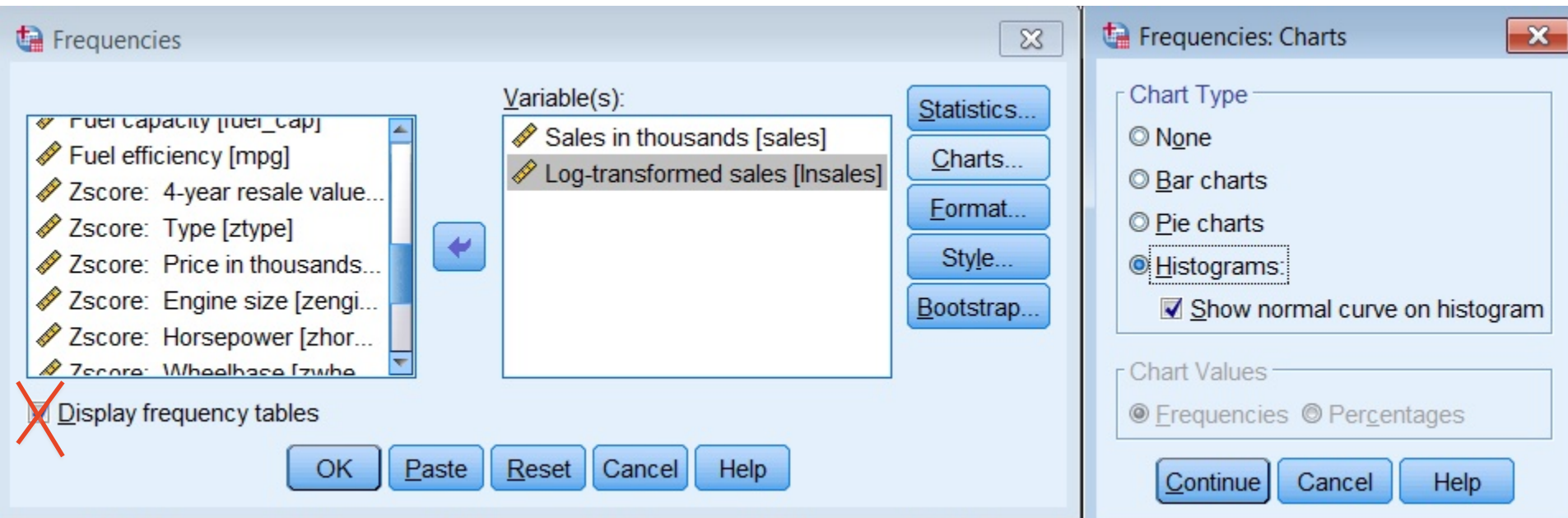
- Not in the tutorial, but we should always plot the data
- Plot histograms of the dependent variable
- Look at the distribution of a key independent variable

3. Linear regression

Normality check

We suspect that the sales are not normally distributed.

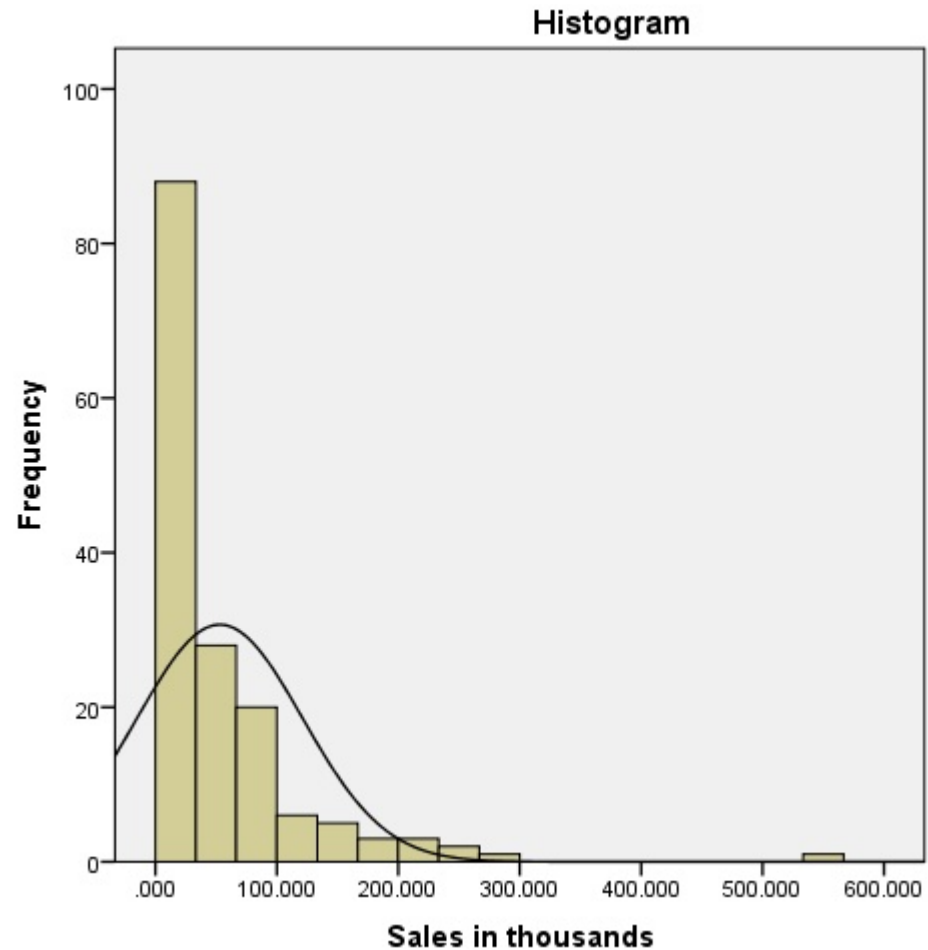
Analyze>Descriptive Statistics>Frequencies



3. Linear regression

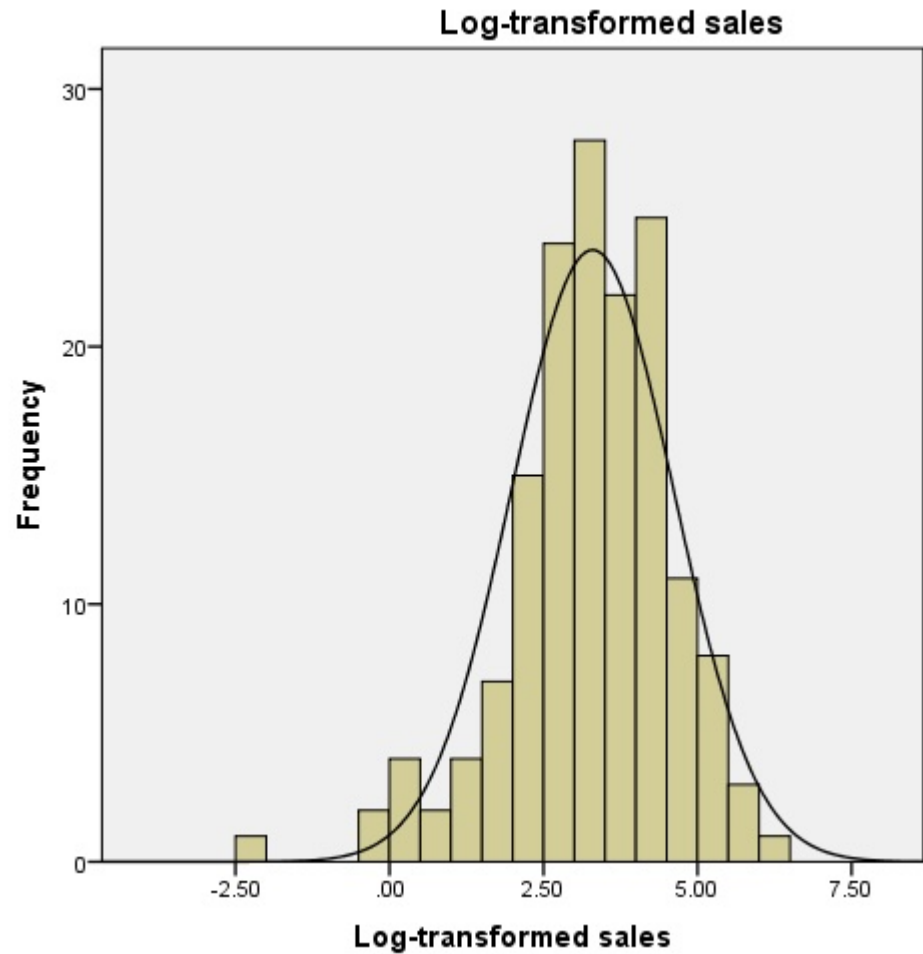
Sales histogram

Skewed to the right



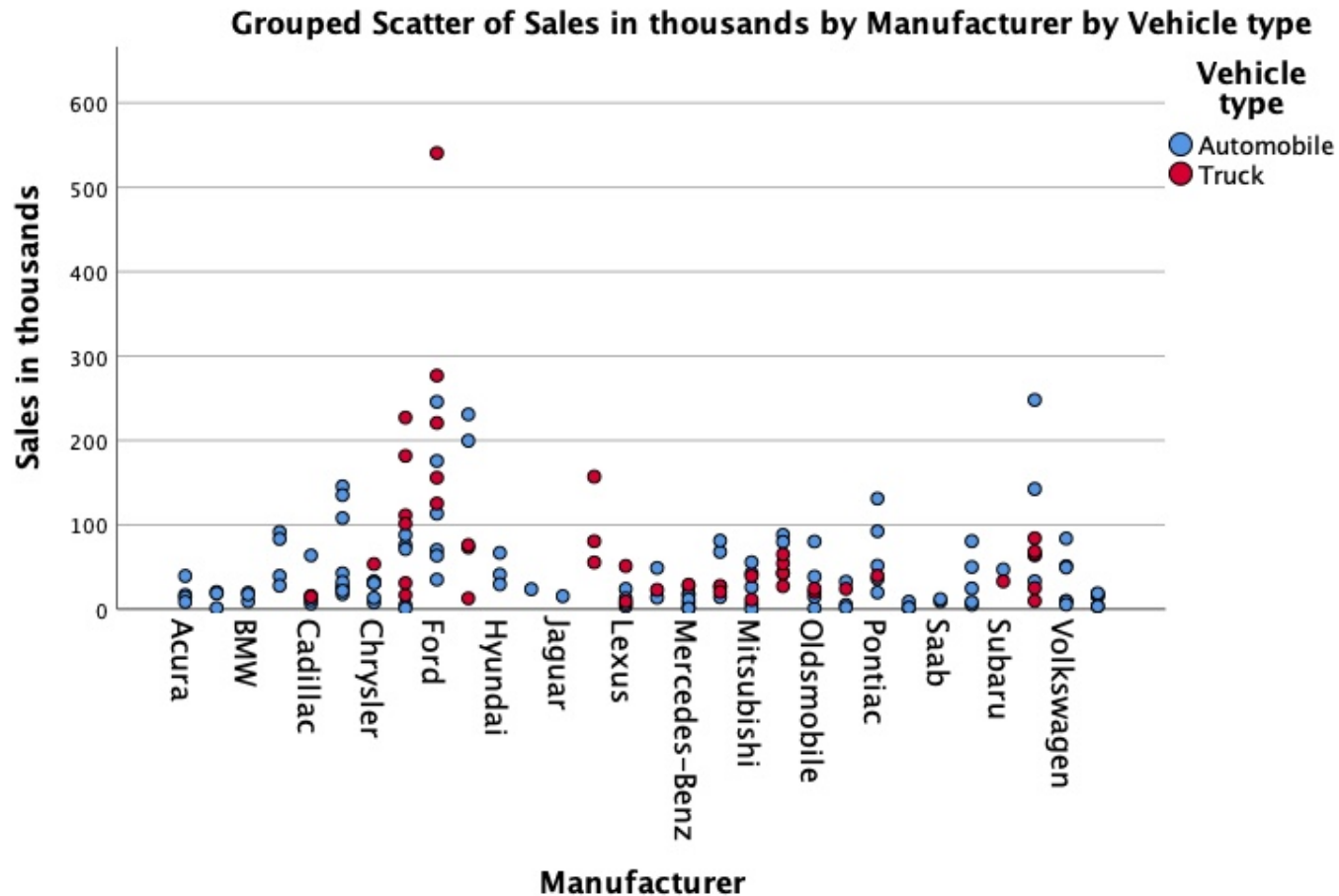
3. Linear regression

log transformed Sales is much better



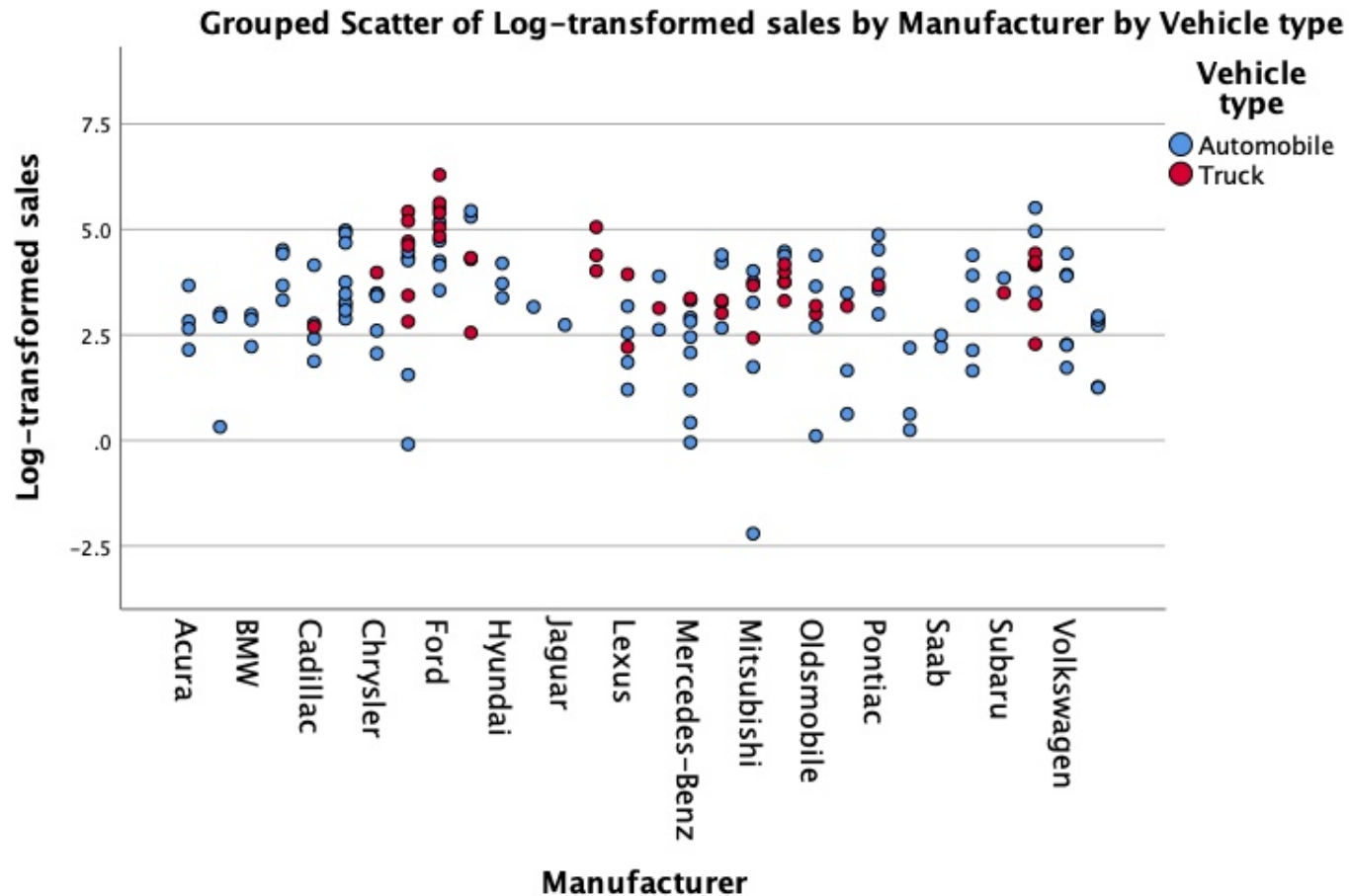
3. Linear regression

Have a look at the vehicle model sales by manufacturer and type



3. Linear regression

Look at the log transformed sales as well



3. Linear regression

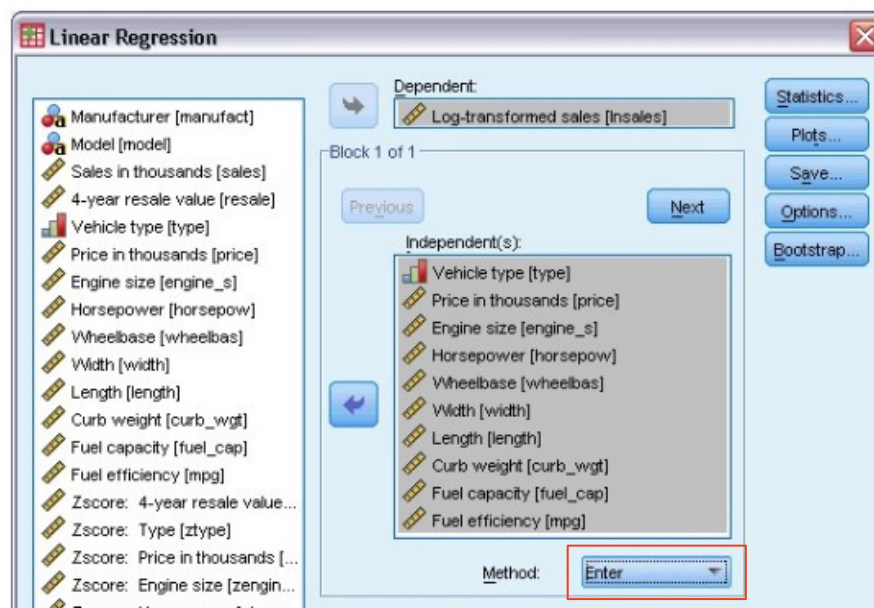
1. Keep tutorial window open alongside your SPSS dataset window for reference.
2. Now follow the steps in the tutorial.

Step 1: Running the analysis

1. To run a linear regression analysis, from the menus choose:

Analyze > Regression > Linear...

Figure 1. Linear Regression dialog box



3. Linear regression

Step 1: Select statistics to output

Linear Regression: Statistics

Regression Coefficients

- ☒ Estimates
- ☐ Confidence intervals
Level(%): 95
- ☐ Covariance matrix

☒ Model fit

☐ R squared change

☐ Descriptives

☒ Part and partial correlations

☒ Collinearity diagnostics

Residuals

- ☐ Durbin-Watson
- ☐ Casewise diagnostics
- ☒ Outliers outside: 3 standard deviations
- ☐ All cases

? Cancel Continue

3. Linear regression

Step 2: Checking the model fit

Figure 1. ANOVA table

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	130.300	10	13.030	13.305	.000 ^a
	Residual	138.082	141	.979		
	Total	268.383	151			

a. Predictors: (Constant), Fuel efficiency, Length, Price in thousands, Vehicle type, Width, Engine size, Fuel capacity, Wheelbase, Curb weight, Horsepower

The ANOVA table reports a significant F statistic, indicating that using the model is better than guessing the mean.

Figure 2. Model summary table

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.697 ^a	.486	.449	.98960

a. Predictors: (Constant), Fuel efficiency, Length, Price in thousands, Vehicle type, Width, Engine size, Fuel capacity, Wheelbase, Curb weight, Horsepower

As a whole, the regression does a good job of modeling sales. Nearly half the variation in sales is explained by the model.

3. Linear regression

Step 3: Coefficients

- Significance of each independent variable

Figure 1. Coefficients table, first half

Model	Variables	Statistics				
		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	-3.017	2.741		-1.101	.273
	Vehicle type	.883	.331	.293	2.670	.008
	Price in thousands	-.046	.013	-.502	-3.596	.000
	Engine size	.356	.190	.281	1.871	.063
	Horsepower	-.002	.004	-.092	-.509	.611
	Wheelbase	.042	.023	.241	1.785	.076
	Width	-.028	.042	-.073	-.676	.500
	Length	.015	.014	.148	1.032	.304
	Curb weight	.156	.350	.075	.447	.655
	Fuel capacity	-.057	.047	-.167	-1.203	.231
	Fuel efficiency	.081	.040	.262	2.023	.045

Figure 2. Coefficients table, second half

- Collinearity
- VIFs <2 are OK
- $2 < \text{VIF} < 10$ need attention
- $\text{VIF} > 10$ is extreme!

Model		Correlations			Collinearity Statistics	
		Zero-order	Partial	Part	Tolerance	VIF
1	Vehicle type	.274	.219	.161	.304	3.293
	Price in thousands	-.552	-.290	-.217	.187	5.337
	Engine size	-.135	.156	.113	.162	6.159
	Horsepower	-.389	-.043	-.031	.112	8.896
	Wheelbase	.292	.149	.108	.200	4.997
	Width	.037	-.057	-.041	.313	3.193
	Length	.215	.087	.062	.178	5.605
	Curb weight	-.041	.038	.027	.131	7.644
	Fuel capacity	-.016	-.101	-.073	.189	5.303
	Fuel efficiency	.121	.168	.122	.217	4.604

3. Linear regression

Step 4: Collinearity diagnostics

Figure 1. Collinearity diagnostics table

Model	Dimension	Eigenvalue	Condition Index
1	1	9.920	1.000
	2	.733	3.678
	3	.259	6.193
	4	.050	14.051
	5	.019	22.589
	6	.008	35.942
	7	.005	44.275
	8	.003	58.480
	9	.002	76.175
	10	.001	130.747
	11	.000	148.267

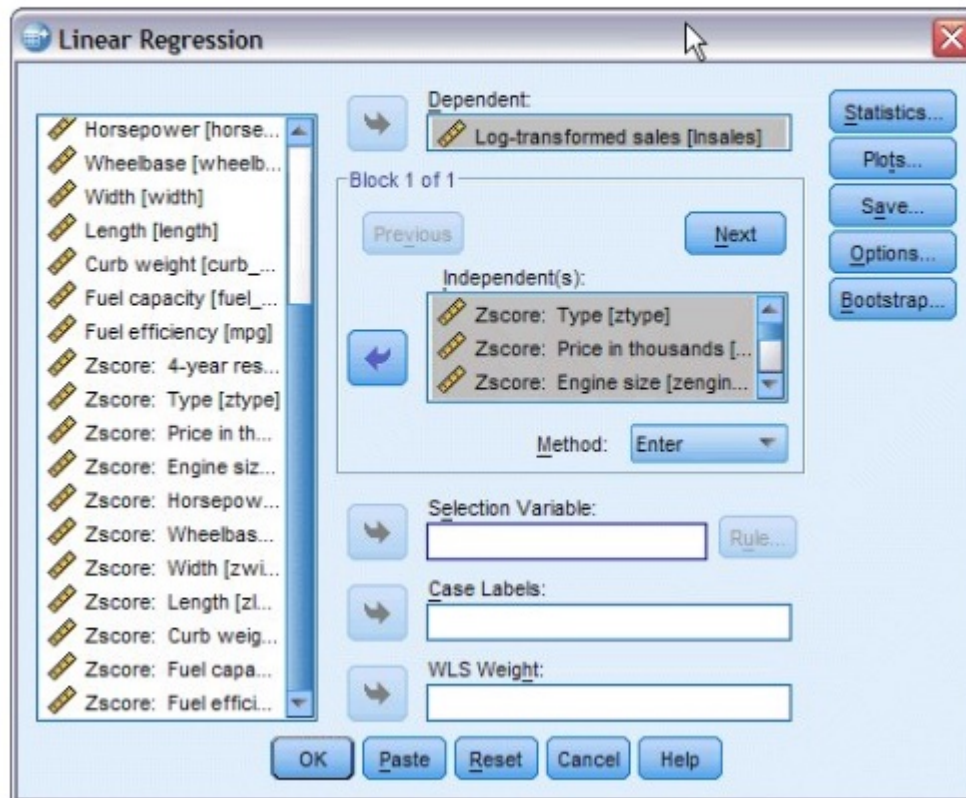
Cond Index >30
indicates serious
collinearity

The collinearity diagnostics confirm that there are serious problems with multicollinearity. Several eigenvalues are close to 0, indicating that the predictors are highly intercorrelated and that small changes in the data values may lead to large changes in the estimates of the coefficients.

3. Linear regression

Step 5: Running a linear regression on the z-scores

Figure 1. Linear Regression dialog box



Z-scores are standardized variables created from the original variables, re-scaled to produce equal influence in the model

3. Linear regression

Step 6: Collinearity Diagnostics

Figure 1. Collinearity diagnostics table

Model	Dimension	Eigenvalue	Condition Index
1	1	5.994	1.000
	2	1.654	1.903
	3	1.127	2.306
	4	.996	2.453
	5	.339	4.206
	6	.254	4.857
	7	.199	5.484
	8	.155	6.225
	9	.130	6.793
	10	.091	8.137
	11	.061	9.874

a. Dependent Variable: Log-transformed sales

The eigenvalues and condition indices are vastly improved relative to the original model.

Figure 2. Coefficients table, second half

Model		Correlations			Collinearity Statistics	
		Zero-order	Partial	Part	Tolerance	VIF
1	(Constant)					
	Zscore: Type	.274	.219	.161	.304	3.293
	Zscore: Price in thousands	-.552	-.290	-.217	.187	5.337
	Zscore: Engine size	-.135	.156	.113	.162	6.159
	Zscore: Horsepower	-.389	-.043	-.031	.112	8.896
	Zscore: Wheelbase	.292	.149	.108	.200	4.997
	Zscore: Width	.037	-.057	-.041	.313	3.193
	Zscore: Length	.215	.087	.062	.178	5.605
	Zscore: Curb weight	-.041	.038	.027	.131	7.644
	Zscore: Fuel capacity	-.016	-.101	-.073	.189	5.303
	Zscore: Fuel efficiency	.121	.168	.122	.217	4.604

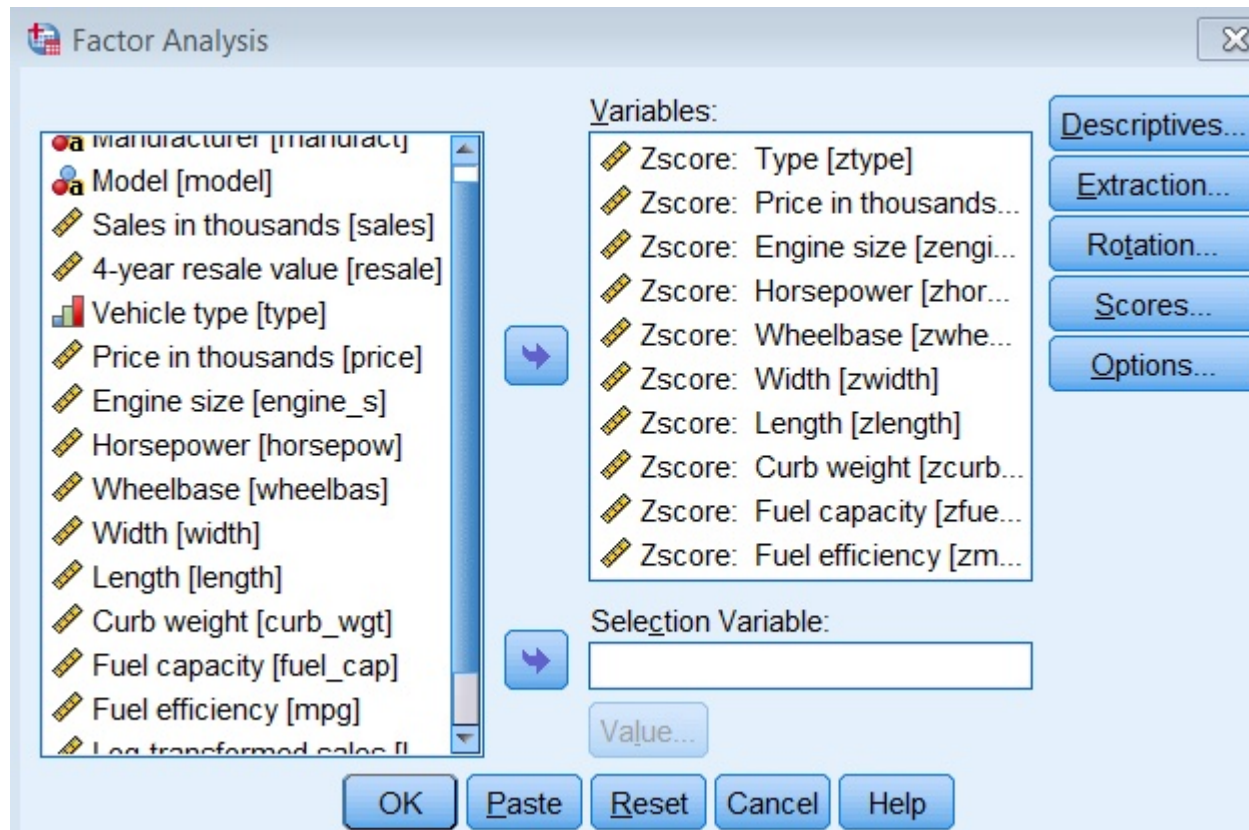
a. Dependent Variable: Log-transformed sales

However, the collinearity statistics reported in the Coefficients table are unimproved. This is because the z-score transformation does not change the correlation between two variables. As a

3. Linear regression

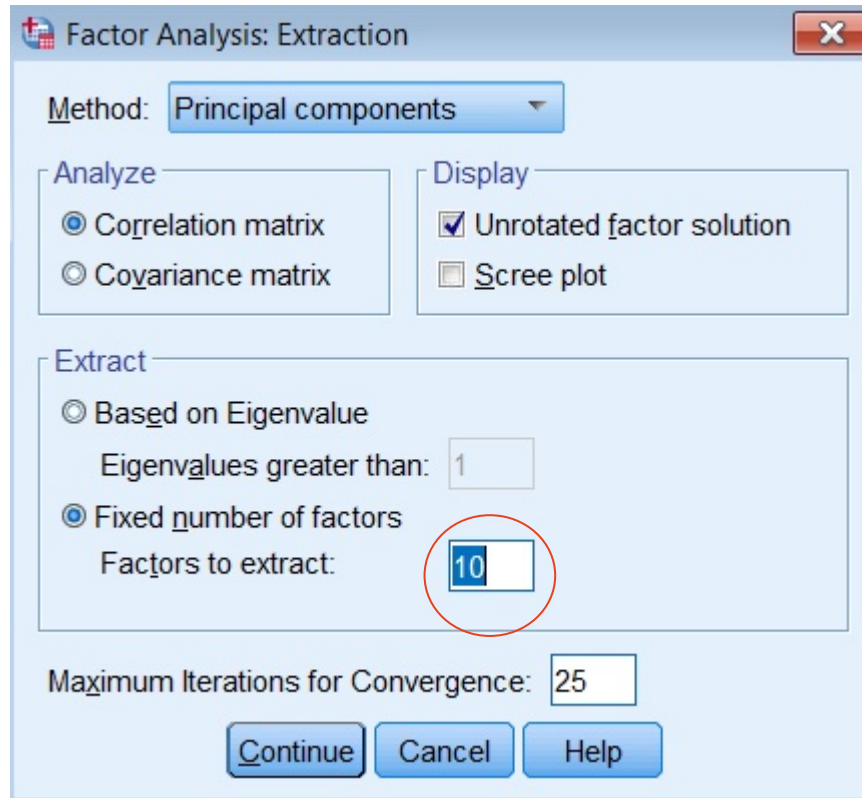
Step 7: Running a linear regression on factor component scores

Analyze>Dimension Reduction>Factor



3. Linear regression

Step 7: Running a linear regression on factor component scores
Extraction



The image shows the 'Factor Analysis: Extraction' dialog box in SPSS. The 'Method' is set to 'Principal components'. Under the 'Analyze' section, 'Correlation matrix' is selected. Under the 'Display' section, 'Unrotated factor solution' is checked, and 'Scree plot' is unchecked. Under the 'Extract' section, 'Fixed number of factors' is selected, and the 'Factors to extract' is set to 10, which is circled in red. The 'Maximum iterations for Convergence' is set to 25. At the bottom are 'Continue', 'Cancel', and 'Help' buttons.

Factor Analysis: Extraction

Method: Principal components

Analyze

- ☒ Correlation matrix
- ☐ Covariance matrix

Display

- ☒ Unrotated factor solution
- ☐ Scree plot

Extract

- ☐ Based on Eigenvalue
Eigenvalues greater than: 1
- ☒ Fixed number of factors
Factors to extract: 10

Maximum iterations for Convergence: 25

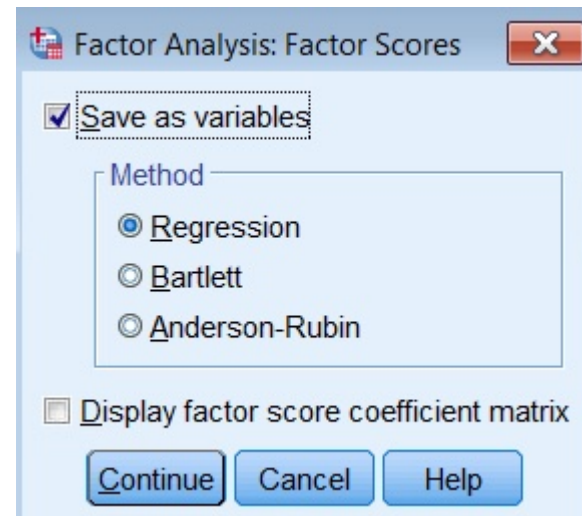
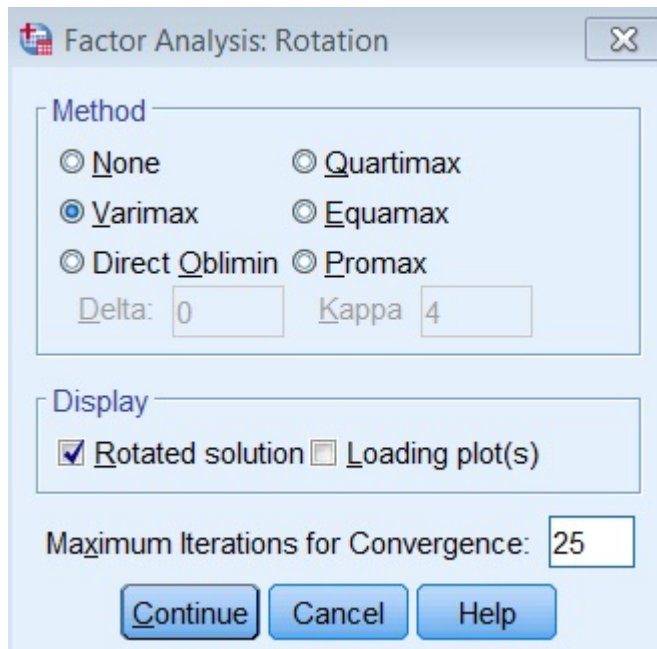
Continue Cancel Help

3. Linear regression

Step 7: Running a linear regression on factor component scores

Rotation

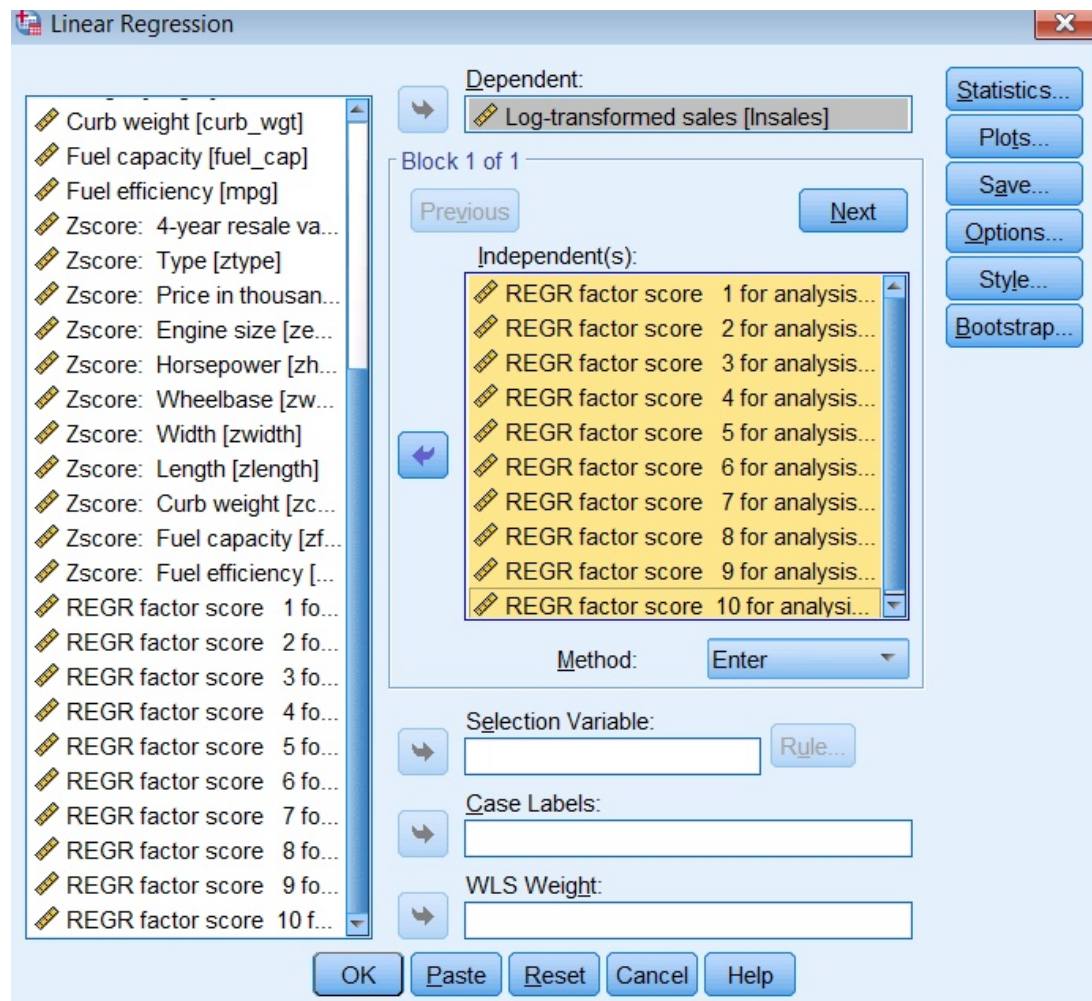
Scores



3. Linear regression

Step 7: Running a linear regression on factor component scores

Analyze>Regression>Linear



3. Linear regression

Step 7: Running a linear regression on factor component scores

Output: ANOVA table

Note: F statistic is the same as for original model (all the same information has been included)

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	130.300	10	13.030	13.305	.000 ^b
	Residual	138.082	141	.979		
	Total	268.383	151			

a. Dependent Variable: Log-transformed sales

b. Predictors: (Constant), REGR factor score 10 for analysis 1, REGR factor score 9 for analysis 1, REGR factor score 8 for analysis 1, REGR factor score 7 for analysis 1, REGR factor score 6 for analysis 1, REGR factor score 5 for analysis 1, REGR factor score 4 for analysis 1, REGR factor score 3 for analysis 1, REGR factor score 2 for analysis 1, REGR factor score 1 for analysis 1

3. Linear regression

Step 7: Running a linear regression on factor component scores

Output: Coefficients table

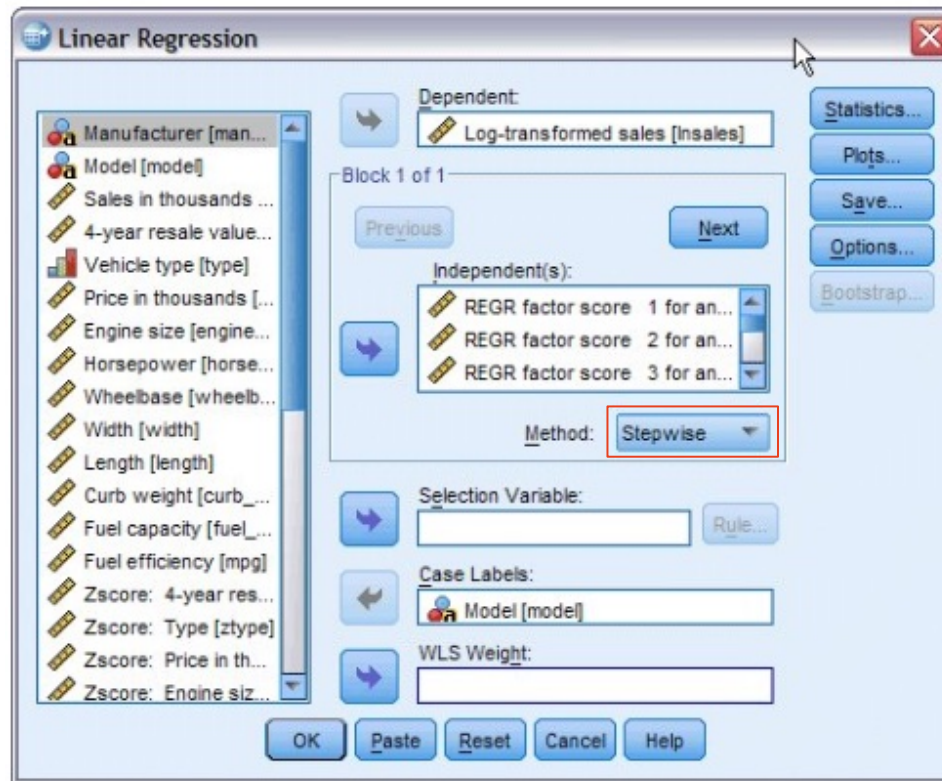
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	Collinearity Statistics	
		B	Std. Error	Beta			Tolerance	VIF
1	(Constant)	3.289	.080		40.976	.000		
	REGR factor score 1 for analysis 1	-.714	.081	-.535	-8.863	.000	1.000	1.000
	REGR factor score 2 for analysis 1	.414	.081	.310	5.138	.000	1.000	1.000
	REGR factor score 3 for analysis 1	.307	.081	.230	3.809	.000	1.000	1.000
	REGR factor score 4 for analysis 1	-.008	.081	-.006	-.100	.921	1.000	1.000
	REGR factor score 5 for analysis 1	-.174	.081	-.131	-2.161	.032	1.000	1.000
	REGR factor score 6 for analysis 1	.182	.081	.136	2.258	.025	1.000	1.000
	REGR factor score 7 for analysis 1	-.135	.081	-.101	-1.677	.096	1.000	1.000
	REGR factor score 8 for analysis 1	-.005	.081	-.004	-.065	.948	1.000	1.000
	REGR factor score 9 for analysis 1	.034	.081	.026	.428	.669	1.000	1.000
	REGR factor score 10 for analysis 1	.072	.081	.054	.898	.371	1.000	1.000

a. Dependent Variable: Log-transformed sales

3. Linear regression

Step 8: Running a stepwise linear regression

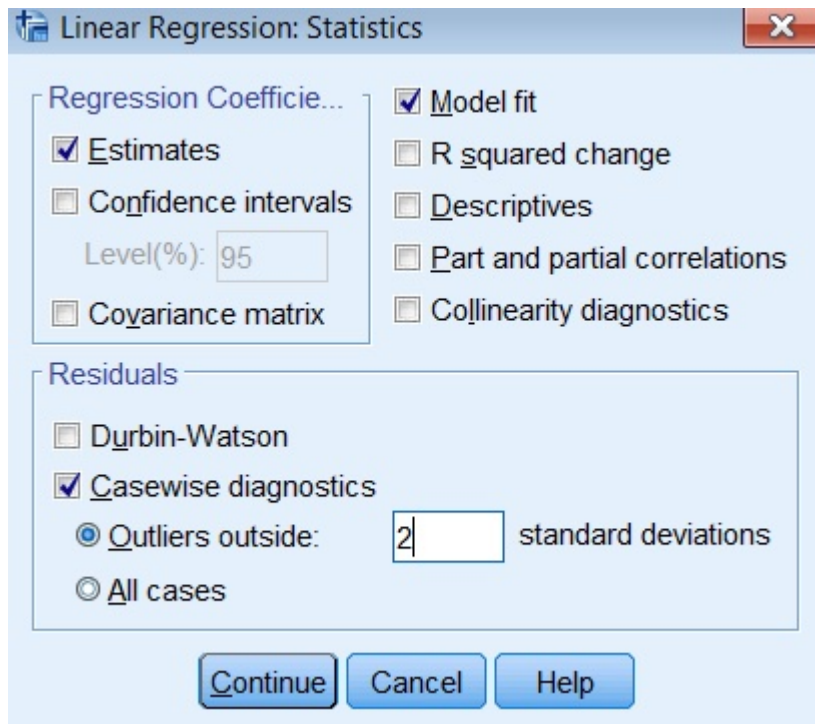
Figure 1. Linear Regression dialog box



1. For example, to run a stepwise Linear Regression on the factor scores,
2. Select **Stepwise** as the entry method.

3. Linear regression

Step 8: Statistics



Linear Regression: Statistics

Regression Coefficients

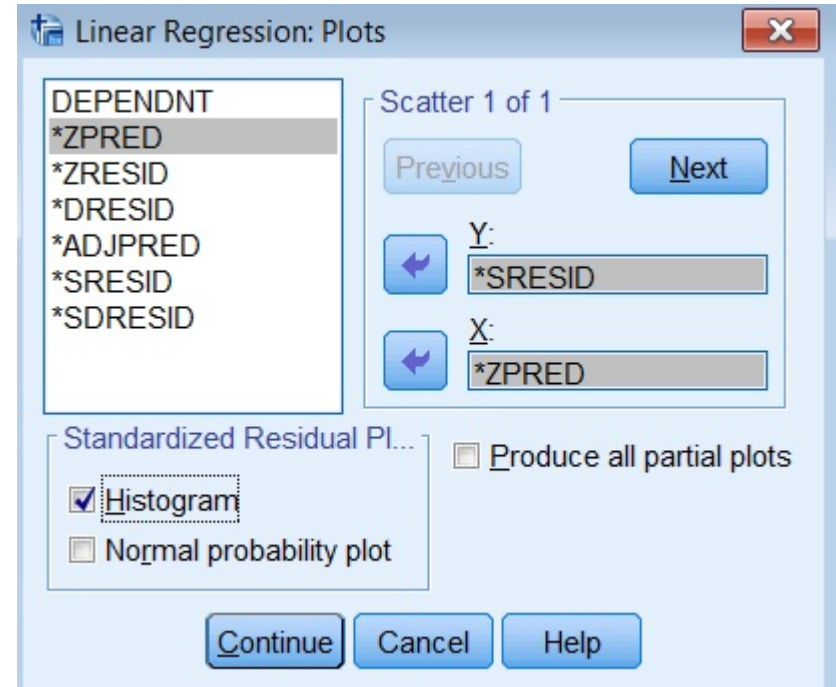
- ☒ Estimates
- ☐ Confidence intervals
Level(%): 95
- ☐ Covariance matrix
- ☒ Model fit
- ☐ R squared change
- ☐ Descriptives
- ☐ Part and partial correlations
- ☐ Collinearity diagnostics

Residuals

- ☐ Durbin-Watson
- ☒ Casewise diagnostics
 - ☒ Outliers outside: 2 standard deviations
 - ☐ All cases

Continue Cancel Help

Plots



Linear Regression: Plots

DEPENDNT

- *ZPRED
- *ZRESID
- *DRESID
- *ADJPRED
- *SRESID
- *SDRESID

Scatter 1 of 1

Previous Next

Y: *SRESID

X: *ZPRED

Standardized Residual Plots

- ☒ Histogram
- ☐ Normal probability plot
- ☐ Produce all partial plots

Continue Cancel Help

3. Linear regression

Step 8: Save

- Predicted values
- Residuals
- Influence statistics



The image shows a 'Linear Regression: Save' dialog box with several sections for saving regression results. The 'Predicted Values' section has checkboxes for 'Unstandardized', 'Standardized' (checked), 'Adjusted', and 'S.E. of mean predictions'. The 'Residuals' section has checkboxes for 'Unstandardized', 'Standardized' (checked), 'Studentized', 'Deleted', and 'Studentized deleted'. The 'Distances' section has checkboxes for 'Mahalanobis', 'Cook's' (checked), and 'Leverage values'. The 'Prediction Intervals' section has checkboxes for 'Mean' and 'Individual', and a 'Confidence Interval' set to 95%. The 'Influence Statistics' section has checkboxes for 'DfBeta(s)', 'Standardized DfBeta(s)', 'DfFit', 'Standardized DfFit', and 'Covariance ratio'. The 'Coefficient statistics' section has a checkbox for 'Create coefficient statistics', which is checked, and two radio buttons: 'Create a new dataset' (selected) and 'Write a new data file'. Below the radio buttons are a 'Dataset name' text box and a 'File...' button. The 'Export model information to XML file' section has a text box and a 'Browse...' button. At the bottom, there is a checkbox for 'Include the covariance matrix' and three buttons: 'Continue', 'Cancel', and 'Help'.

Linear Regression: Save

Predicted Values

- ☐ Unstandardized
- ☒ Standardized
- ☐ Adjusted
- ☐ S.E. of mean predictions

Residuals

- ☐ Unstandardized
- ☒ Standardized
- ☐ Studentized
- ☐ Deleted
- ☐ Studentized deleted

Distances

- ☐ Mahalanobis
- ☒ Cook's
- ☒ Leverage values

Prediction Intervals

- ☐ Mean ☐ Individual
- Confidence Interval: 95 %

Influence Statistics

- ☐ DfBeta(s)
- ☐ Standardized DfBeta(s)
- ☐ DfFit
- ☐ Standardized DfFit
- ☐ Covariance ratio

Coefficient statistics

- ☒ Create coefficient statistics
 - ☒ Create a new dataset
 - Dataset name:
 - ☐ Write a new data file
 - File...

Export model information to XML file

☐ Include the covariance matrix

3. Linear regression

Step 9: Checking the model fit

Figure 1. Model summary table, model with *all predictors*

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.697 ^a	.486	.449	.98960

a. Predictors: (Constant), Fuel efficiency, Length, Price in thousands, Vehicle type, Width, Engine size, Fuel capacity, Wheelbase, Curb weight, Horsepower

Figure 2. Model summary table, model with *stepwise-selected predictors*

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.535 ^a	.287	.282	1.12975
2	.619 ^b	.383	.375	1.05423
3	.660 ^c	.436	.424	1.01138
4	.674 ^d	.455	.440	.99793
5	.687 ^e	.472	.453	.98558

1. Adjusted R square is now higher because model complexity is penalised.

3. Linear regression

Step 10: Stepwise Coefficients

Figure 1. Coefficients

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	3.289	.092		35.892	.000
	REGR factor score 1 for analysis 1	-.714	.092	-.535	-7.764	.000
2	(Constant)	3.289	.086		38.464	.000
	REGR factor score 1 for analysis 1	-.714	.086	-.535	-8.320	.000
	REGR factor score 2 for analysis 1	.414	.086	.310	4.823	.000
3	(Constant)	3.289	.082		40.093	.000
	REGR factor score 1 for analysis 1	-.714	.082	-.535	-8.672	.000
	REGR factor score 2 for analysis 1	.414	.082	.310	5.027	.000
	REGR factor score 3 for analysis 1	.307	.082	.230	3.727	.000
4	(Constant)	3.289	.081		40.633	.000
	REGR factor score 1 for analysis 1	-.714	.081	-.535	-8.789	.000
	REGR factor score 2 for analysis 1	.414	.081	.310	5.095	.000
	REGR factor score 3 for analysis 1	.307	.081	.230	3.778	.000
	REGR factor score 6 for analysis 1	.182	.081	.136	2.240	.027
5	(Constant)	3.289	.080		41.143	.000
	REGR factor score 1 for analysis 1	-.714	.080	-.535	-8.899	.000
	REGR factor score 2 for analysis 1	.414	.080	.310	5.159	.000
	REGR factor score 3 for analysis 1	.307	.080	.230	3.825	.000
	REGR factor score 6 for analysis 1	.182	.080	.136	2.268	.025
	REGR factor score 5 for analysis 1	-.174	.080	-.131	-2.170	.032

a. Dependent Variable: Log-transformed sales

3. Linear regression

Step 10: Factor Analysis – Rotated Component Matrix

Shows loadings of each original variable on the factor scores

Rotated Component Matrix^a

	Component									
	1	2	3	4	5	6	7	8	9	10
Zscore: Type	-.061	.112	.976	.062	.117	.051	.094	.043	.013	.000
Zscore: Price in thousands	.973	.011	-.011	.065	.078	-.012	.112	.086	.011	-.151
Zscore: Engine size	.594	.306	.188	.283	.203	.614	.115	.083	-.005	.016
Zscore: Horsepower	.873	.184	-.001	.190	.165	.267	.057	.008	-.020	.258
Zscore: Wheelbase	.066	.894	.268	.186	.082	.042	.120	.026	.258	-.005
Zscore: Width	.252	.501	.135	.775	.141	.153	.129	.075	.003	.013
Zscore: Length	.117	.919	.012	.204	.112	.146	.092	.085	-.225	.029
Zscore: Curb weight	.434	.439	.416	.272	.244	.155	.259	.471	-.006	-.002
Zscore: Fuel capacity	.324	.408	.476	.226	.231	.113	.610	.115	.006	.004
Zscore: Fuel efficiency	-.415	-.247	-.465	-.206	-.658	-.173	-.190	-.097	-.001	-.014

Extraction Method: Principal Component Analysis.

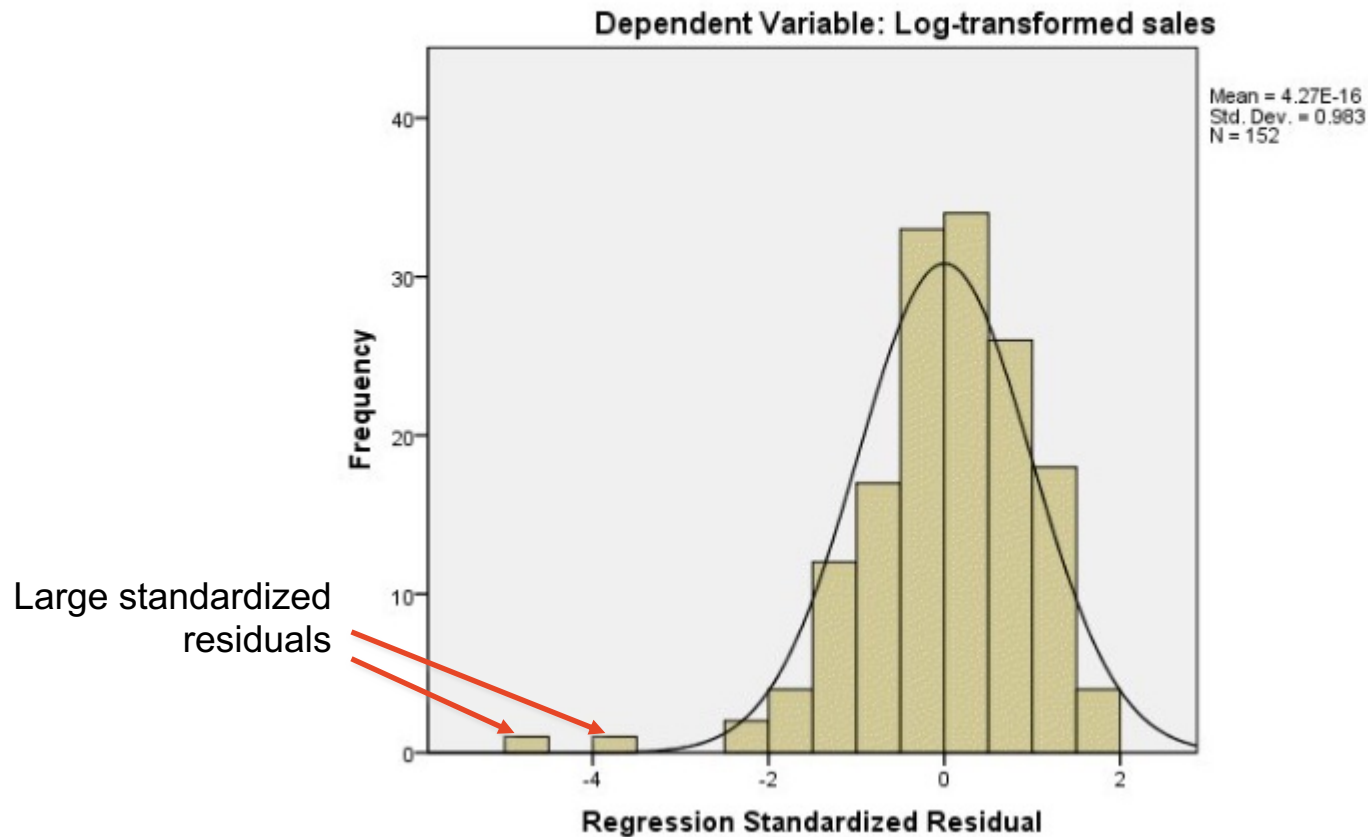
Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 6 iterations.

3. Linear regression

Step 11: Checking the Normality of the Error Term

Figure 1. Histogram of residuals with normal curve



3. Linear regression

Step 12: Casewise diagnostics

Figure 1. Casewise diagnostics table

Case Number	Model	Std. Residual	Log-transformed sales	Predicted Value	Residual
84	3000GT	-4.896	-2.21	2.6185	-4.82576
109	Cutlass	-3.828	.11	3.8792	-3.77301
116	Breeze	-2.025	1.66	3.6520	-1.99570
132	SW	-2.191	1.65	3.8121	-2.15907

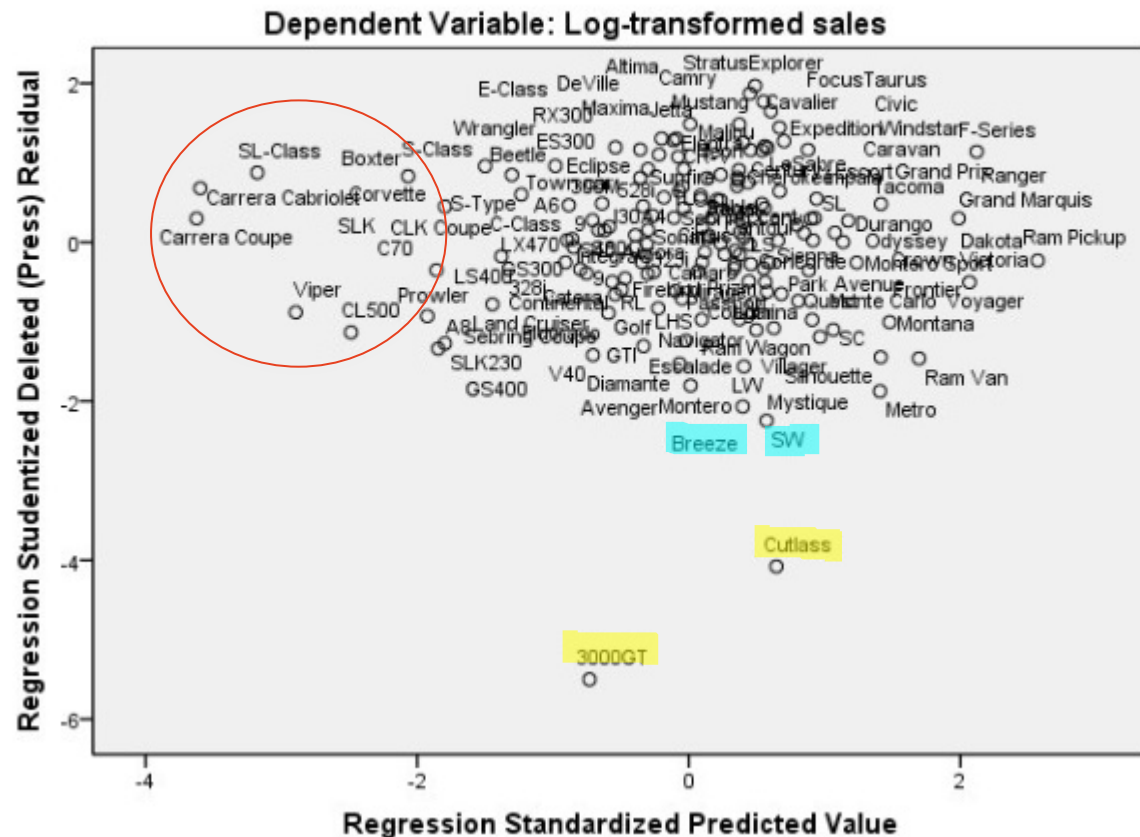
a. Dependent Variable: Log-transformed sales

This table identifies the cases with large negative residuals as the 3000GT and the Cutlass. This means that, based on the expected sales predicted by the regression model, these two models underperformed in the market. The Breeze and SW also appear to have underperformed to a lesser extent.

3. Linear regression

Step 13: Residual scatterplots

Figure 1. Residuals by predicted values



3. Linear regression

Step 13: Residual scatterplots

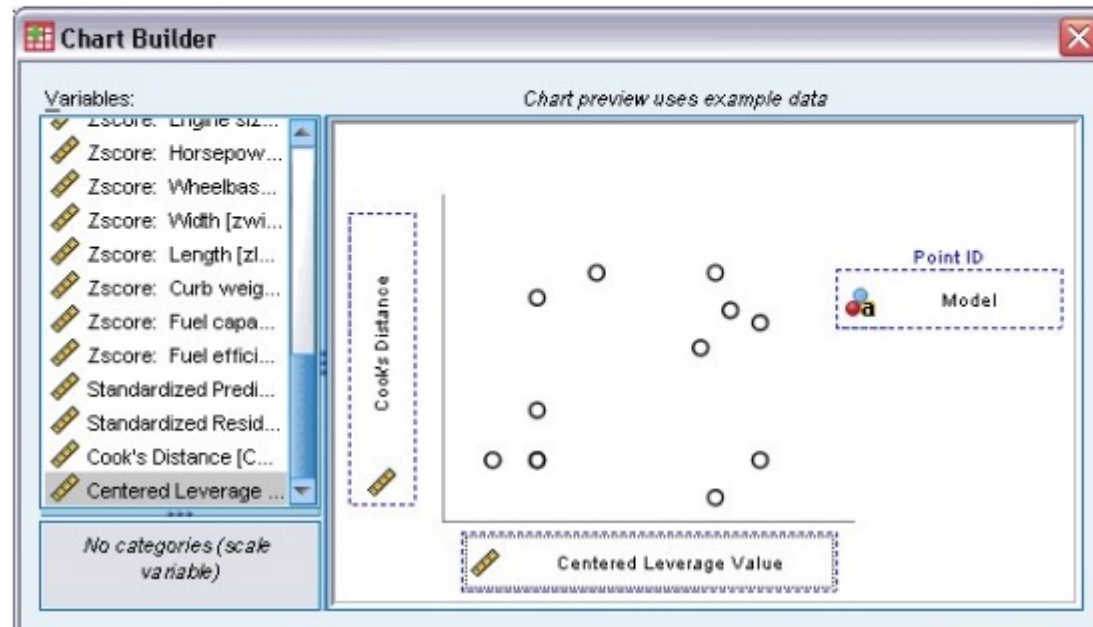
You can also check residuals by each individual factor score. This may identify other cases that have a large influence in the model. Refer to the SPSS case study for details on this step.

3. Linear regression

Step 14: Identifying influential points

1. To check Cook's distance by the centered leverage value, recall the Chart Builder.

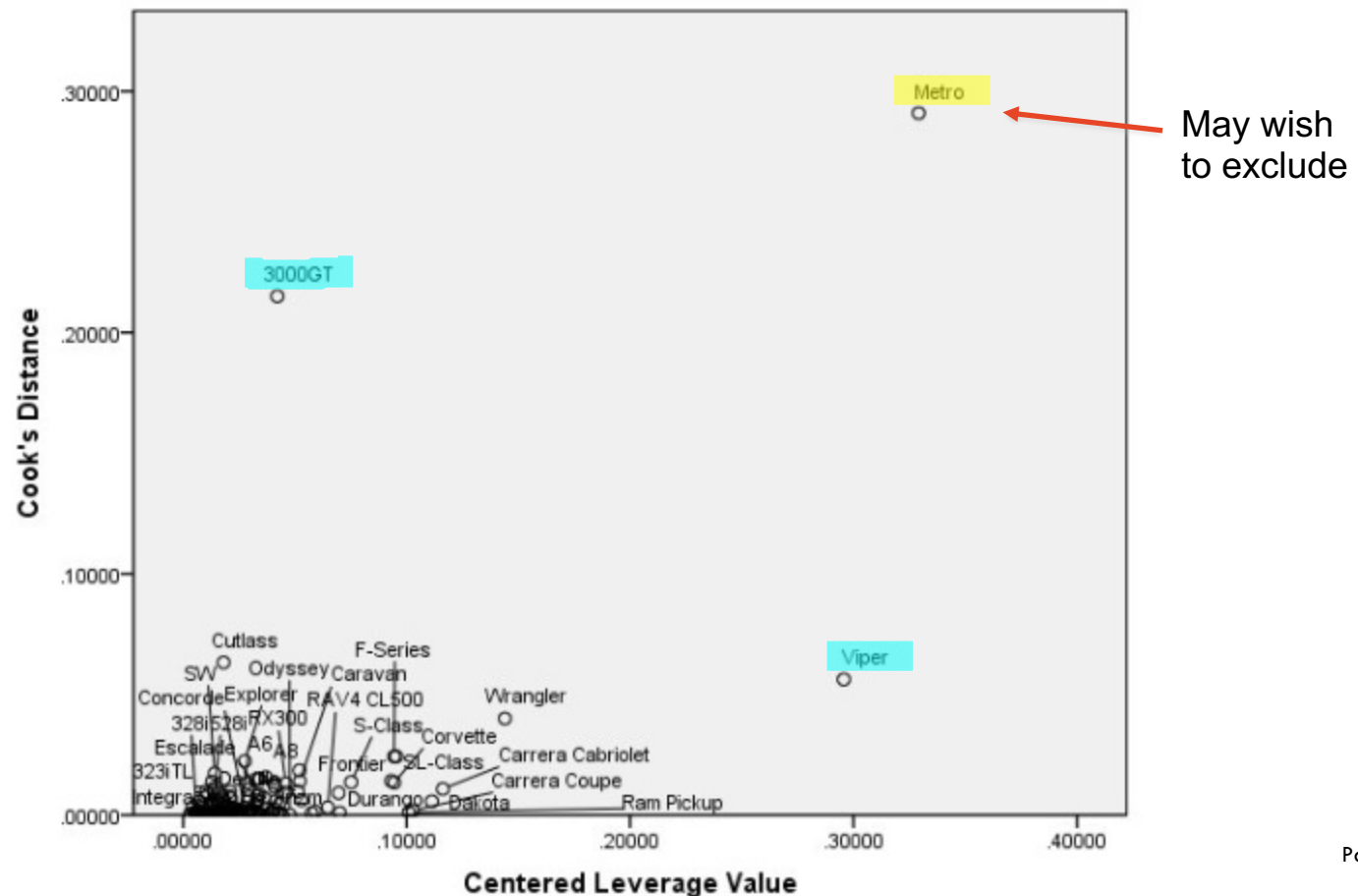
Figure 1. Chart Builder



3. Linear regression

Step 14: Cook's distance by centred leverage value

Figure 2. Cook's distance by centered leverage value



3. Linear regression

Step 15: Summary

Using stepwise methods in Linear Regression, you have selected a "best" model for predicting motor-vehicle sales. With this model, you found two vehicle models that were underperforming in the market, while no vehicle was clearly overperforming.

Diagnostic plots of residuals and influence statistics indicated that your regression model may be adversely affected by the Metro. Removing this case and rerunning the analysis to see the difference in the results would be a good next step.

3. Linear regression

Thank God that's over! Can't we make it easier?

Let's try a simpler approach.

3. Linear regression

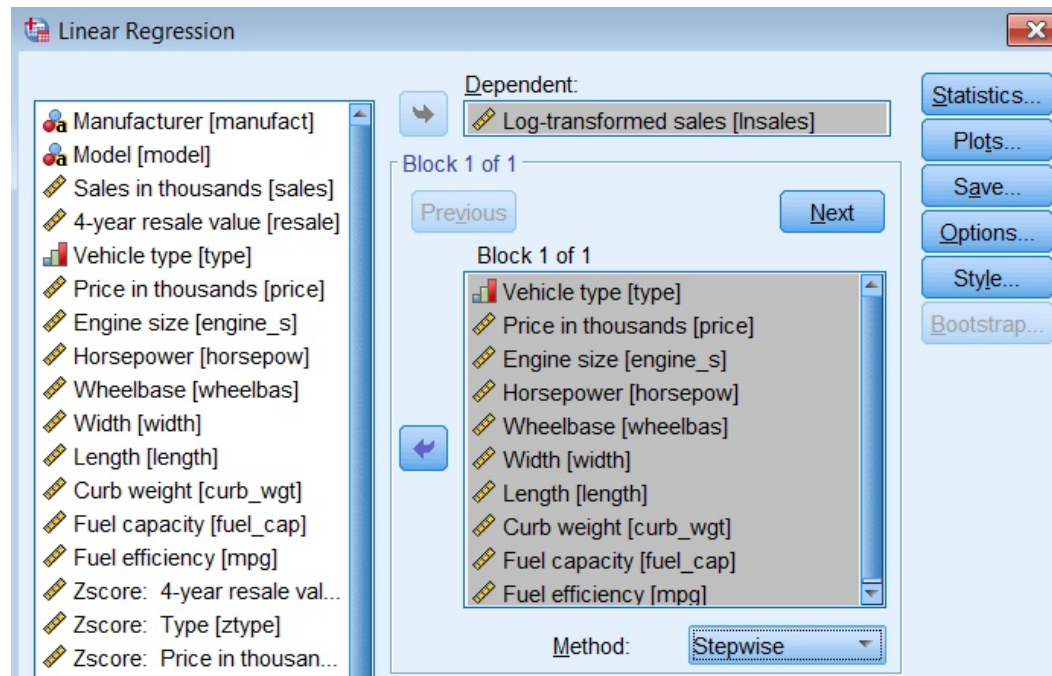
Second method:

Step 1: Analyze>Regression>Linear (reset values)

Insales == Dependent variable

Vehicle type~fuel efficiency == independent

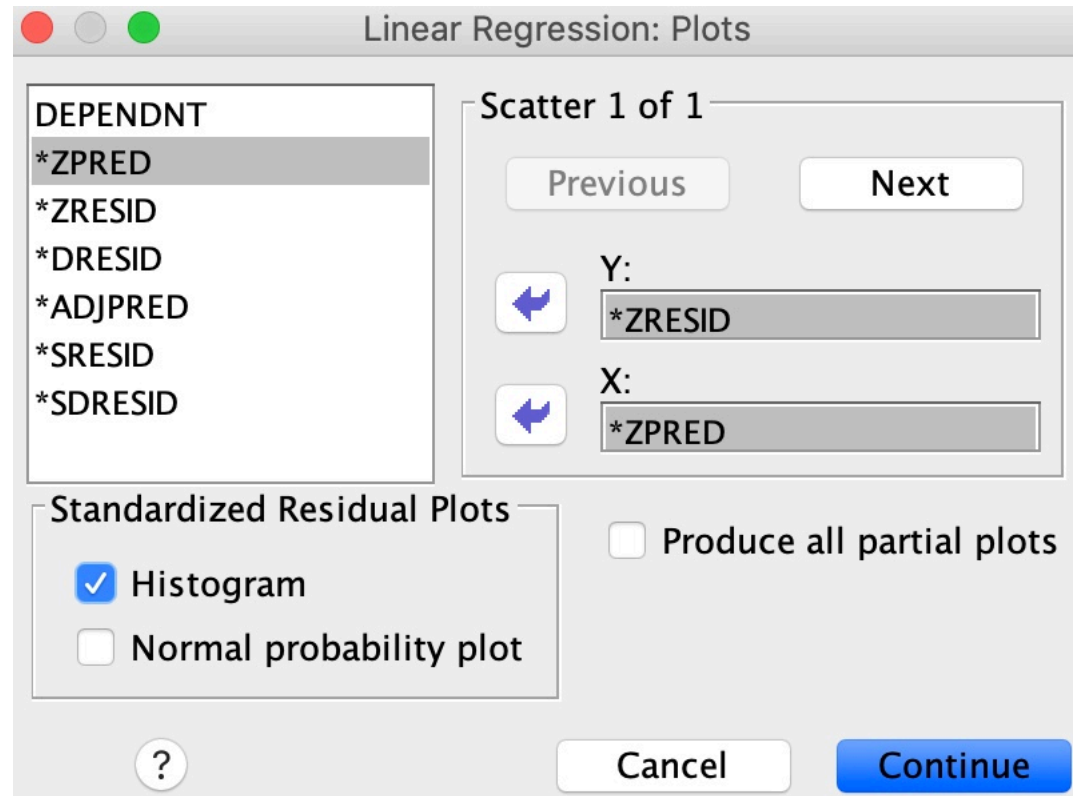
Method > Stepwise



3. Linear regression

Step 2: Plot residuals

Produce the scatter plot to check variance and independence and a histogram to check normality.



The image shows the 'Linear Regression: Plots' dialog box in SPSS. On the left, a list of dependent variables includes *ZPRED, *ZRESID, *DRESID, *ADJPRED, *SRESID, and *SDRESID. *ZPRED is selected. On the right, under 'Scatter 1 of 1', the Y-axis is set to *ZRESID and the X-axis is set to *ZPRED. Below this, the 'Standardized Residual Plots' section has 'Histogram' checked and 'Normal probability plot' unchecked. At the bottom right, there is a checkbox for 'Produce all partial plots' which is unchecked. The dialog has 'Previous' and 'Next' buttons, a '?' help button, and 'Cancel' and 'Continue' buttons at the bottom.

Linear Regression: Plots

DEPENDNT

- *ZPRED
- *ZRESID
- *DRESID
- *ADJPRED
- *SRESID
- *SDRESID

Scatter 1 of 1

Previous Next

Y: *ZRESID

X: *ZPRED

Standardized Residual Plots

- ☒ Histogram
- ☐ Normal probability plot

☐ Produce all partial plots

? Cancel Continue

3. Linear regression

Step 2: Check output

Variables included are:

- Wheelbase
- Price

Factor model $R^2_{adj} = 0.453$

Simple model $R^2_{adj} = 0.422$

We have achieved a model with almost as much explanatory power as the complicated first model

Model	Variables Entered	Variables Removed	Method
1	Price in thousands	.	Stepwise (Criteria: Probability-of-F-to-enter <= .050, Probability-of-F-to-remove >= .100).
2	Wheelbase	.	Stepwise (Criteria: Probability-of-F-to-enter <= .050, Probability-of-F-to-remove >= .100).

a. Dependent Variable: Log-transformed sales

Model Summary^c

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.552 ^a	.304	.300	1.11553
2	.655 ^b	.430	.422	1.01357

a. Predictors: (Constant), Price in thousands

b. Predictors: (Constant), Price in thousands, Wheelbase

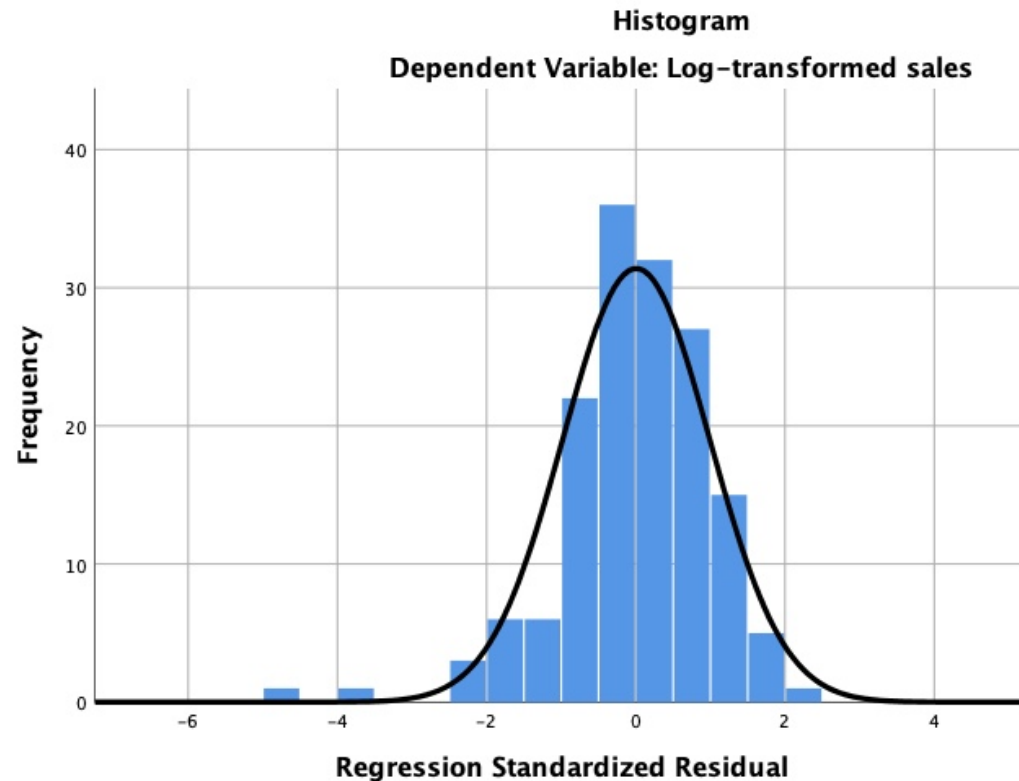
c. Dependent Variable: Log-transformed sales

Was
0.98558

3. Linear regression

Better just check the residuals as well before we get too excited.

Histogram shows normality is OK, noting a couple of large negative residuals.

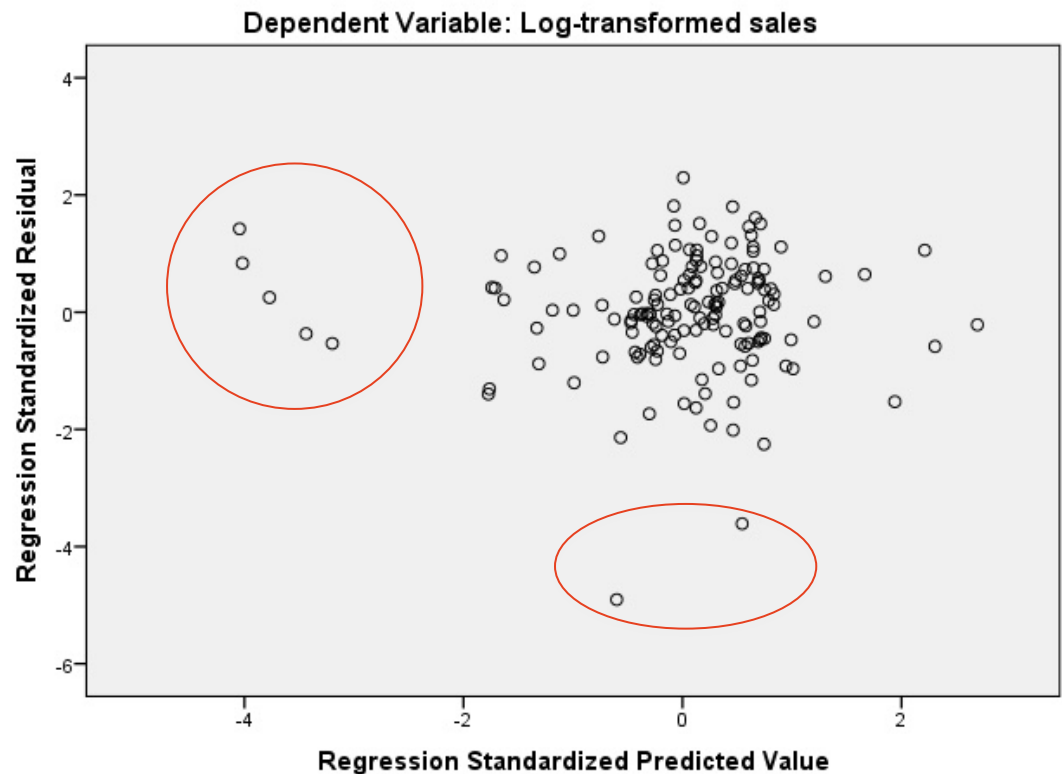


3. Linear regression

Better just check the residuals as well before we get too excited.

It looks OK now. The log transformation on sales has worked.

There are some large negative residuals and perhaps influential points so it could be worth checking Cooks distance as we did before.



3. Linear regression

What to report?

Main thing is to report the regression coefficients with their standard errors

Coefficients ^a						
		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
2	(Constant)	-1.822	1.151		-1.583	.116
	Price in thousands	-.055	.006	-.590	-9.487	.000
	Wheelbase	.061	.011	.356	5.718	.000

a. Dependent Variable: Log-transformed sales

3. Linear regression

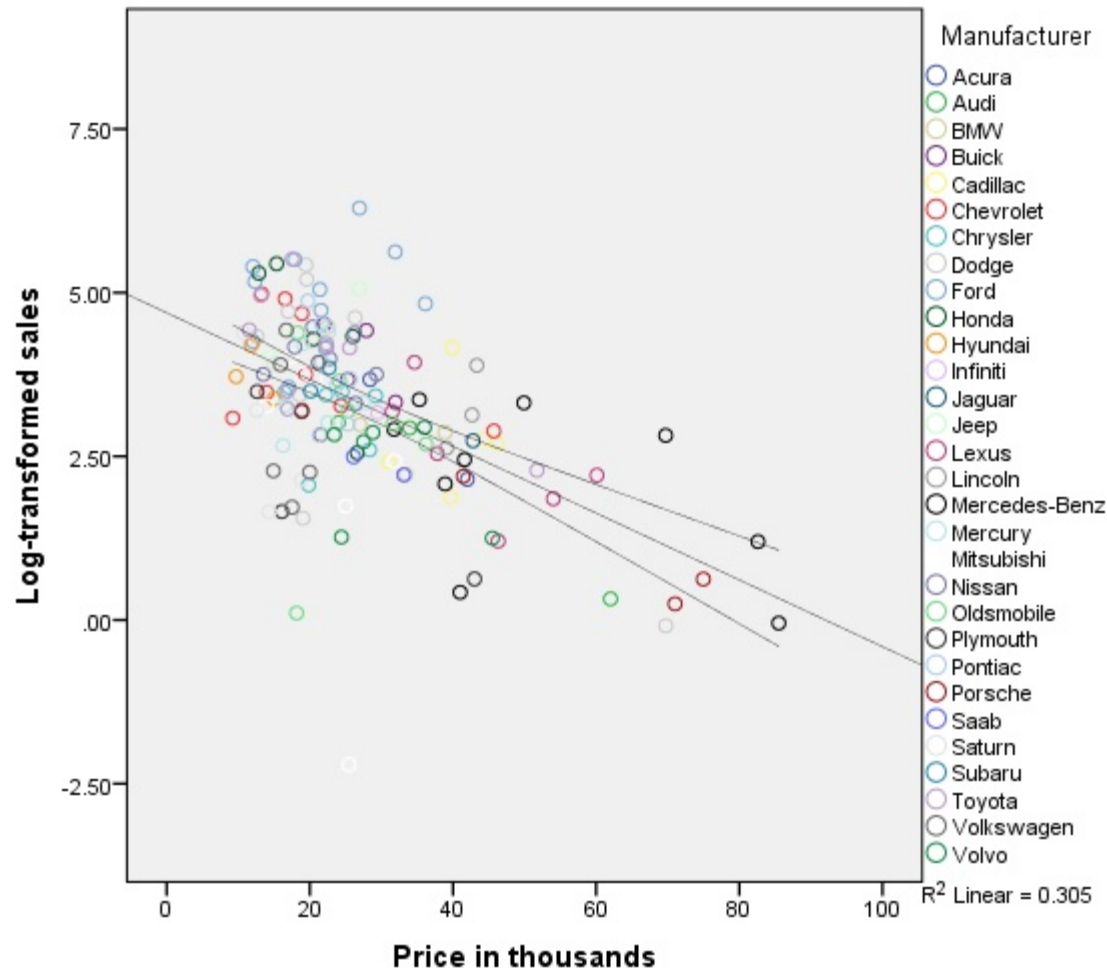
What to report?

Visual depiction of the data is also great.

Steps:
Graphs>Chart Builder
Scatter/Dot>Grouped
Scatter
Y-axis= $\ln(\text{sales})$
X-axis=price
Set Color=Manufacturer
“OK”

Double-click to edit
Chart Editor Steps:
X-axis>Number Format
Decimal Places=0, apply, cl
Add fit line at total
Fit Method=Linear
Confidence Int=Mean
Attach label to line=no
“Apply”
Lines/Weight=0.5
“Apply” Close, close

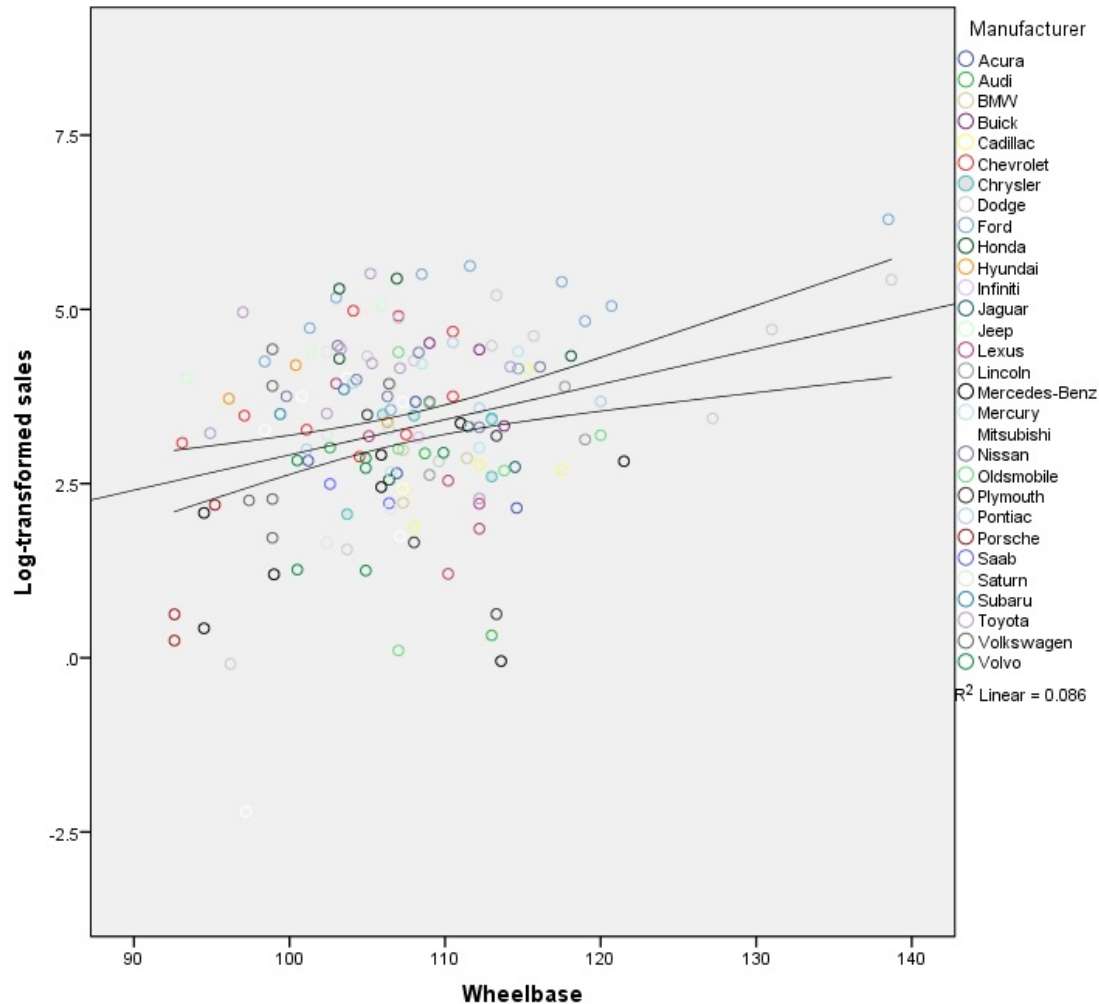
The University of Sydney



3. Linear regression

What to report?

Second independent variable – “wheelbase”



3. Linear regression

Second regression model is:

$$\ln \text{ sales} = \text{intercept} + \text{price} + \text{wheelbase}$$

With coefficients:

$$\ln(\text{sales } \$/1000) = -1.822 - 0.055 * \text{price}(\$/1000) + 0.061 * \text{wheelbase}(\text{inches})$$

Tutorial model with factor analysis is:

$$\ln \text{ sales} = \text{intercept} + \text{Factor_Score_1} + \text{Factor_Score_2} + \text{Factor_Score_3} + \text{Factor_Score_6} + \text{Factor_Score_5}$$

Both models are valid but the factor analysis seems overly complicated. If the factor score variables could be defined in a meaningful way, then perhaps it would be more interesting.

SPSS Syntax

What is syntax?

- SPSS commands can be run either through menus and dialog boxes, or through commands (on a command line).
- Most (but not all) commands are accessible from the menus.
- A command syntax file is a text file containing the SPSS syntax commands.
- You can save your work as a syntax file so that the analysis can be performed in exactly the same way (or with modifications) at a later time.
- You can share this file to allow others to reproduce the analysis.

SPSS Syntax

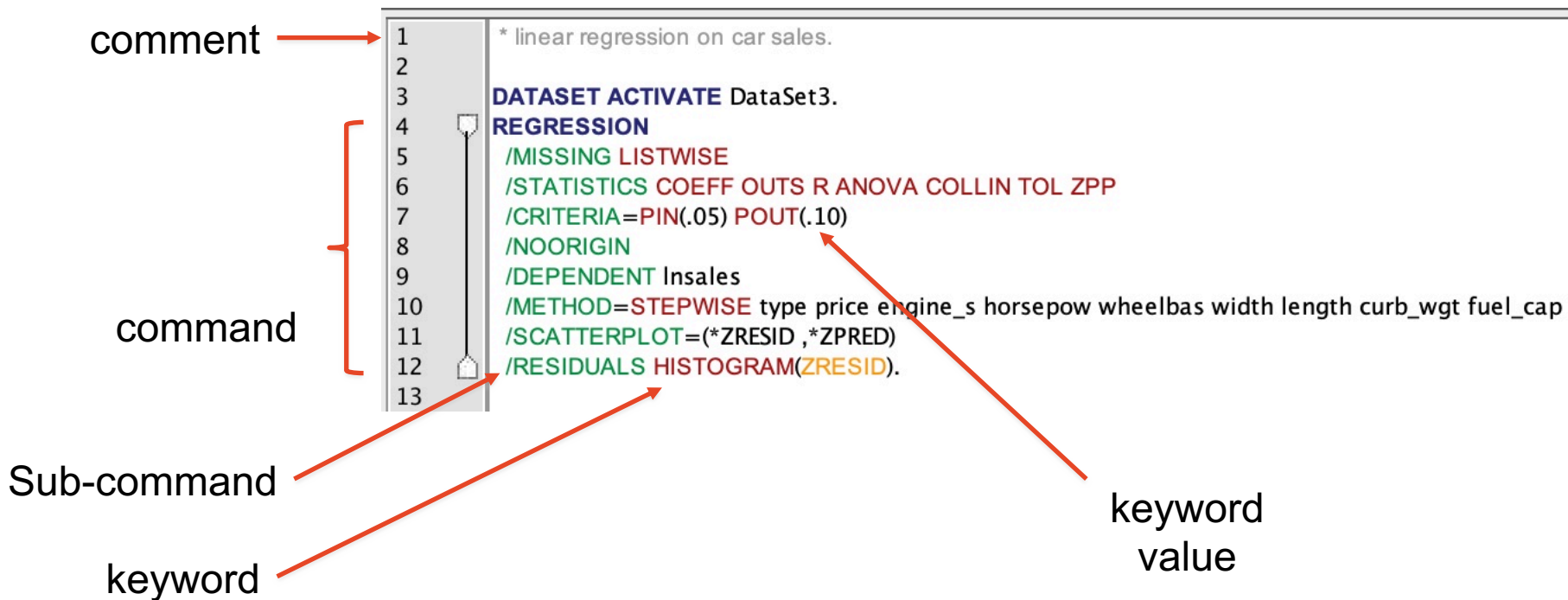
How to Save a syntax file

- Use previous example
- Recall recently used dialogs
- Open linear regression dialog box
- Click "paste"
- A new syntax file will open
- Add a title as a comment by using * at the start of the line and end with a period.
- Eg * Linear regression on car sales.
- Make sure the comment is greyed out and nothing else.

SPSS Syntax

How to Save a syntax file

- File > Save
- Name and save your syntax file



SPSS Syntax

How to Run a syntax file

- File > Open > Syntax
- Run > All
- OR
- Highlight command(s), choose Run > Selection

How are we going?

Last analysis example coming up.
Do we need to pause for breath?



4. Binary Logistic Regression

- One categorical dependent variable with 2 values (often a yes/no or success/fail)
- One or more continuous independent variables

In SPSS we use the procedure:

Regression – Binary Logistic

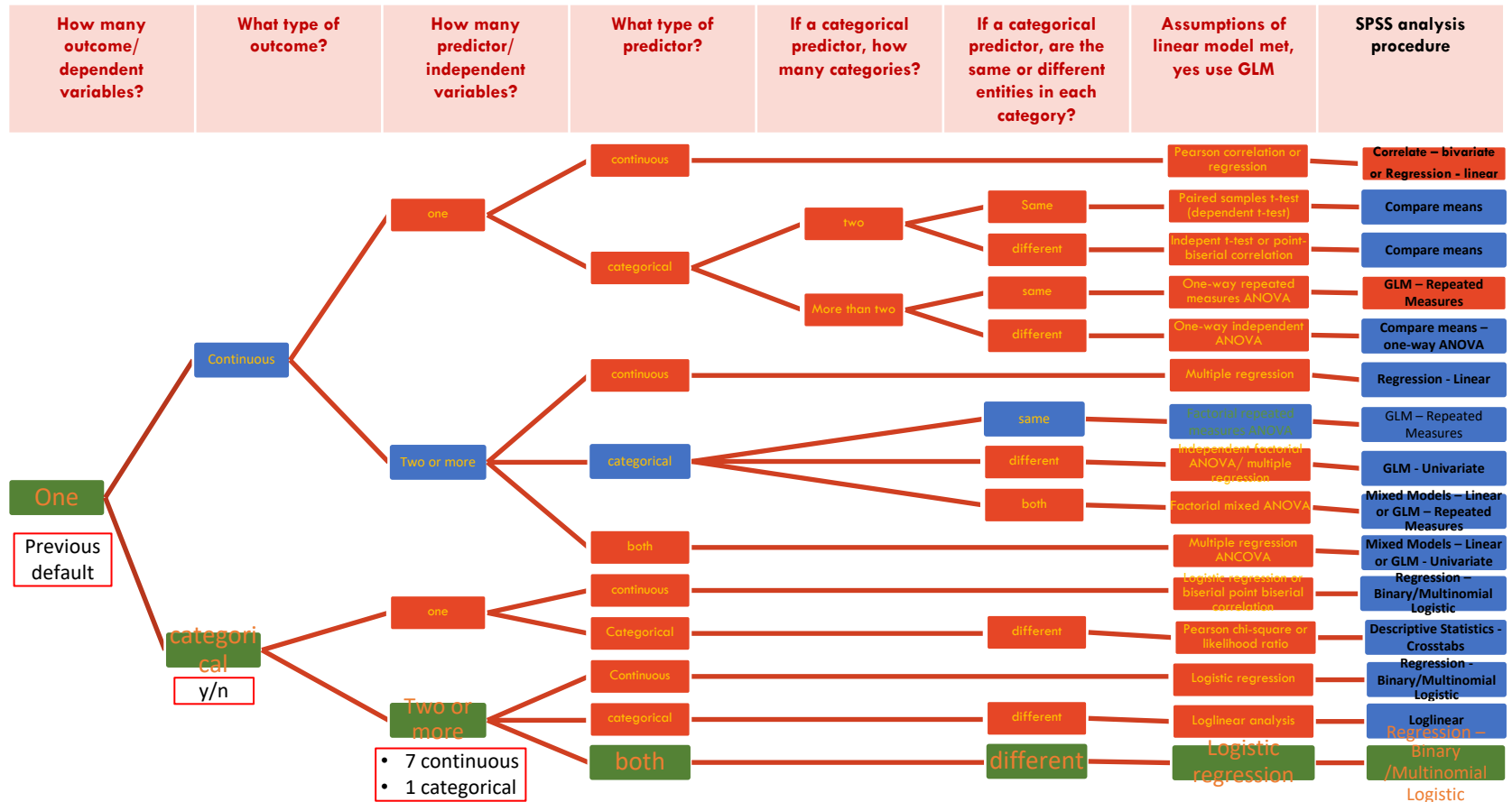
(can also use the Generalized Linear Model procedure)

We will follow the SPSS case study assessing credit risk using data file “**bankloan.sav**”.

The independent variables are both continuous and categorical

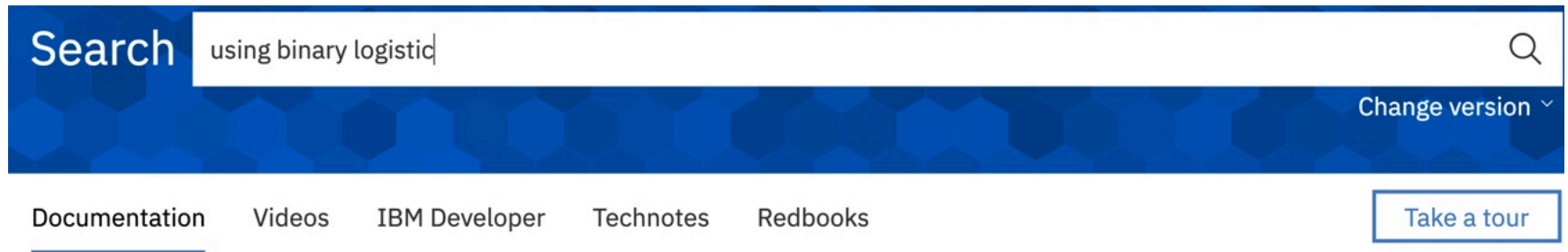
The dependent variable is “previously defaulted” yes/no. We will use this history to predict future default probability.

4. Binary Logistic Regression



4. Binary Logistic Regression

1. Open SPSS
2. Open another file > choose “bankloan.sav”
3. Go to help/topics, then search “using binary logistic”
4. Choose “Using Binary Logistic Regression to Assess Credit Risk”



61 results

+ Binary Logistic Regression

Binary Logistic Regression **Binary logistic** regression is most useful when you want to model the event probability for a categorical response variable with two outcomes. For example: A catalog company wants to increase the proportion of mailings that result in sales. A doctor wants to accurately diagnose

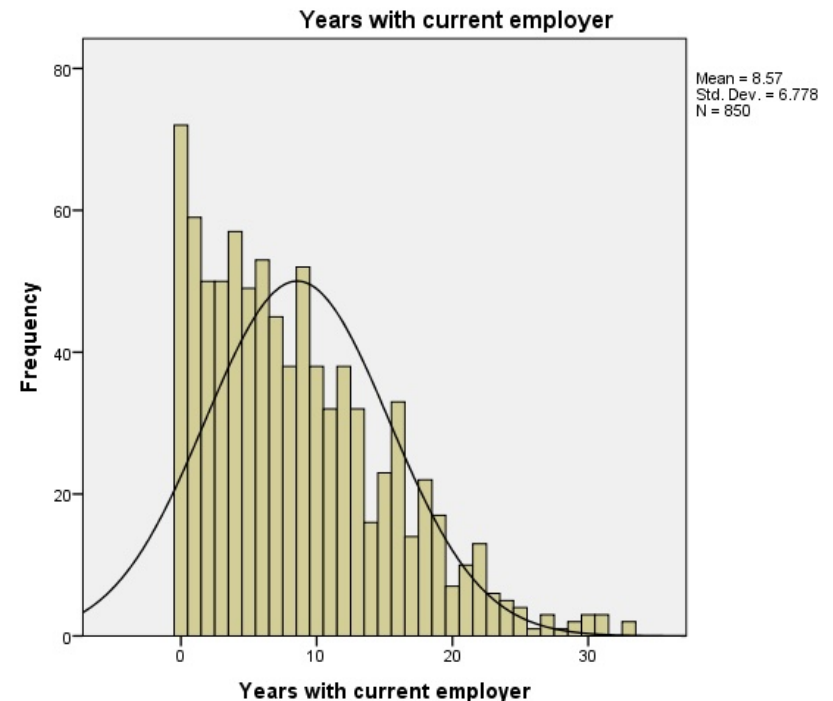
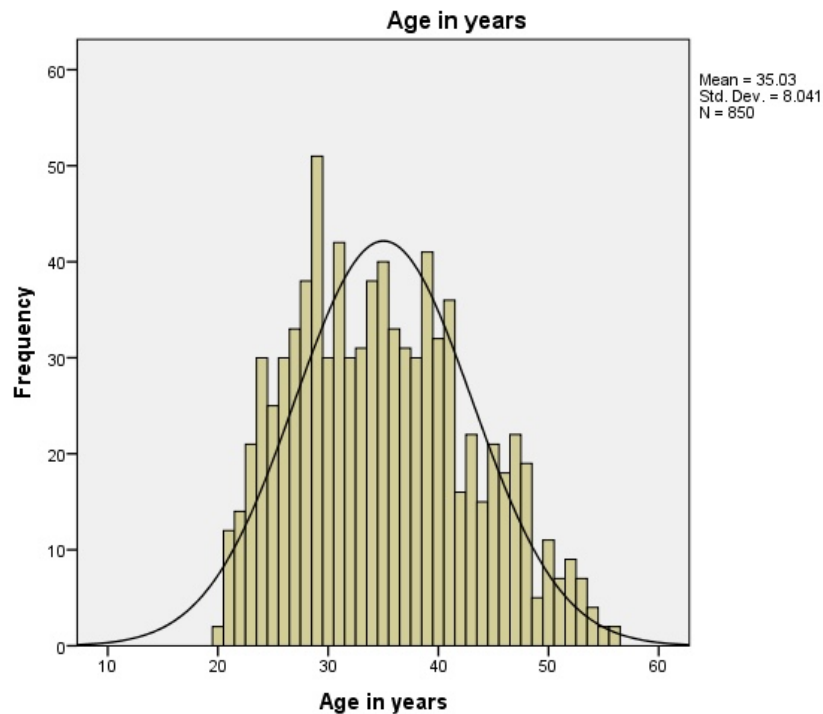
+ Using Binary Logistic Regression to Assess Credit Risk

Using **Binary Logistic Regression** to Assess Credit Risk If you are a loan officer at a bank, then you want to be able to identify characteristics that are indicative of people who are likely to default on loans, and use those characteristics to identify good and bad credit risks. Suppose information on

4. Binary Logistic Regression

Before the tutorial steps

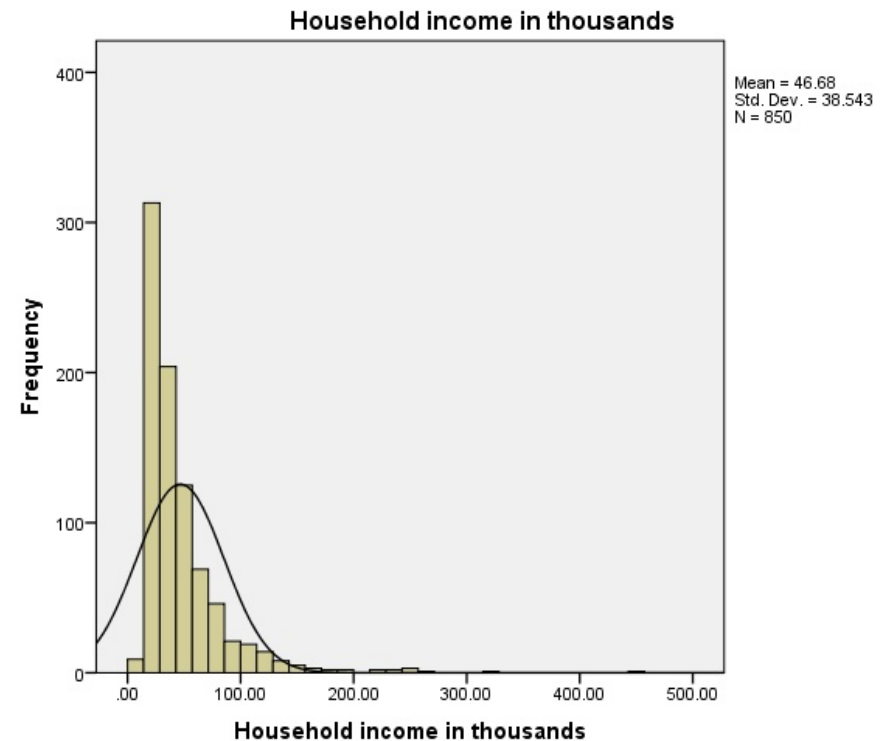
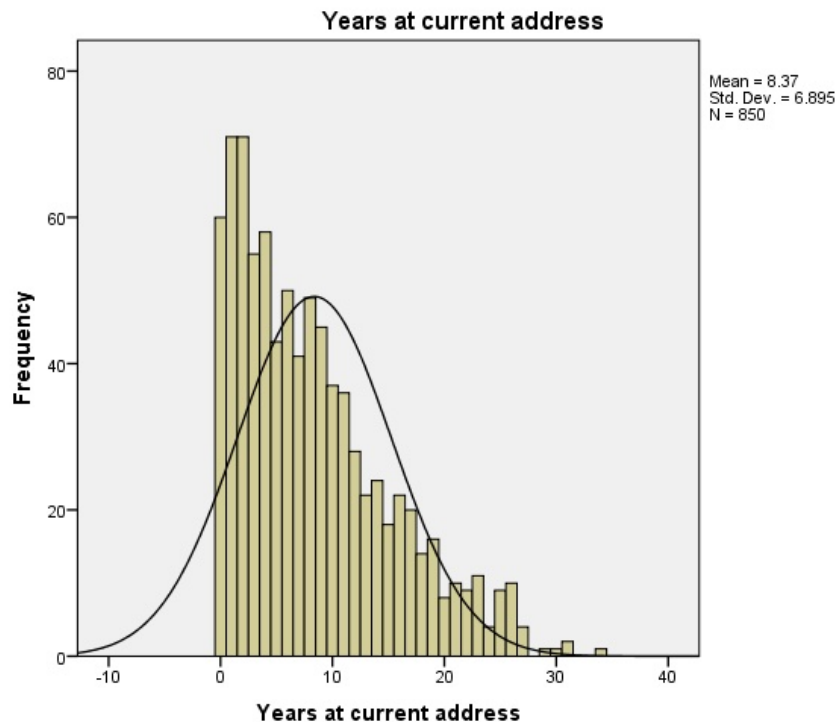
- Visually inspect the data



4. Binary Logistic Regression

Before the tutorial steps

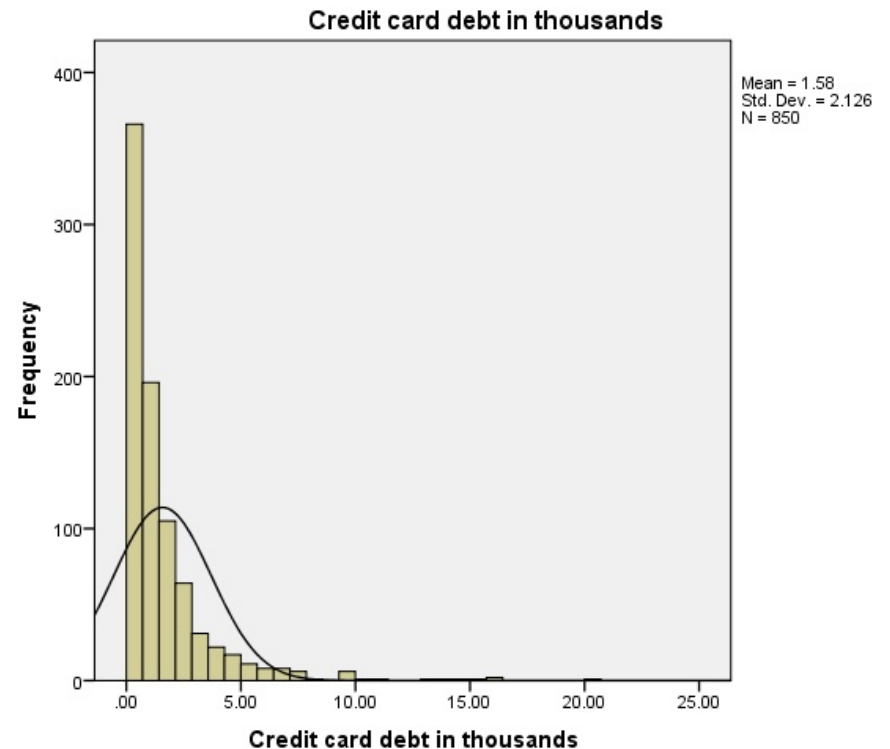
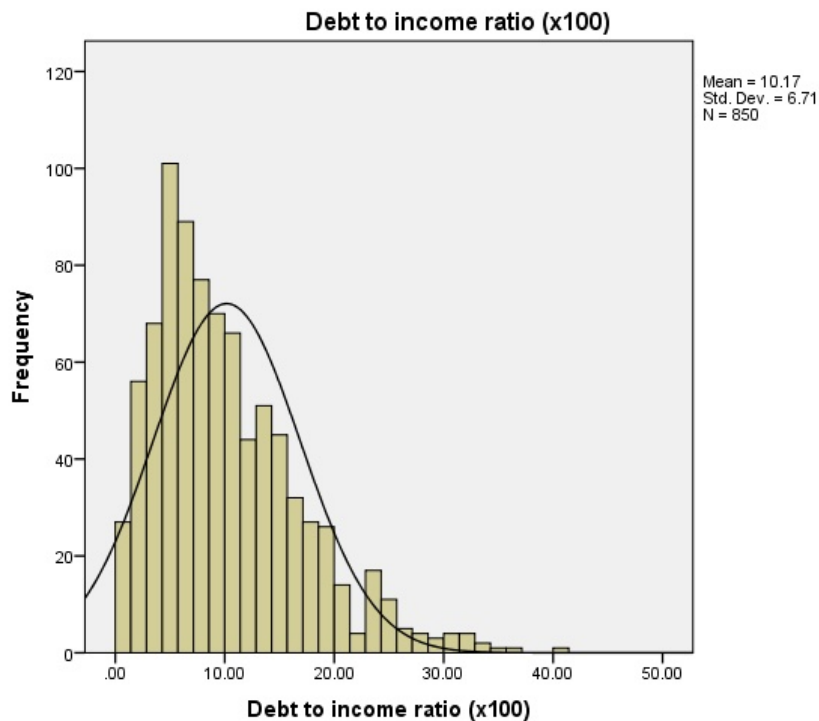
- Visually inspect the data



4. Binary Logistic Regression

Before the tutorial steps

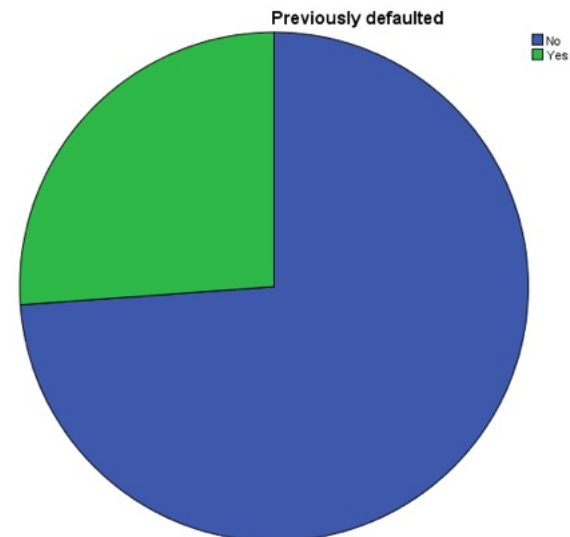
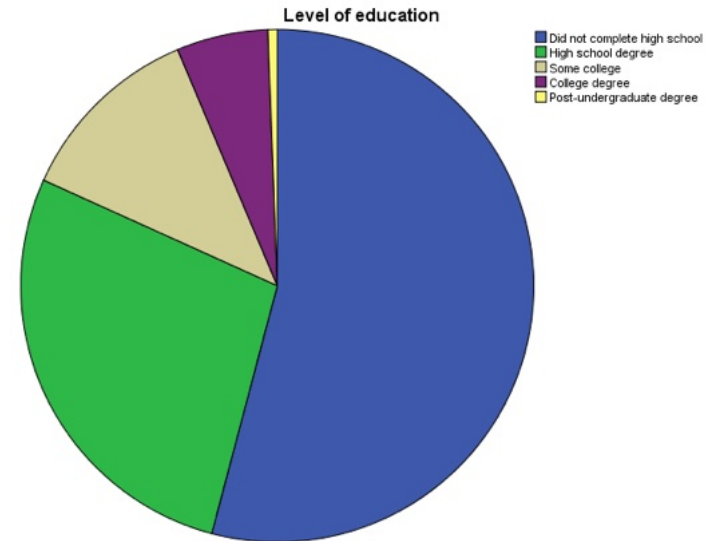
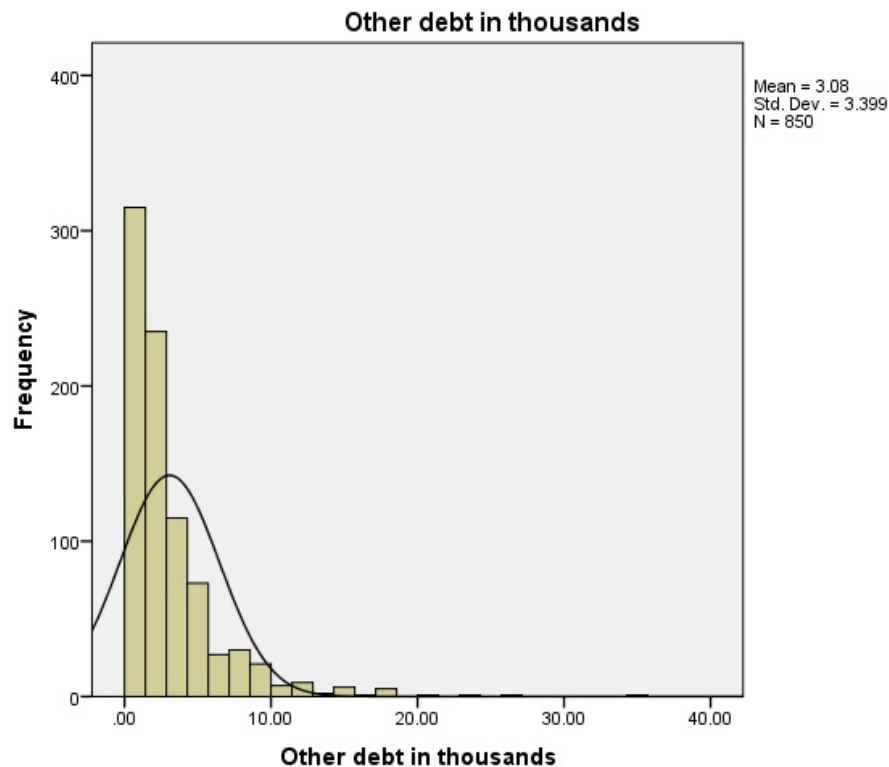
- Visually inspect the data



4. Binary Logistic Regression

Before the tutorial steps

- Visually inspect the data



4. Binary Logistic Regression

1. Keep the tutorial window open for reference.
2. Follow the steps in the tutorial, starting from “Running the Analysis”. (skip the data sampling step)

Step 1: Running the Analysis *(ignore items 5-7 in this step)*

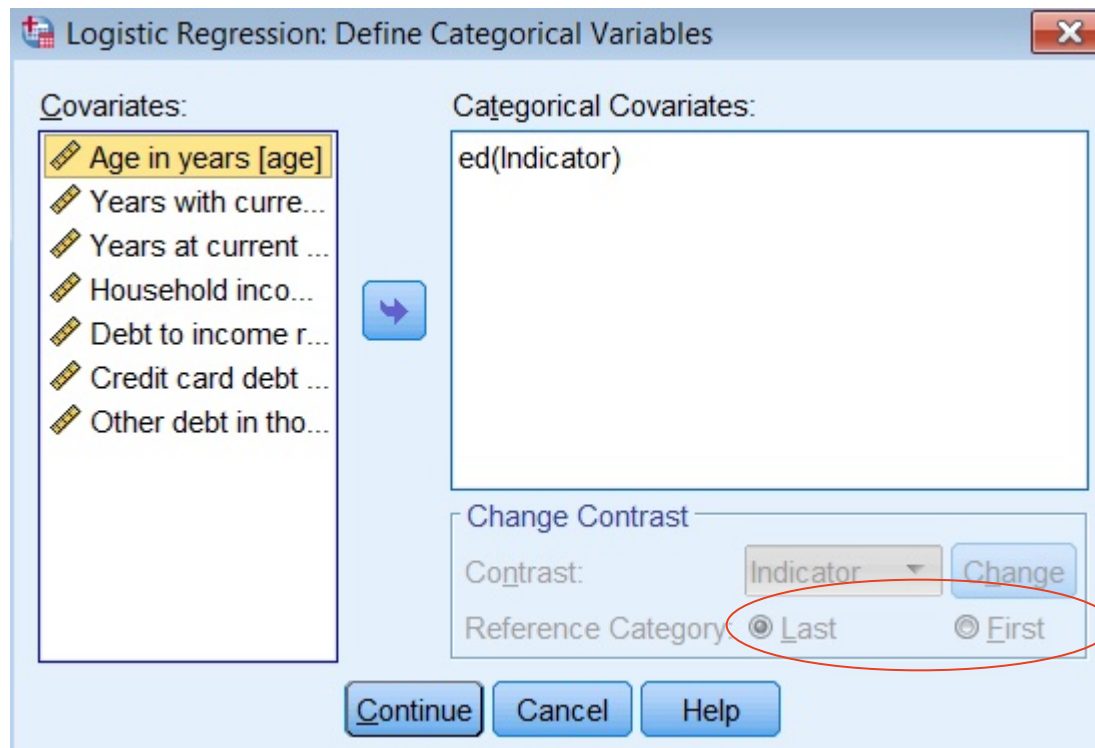
Analyze>
Regression>
Binary Logistic

The screenshot shows the 'Logistic Regression' dialog box in SPSS. On the left is a list of variables: 'Age in years [age]', 'Level of education [ed]', 'Years with current employer [empl...]', 'Years at current address [address]', 'Household income in thousands [i...]', 'Debt to income ratio (x100) [debtinc]', 'Credit card debt in thousands [cre...]', 'Other debt in thousands [othdebt]', 'Predicted default, model 1 [predde...]', 'Predicted default, model 2 [predde...]', and 'Predicted default, model 3 [predde...]'. The 'Dependent' variable is set to 'Previously defaulted [default]'. Below this, 'Block 1 of 1' is shown with a list of 'Covariates': 'age', 'ed', 'employ', 'address', 'income', 'debtinc', 'creddebt', and 'othdebt'. The 'Method' is set to 'Forward: LR'. At the bottom, there is a 'Selection Variable' field and a 'Rule...' button. On the right side of the dialog, there are buttons for 'Categorical...', 'Save...', 'Options...', 'Style...', and 'Bootstrap...'. At the bottom of the dialog are 'OK', 'Paste', 'Reset', 'Cancel', and 'Help' buttons.

4. Binary Logistic Regression

Step 1: Running the Analysis

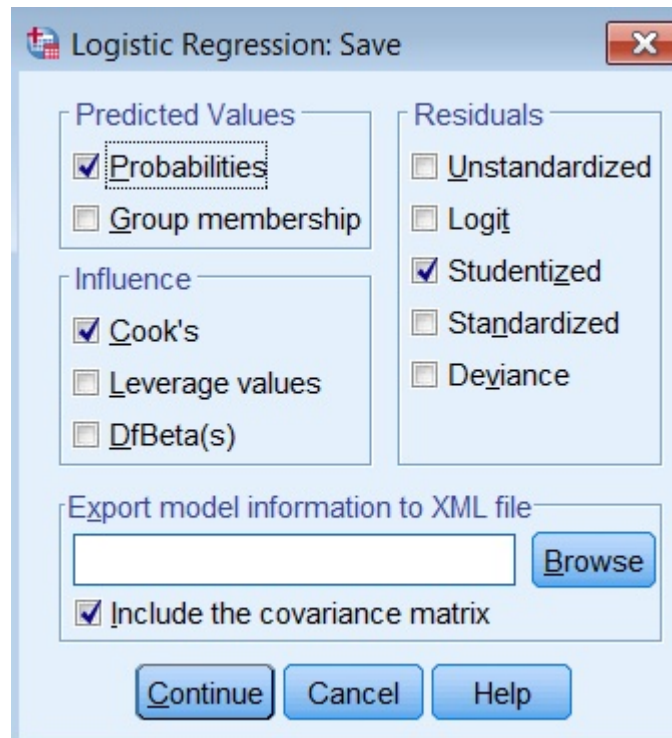
Categorical



4. Binary Logistic Regression

Step 1: Running the Analysis

Save



The image shows the 'Logistic Regression: Save' dialog box in SPSS. It has a title bar with the text 'Logistic Regression: Save' and a close button (X). The dialog is divided into several sections with expandable/collapsible headers. The 'Predicted Values' section is expanded, showing 'Probabilities' checked and 'Group membership' unchecked. The 'Influence' section is also expanded, showing 'Cook's' checked, and 'Leverage values' and 'DfBeta(s)' unchecked. The 'Residuals' section is expanded, showing 'Unstandardized', 'Logit', and 'Standardized' unchecked, while 'Studentized' and 'Deviance' are checked. At the bottom, there is a section 'Export model information to XML file' with an empty text box and a 'Browse' button. Below this, 'Include the covariance matrix' is checked. At the very bottom are three buttons: 'Continue', 'Cancel', and 'Help'.

Logistic Regression: Save

Predicted Values

- ☒ Probabilities
- ☐ Group membership

Influence

- ☒ Cook's
- ☐ Leverage values
- ☐ DfBeta(s)

Residuals

- ☐ Unstandardized
- ☐ Logit
- ☒ Studentized
- ☐ Standardized
- ☐ Deviance

Export model information to XML file

☒ Include the covariance matrix

4. Binary Logistic Regression

Step 1: Running the Analysis

Options

Logistic Regression: Options

Statistics and Plots

- ☒ Classification plots
- ☒ Hosmer-Lemeshow goodness-of-fit
- ☐ Casewise listing of residuals
- ☐ Correlations of estimates
- ☐ Iteration history
- ☐ CI for exp(B): 95 %
- ☒ Outliers outside 2 std. dev.
- ☒ All cases

Display

- ☒ At each step
- ☐ At last step

Probability for Stepwise

Entry: 0.05 Removal: 0.10

Classification cutoff: 0.5

Maximum iterations: 20

☐ Conserve memory for complex analyses or large datasets

☒ Include constant in model

Continue **Cancel** **Help**

4. Binary Logistic Regression

Step 2: Model Diagnostics

Model Diagnostics

After building a model, you need to determine whether it reasonably approximates the behavior of your data.

Tests of Model Fit. The Binary Logistic Regression procedure reports the Hosmer-Lemeshow goodness-of-fit statistic.

Residual Plots. Using variables specified in the Save dialog box, you can construct various diagnostic plots. Two helpful plots are the change in deviance versus predicted probabilities and Cook's distances versus predicted probabilities.

4. Binary Logistic Regression

Step 3: Tests of model fit

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	3.160	8	.924
2	4.158	8	.843
3	6.418	8	.600
4	8.556	8	.381

← P>0.05, model fit is OK

Contingency Table for Hosmer and Lemeshow Test

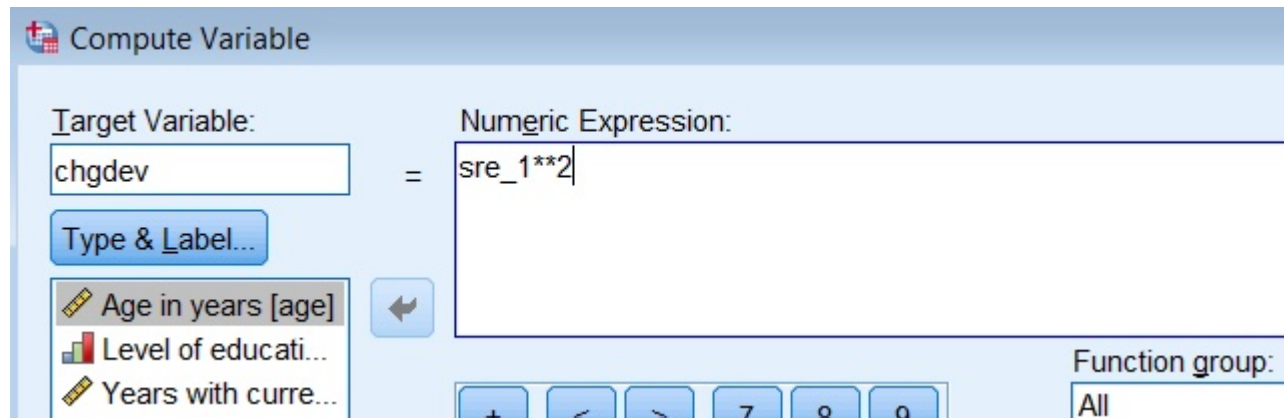
		Previously defaulted = No		Previously defaulted = Yes		Total
		Observed	Expected	Observed	Expected	
Step 4	1	70	69.669	0	.331	70
	2	69	68.554	1	1.446	70
	3	64	66.539	6	3.461	70
	4	64	63.521	6	6.479	70
	5	65	59.692	5	10.308	70
	6	50	55.141	20	14.859	70
	7	48	49.016	22	20.984	70
	8	43	41.000	27	29.000	70
	9	32	30.470	38	39.530	70
	10	12	13.397	58	56.603	70

4. Binary Logistic Regression

Step 4: Change in deviance vs predicted probabilities

Transform>Compute Variable...

(calculate the square of the standardized residual variable: sre_1)



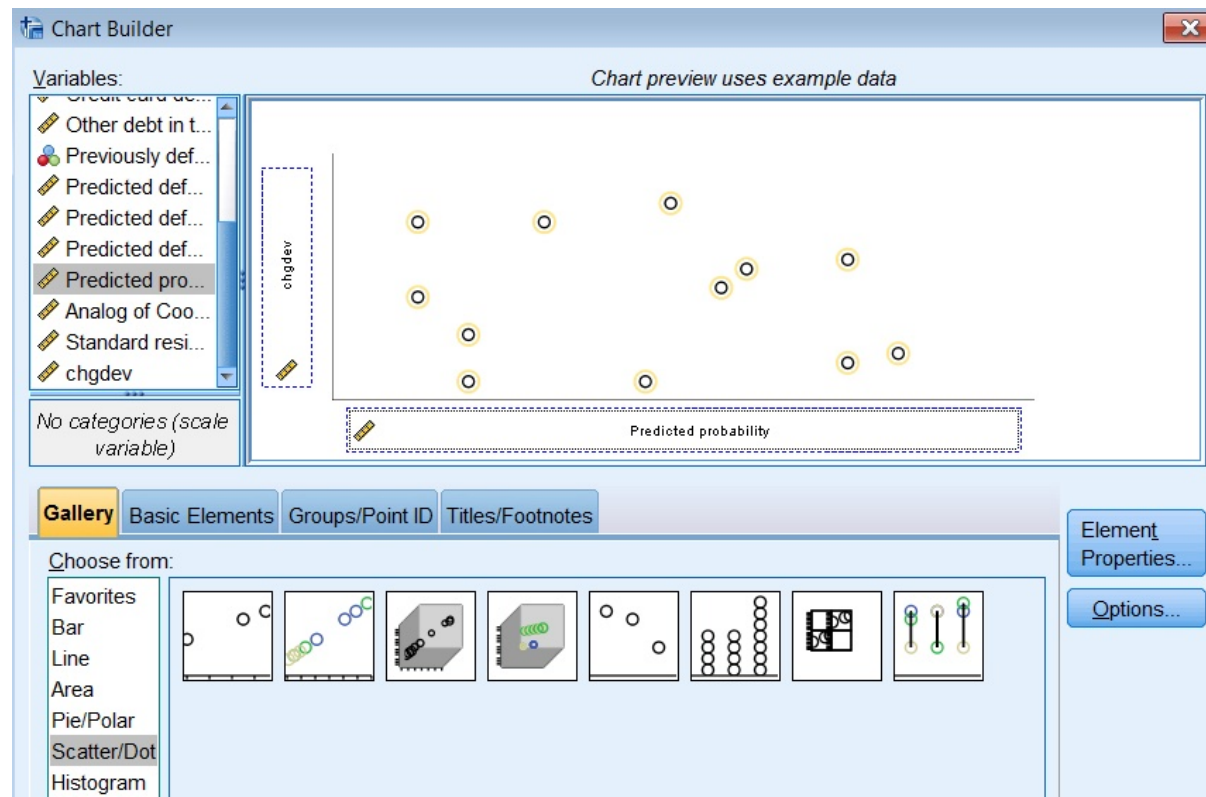
4. Binary Logistic Regression

Step 4: Change in deviance vs predicted probabilities

Plot “chgdev” vs “predicted probabilities”

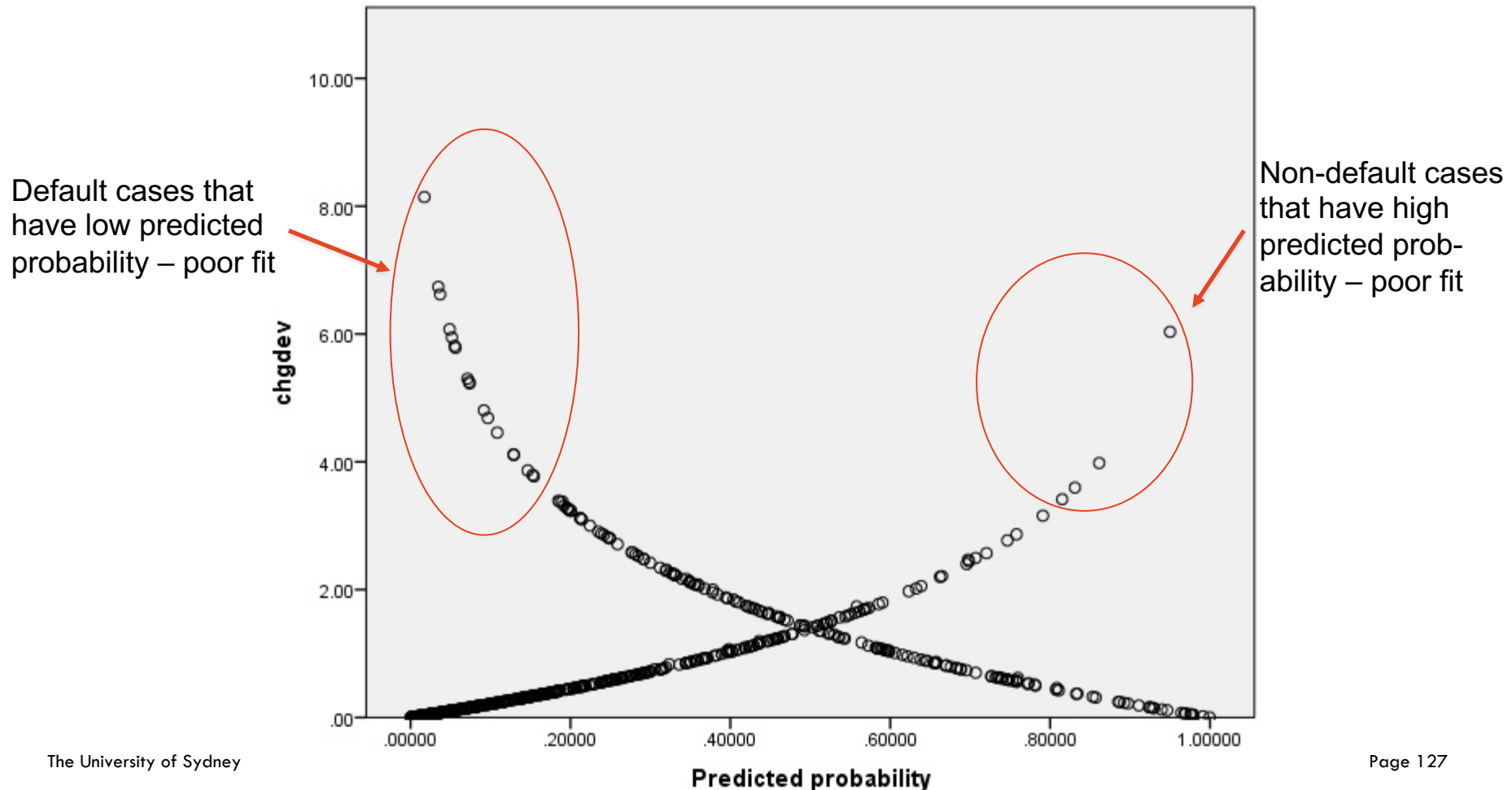
Graphs>Chart Builder

Simple Scatter



4. Binary Logistic Regression

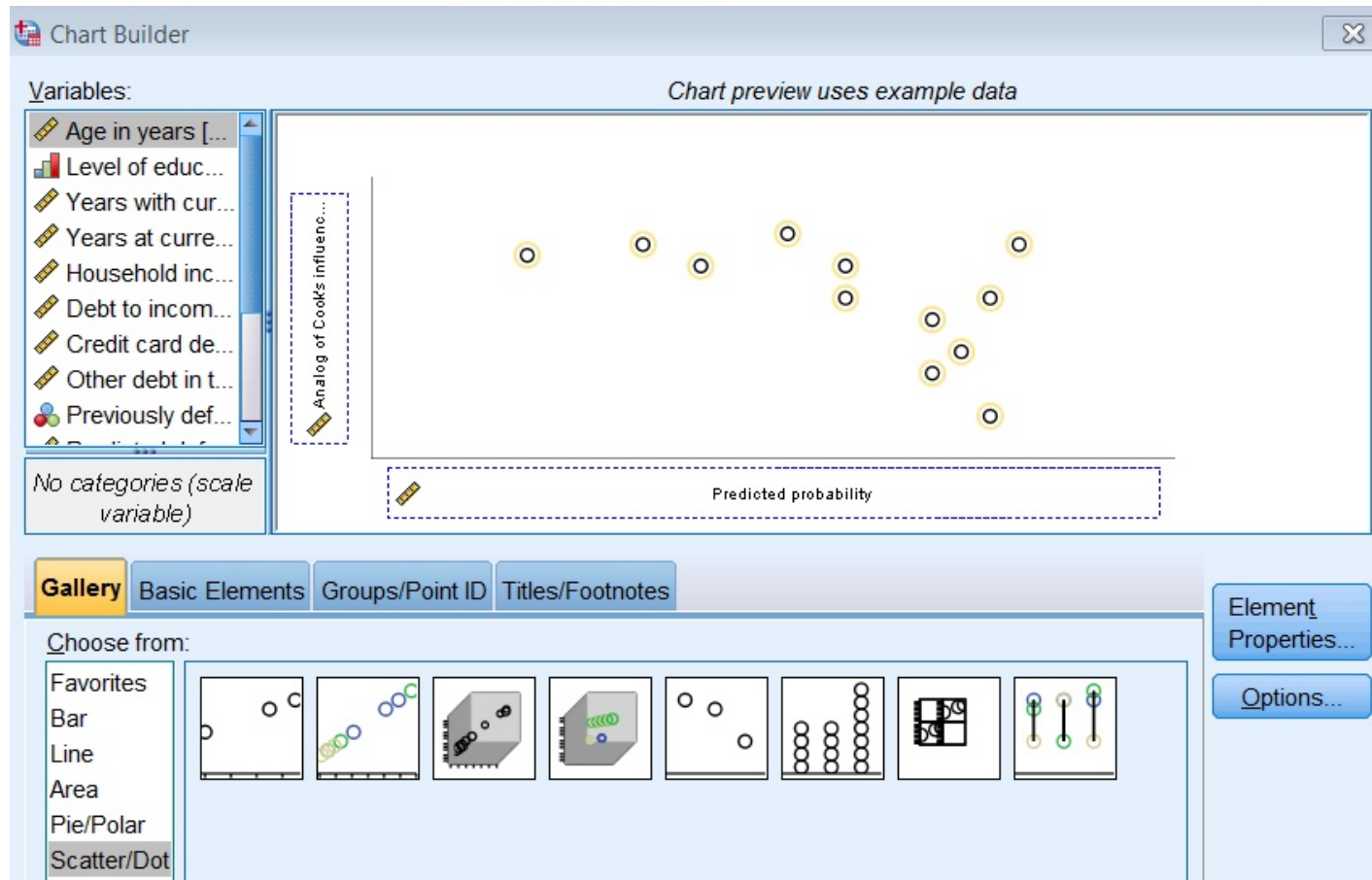
Step 4: Change in deviance vs predicted probabilities



4. Binary Logistic Regression

Step 5: Cook's Distance

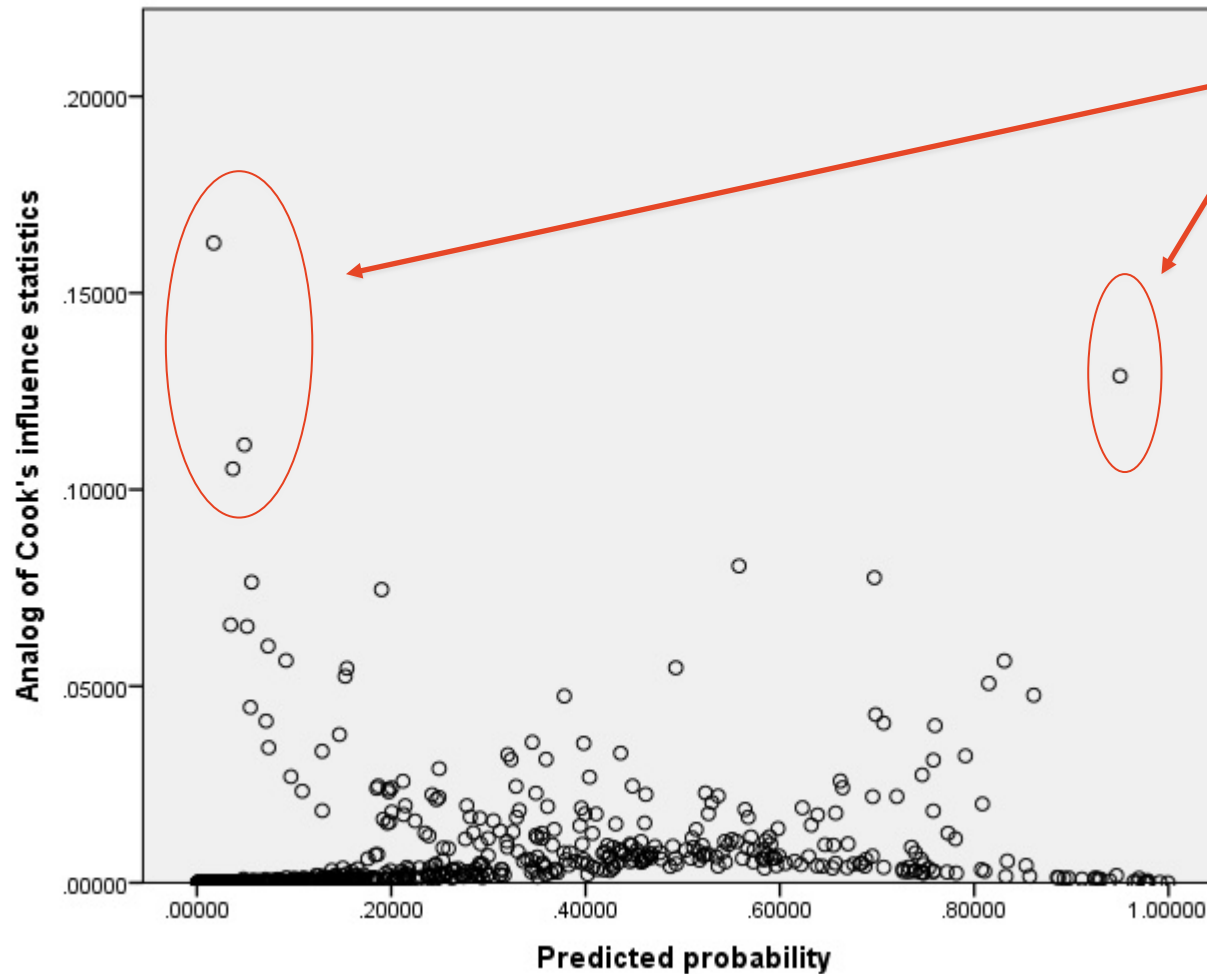
Figure 1. Chart Builder



1. Recall the Chart Builder.
2. Select *Analog of Cook's influence statistics* as the variable to plot on the Y axis.
3. Select *Predicted probability* as the variable to plot on the X axis.
4. Click **OK**.

4. Binary Logistic Regression

Step 5: Cook's Distance



Influential values – worth reviewing these cases to understand them better.

4. Binary Logistic Regression

Step 6: Choosing the right model

There are usually several models that pass the diagnostic checks, so you need tools to choose between them.

- **Automated Variable Selection.** When constructing a model, you generally want to only include predictors that contribute significantly to the model. The Binary Logistic Regression procedure offers several methods for stepwise selection of the "best" predictors to include in the model.
- **Pseudo R-Squared Statistics.** The r-squared statistic, which measures the variability in the dependent variable that is explained by a linear regression model, cannot be computed for logistic regression models. The pseudo r-squared statistics are designed to have similar properties to the true r-squared statistic.
- **Classification and Validation.** Crosstabulating observed response categories with predicted categories helps you to determine how well the model identifies defaulters.

4. Binary Logistic Regression

Step 7: Variable Selection (Block 0)

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	Age in years	13.265	1	.000
		Level of education	11.492	4	.022
		Level of education(1)	9.898	1	.002
		Level of education(2)	1.910	1	.167
		Level of education(3)	3.579	1	.059
		Level of education(4)	2.382	1	.123
		Years with current employer	56.054	1	.000
		Years at current address	18.931	1	.000
		Household income in thousands	3.526	1	.060
		Debt to income ratio (x100)	106.238	1	.000
		Credit card debt in thousands	41.928	1	.000
		Other debt in thousands	14.863	1	.000
	Overall Statistics		203.867	11	.000

Step 0 is before adding any variables to the model

4. Binary Logistic Regression

Step 7: Variable Selection

Model if Term Removed					
Variable		Model Log Likelihood	Change in -2 Log Likelihood	df	Sig. of the Change
Step 1	Debt to income ratio (x100)	-402.182	102.935	1	.000
Step 2	Years with current employer	-350.714	70.346	1	.000
	Debt to income ratio (x100)	-369.708	108.332	1	.000
Step 3	Years with current employer	-349.577	123.518	1	.000
	Debt to income ratio (x100)	-299.710	23.783	1	.000
	Credit card debt in thousands	-315.541	55.446	1	.000
Step 4	Years with current employer	-333.611	110.490	1	.000
	Years at current address	-287.818	18.905	1	.000
	Debt to income ratio (x100)	-290.006	23.281	1	.000
	Credit card debt in thousands	-311.176	65.621	1	.000

The variables chosen by the forward stepwise method should all have significant changes in -2 log-likelihood. The change in -2 log-likelihood is generally more reliable than the Wald statistic. If the two disagree as to whether a predictor is useful to the model, trust the change in -2 log-likelihood. See the [regression coefficients table](#) for the Wald statistic.

4. Binary Logistic Regression

Step 8: Pseudo R-squared statistics

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	701.429 ^a	.137	.200
2	631.083 ^b	.219	.321
3	575.636 ^b	.279	.408
4	556.732 ^c	.298	.436

- a. Estimation terminated at iteration number 4 because parameter estimates changed by less than .001.
- b. Estimation terminated at iteration number 5 because parameter estimates changed by less than .001.
- c. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

What constitutes a “good” R^2 value varies between different areas of application. While these statistics can be suggestive on their own, they are most useful when comparing competing models for the same data. The model with the largest R^2 statistic is “best” according to this measure.

4. Binary Logistic Regression

Step 9: Classification

Figure 1. Classification table, block 1

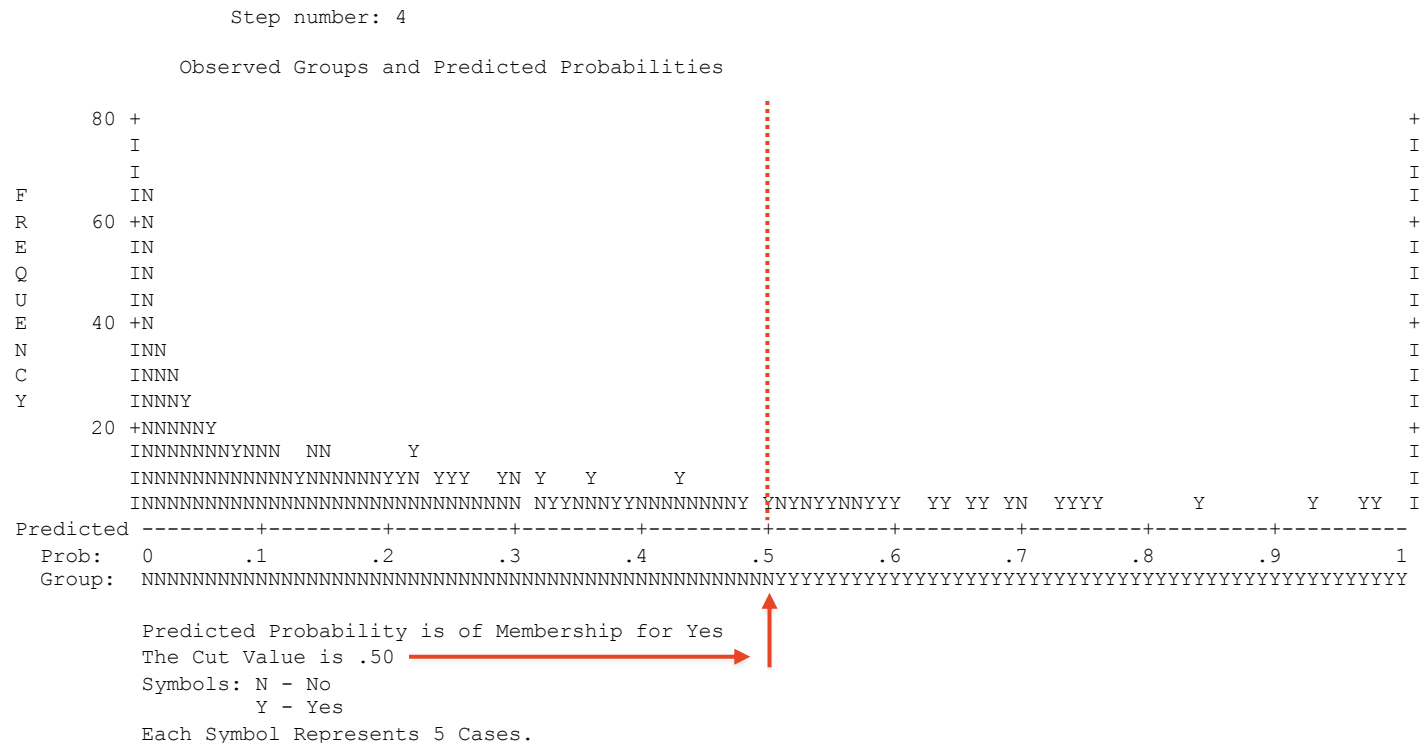
Classification Table ^a					
Observed			Predicted		Percentage Correct
			Previously defaulted No	Yes	
Step 1	Previously defaulted	No	490	27	94.8
		Yes	137	46	25.1
	Overall Percentage				76.6
Step 2	Previously defaulted	No	481	36	93.0
		Yes	110	73	39.9
	Overall Percentage				79.1
Step 3	Previously defaulted	No	477	40	92.3
		Yes	99	84	45.9
	Overall Percentage				80.1
Step 4	Previously defaulted	No	478	39	92.5
		Yes	91	92	50.3
	Overall Percentage				81.4

a. The cut value is .500

Model is much more accurate predicting non-defaults compared to defaults

4. Binary Logistic Regression

Step 9: Classification (not in the tutorial steps)



4. Binary Logistic Regression

Step 10: Logistic regression coefficients

Parameter estimates
(Variables in the
equation)



		Variables in the Equation					
		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	Debt to income ratio (x100)	.132	.014	85.377	1	.000	1.141
	Constant	-2.531	.195	168.524	1	.000	.080
Step 2 ^b	Years with current employer	-.141	.019	53.755	1	.000	.868
	Debt to income ratio (x100)	.145	.016	87.231	1	.000	1.156
	Constant	-1.693	.219	59.771	1	.000	.184
Step 3 ^c	Years with current employer	-.244	.027	80.262	1	.000	.783
	Debt to income ratio (x100)	.088	.018	23.328	1	.000	1.092
	Credit card debt in thousands	.503	.081	38.652	1	.000	1.653
	Constant	-1.227	.231	28.144	1	.000	.293
Step 4 ^d	Years with current employer	-.243	.028	74.761	1	.000	.785
	Years at current address	-.081	.020	17.183	1	.000	.922
	Debt to income ratio (x100)	.088	.019	22.659	1	.000	1.092
	Credit card debt in thousands	.573	.087	43.109	1	.000	1.774
	Constant	-.791	.252	9.890	1	.002	.453

a. Variable(s) entered on step 1: Debt to income ratio (x100).

b. Variable(s) entered on step 2: Years with current employer.

c. Variable(s) entered on step 3: Credit card debt in thousands.

d. Variable(s) entered on step 4: Years at current address.

4. Binary Logistic Regression

Where did $\exp(B)$ come from?

We are not modelling the binary outcome directly.

We are modelling the logarithm of the odds of the event.

$$\text{logit } p = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \dots$$

Where: p = probability of event happening
 $1-p$ = probability of event not happening

4. Binary Logistic Regression

Step 10: Logistic regression coefficients

The meaning of a logistic regression coefficient is not as straightforward as that of a linear regression coefficient. While B is convenient for testing the usefulness of predictors, $\text{Exp}(B)$ is easier to interpret. $\text{Exp}(B)$ represents the ratio-change in the odds of the event of interest for a one-unit change in the predictor. For example, $\text{Exp}(B)$ for employ is equal to 0.781, which means that the odds of default for a person who has been employed at their current job for two years are 0.781 times the odds of default for a person who has been employed at their current job for 1 year, all other things being equal.

4. Binary Logistic Regression

Step 11: Summary

Using the Logistic Regression Procedure, you have constructed a model for predicting the probability a given customer will default on their loan.

A critical issue for loan officers is the cost of Type I and Type II errors. That is, what is the cost of classifying a defaulter as a non-defaulter (Type I)? What is the cost of classifying a non-defaulter as a defaulter (Type II)? If bad debt is the primary concern, then you want to lower your Type I error and maximize your *sensitivity*. If growing your customer base is the priority, then you want to lower your Type II error and maximize your *specificity*. Usually both are major concerns, so you have to choose a decision rule for classifying customers that gives the best mix of sensitivity and specificity.

4. Binary Logistic Regression

Step 12: Related Procedures

ROC Curve

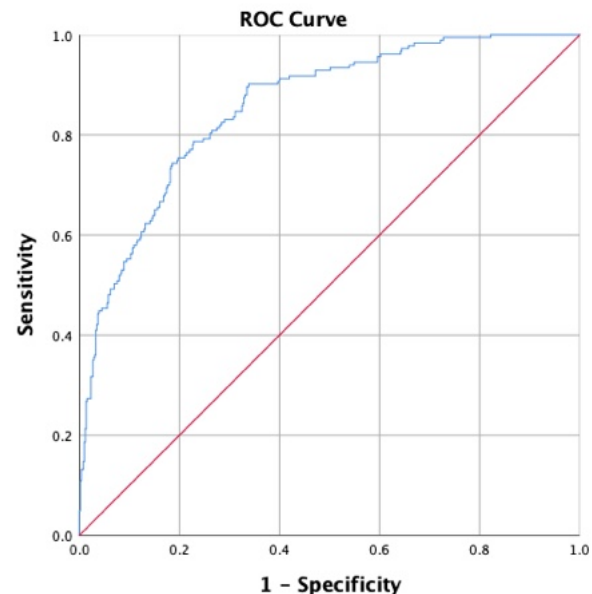
An ROC curve is often created to graphically display the index of accuracy of the model for a range of cutoff values.

Use Analyze>ROC

Test Variable = Predicted default
(from one or more models)

State 1=yes

State Variable = Previously
defaulted (default)



AUC = 0.856

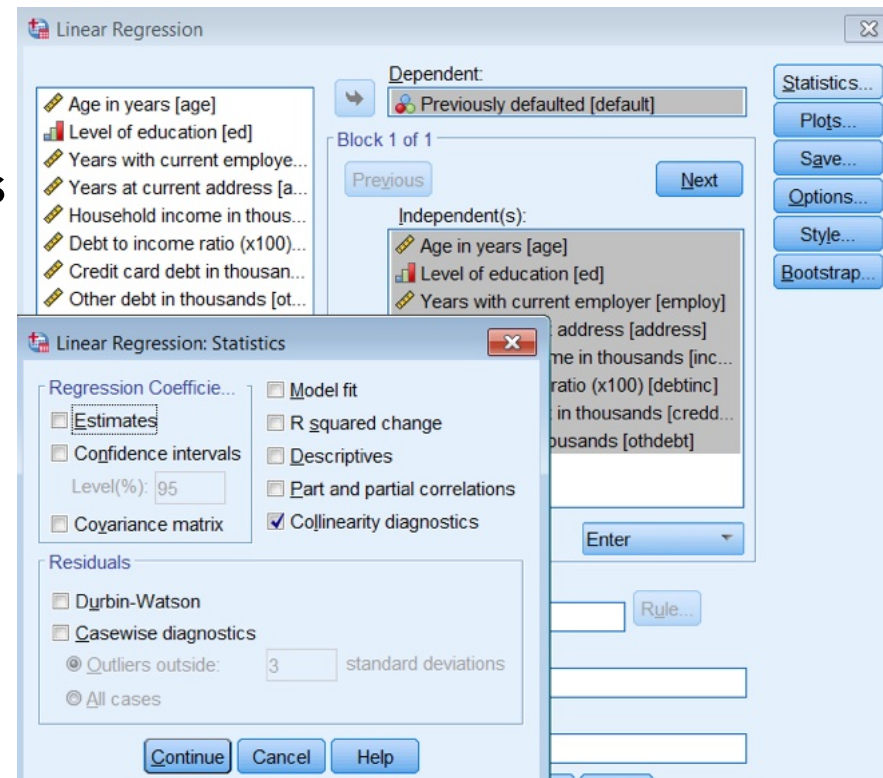
4. Binary Logistic Regression

Step 13: Multicollinearity (not in the SPSS tutorial)

The SPSS example does not mention checking for multicollinearity and the diagnostic tests are not available in the binary logistic regression procedure.

To check for collinearity of predictor variables, run the linear regression procedure and select “collinearity diagnostics”.

Check as per linear regression.



4. Binary Logistic Regression

What to report?

- Binary logistic regression model, forward LR selection method
- Model statement (optional)
- $\log\left(\frac{P(\text{default})}{P(\text{no default})}\right) = \beta_0 + \beta_1 \text{Emp}(\text{yr}) + \beta_2 \text{Address}(\text{yr}) + \beta_3 \text{DebtIncRatio}(x100) + \beta_4 \text{CreditCard}(x\$1000)$
- Table of regression coefficients

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 4 ^d	Years with current employer	-.243	.028	74.761	1	.000	.785
	Years at current address	-.081	.020	17.183	1	.000	.922
	Debt to income ratio (x100)	.088	.019	22.659	1	.000	1.092
	Credit card debt in thousands	.573	.087	43.109	1	.000	1.774
	Constant	-.791	.252	9.890	1	.002	.453

4. Binary Logistic Regression

What to report?

Interpretation of regression coefficients

- Odds of default change (decrease) by 0.78 times for each additional year with current employer
- Odds of default change (decrease) by 0.92 times for each additional year at current address
- Odds of default change (increase) by 1.09 times for each increase of 1% point of debt to income ratio
- Odds of default change (increase) by 1.77 times for each additional \$1000 of credit card debt

Also report beta coefficients and standard errors

4. Binary Logistic Regression

What to report?

Classification Table

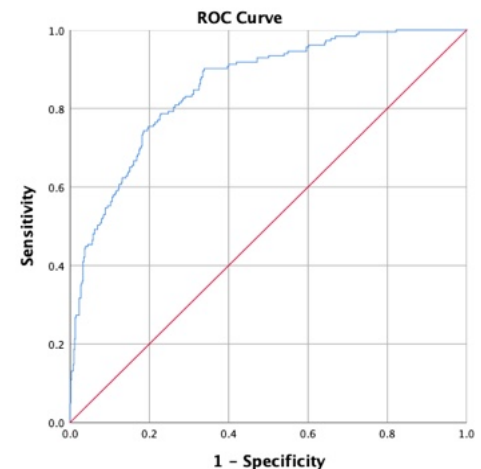
(with choice of cutoff value)

Classification Table^a

Observed			Predicted		Percentage Correct
			Previously defaulted No	Yes	
Step 4	Previously defaulted	No	478	39	92.5
		Yes	91	92	50.3
	Overall Percentage				81.4

a. The cut value is .500

Also include ROC curve with AUC.



AUC = 0.856

4. Binary Logistic Regression

Binary logistic regression - so easy and so impressive!



Generalized Linear Models

The Generalized Linear Models procedure expands the general linear model so that the dependent variable is linearly related to the factors and covariates via a specified link function. Moreover, the model allows for the dependent variable to have a non-normal distribution.

The generalized linear model covers widely used statistical models such as linear regression for normally distributed responses, logistic models for binary data, and loglinear models for count data, plus many other statistical models through its very general model formulation.

Generalized Linear Models

Common types of **GLM**:

- Logistic regression – binary, ordinal
- Poisson loglinear
- Negative binomial
- Interval censored survival

Generalized Linear Models

Generalized Linear Models – main dialog box

Generalized Linear Models

Type of Model | Response | Predictors | Model | Estimation | Statistics | EM Means | Save | Export

Choose one of the model types listed below or specify a custom combination of distribution and link function.

Scale Response

- ☒ Linear
- ☐ Gamma with log link

Ordinal Response

- ☐ Ordinal logistic
- ☐ Ordinal probit

Counts

- ☐ Poisson loglinear
- ☐ Negative binomial with log link

Binary Response or Events/Trials Data

- ☒ Binary logistic
- ☐ Binary probit
- ☐ Interval censored survival

Mixture

- ☐ Tweedie with log link
- ☐ Tweedie with identity link

Custom

- ☐ Custom

Distribution: Normal Link function: Identity

Alternative procedure to Regression>binary logistic

Generalized Linear Models

Generalized Estimating Equations – main dialog box

Generalized Estimating Equations

Repeated Type of Model Response Predictors Model Estimation Statistics EM Means Save Export

Variables:

- Age in years [a...]
- Age category [a...]
- Gender [gender]
- History of diabe...
- Blood pressure ...
- Smoker [smoker]
- Cholesterol [ch...]
- Physically activ...
- Obesity [obesity]
- History of angin...
- History of myoc...
- Prescribed nitro...
- Taking anti-clott...
- Hospital ID [site]
- Attending physi...
- Time to hospital...
- Dead on arrival ...
- EKG result [ekg]
- CPK blood resu...
- Troponin T bloo...
- Clot-dissolving ...
- Hemorrhaging [...]
- Magnesium [m...]
- Digitalis [digi]
- Beta blockers [...]

Subject variables:

Within-subject variables:

☒ Sort cases by subject and within-subject variables

Covariance Matrix

☒ Robust estimator ☐ Model-based estimator

Working Correlation Matrix

Structure: Independent M:

☒ Adjust estimator by number of non-redundant parameters

Maximum iterations: 100

☒ Update matrix Iterations between updates: 1

Convergence Criteria

At least one convergence criterion must be specified with a minimum greater than zero.

☒ Change in parameter estimates Minimum: 1E-006 Type: Absolute

☐ Hessian convergence Minimum: Type: Absolute

OK Paste Reset Cancel Help

Alternative procedure to GLM Repeated Measures

Linear models in SPSS - overview

Summary...

- Linear models (including Generalized LMs) can suit a wide variety of experimental conditions.
- Selection of the right model allows the researcher to make the best use of the available experimental data.
- Sometimes the selection of a choice of model is very much a value judgement, and a simple model may be preferred even if some information is lost.

Linear models in SPSS – other resources

Tools to help and guide your analysis

- The roadmap
- SPSS case studies and online help
- Sydney Informatics Hub
- Good textbooks, eg Andy Field

General Linear Models using SPSS – end of workshop

Jim Matthews BEng MStat | Senior Consultant: Statistics
The University of Sydney
Sydney Informatics Hub | Core Research Facilities
32 Queen St Chippendale | The University of Sydney | NSW | 2006
+61 412 246 271
Jim.Matthews@sydney.edu.au | sydney.edu.au